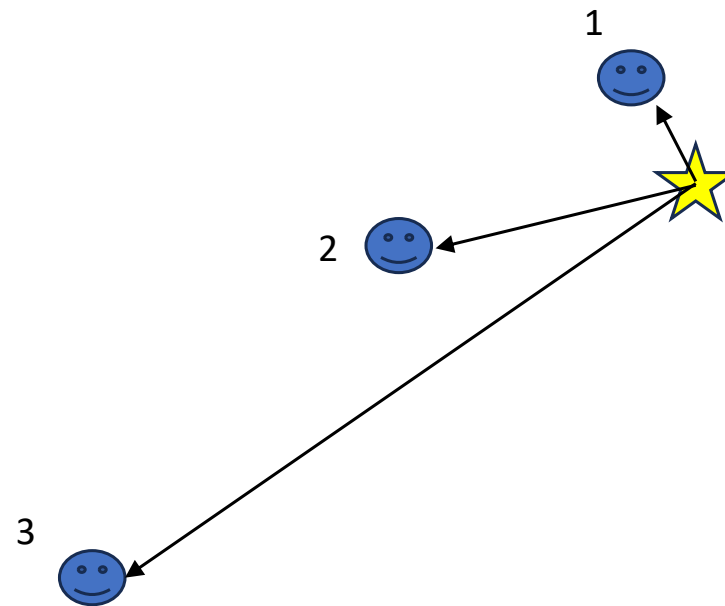


An Introduction to Radio Interferometry

線上英語課程(<https://github.com/baobabyoo/interferometryEMI>)
(<https://baobabyoo.github.io/pages/lectures.html#radio-interferometry>)
(https://baobabyoo.github.io/pages/students_topics/UsingTelescope.html)

呂浩宇(中山大學)、賴品憲(群聯電子)、陳亭蓁(清華大學)

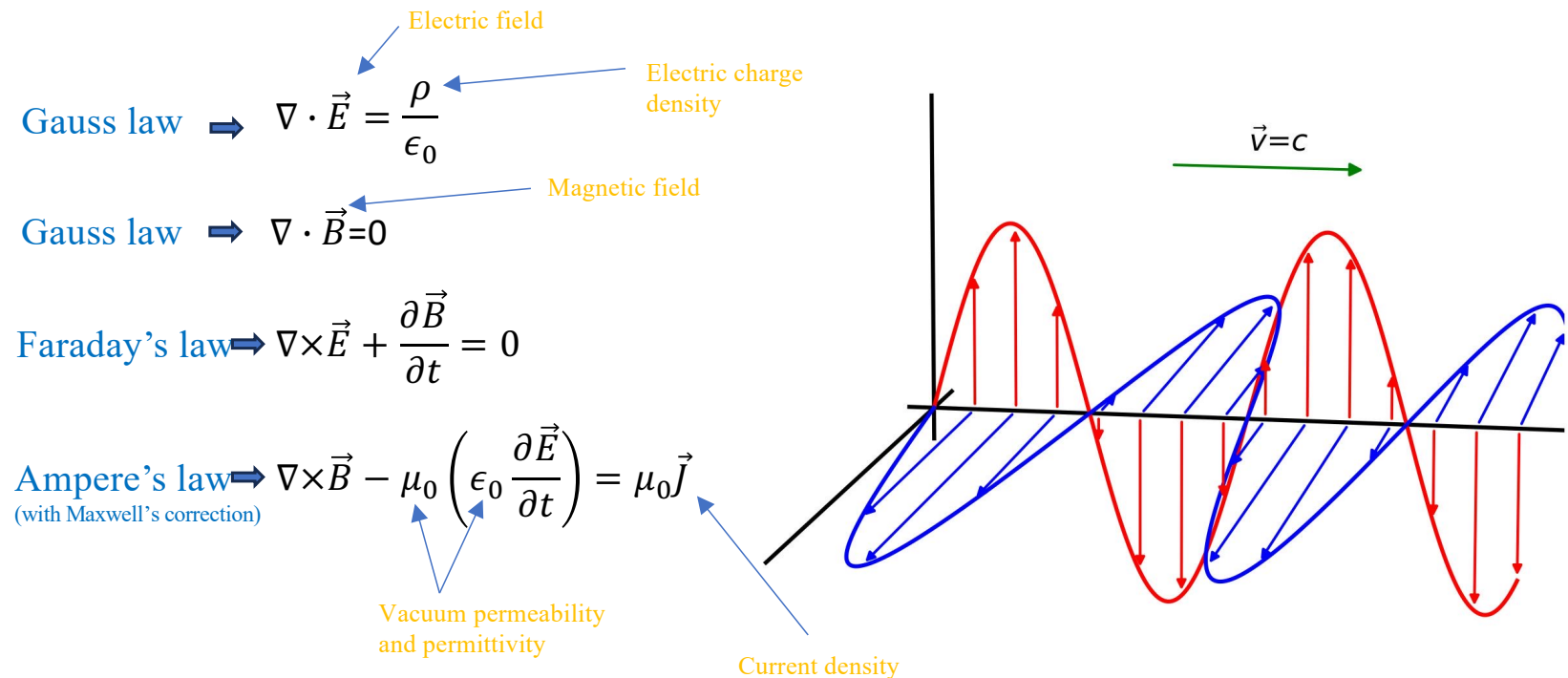
Emergence of electric (and magnetic field) in the observational devices.



Classically, light can be described as **electromagnetic waves (EM waves)**:

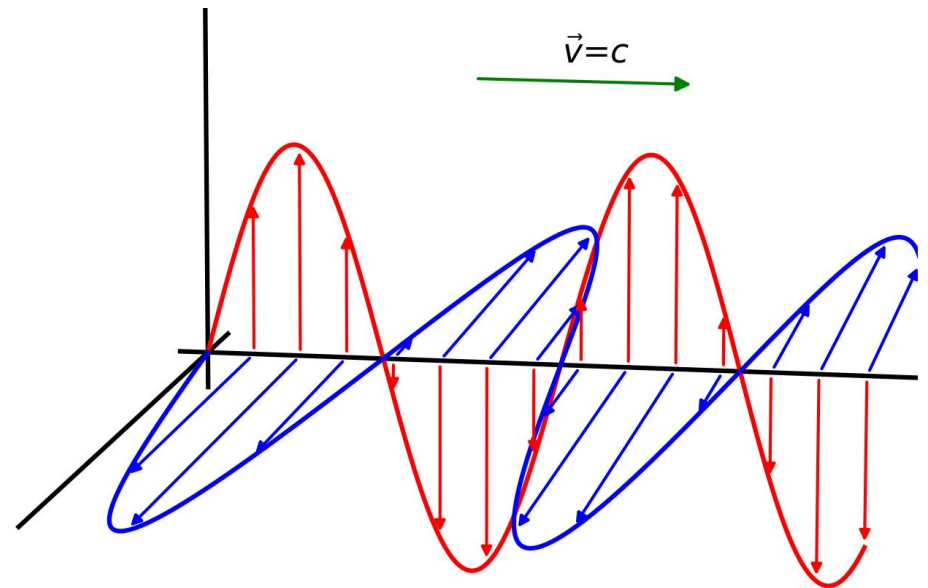
(i) linear perturbations of electric and magnetic fields

(ii) propagation obeys Maxwell's equations



Monochromatic plane wave

$$\begin{cases} E = E_m \sin(kx - \omega t + \phi) \\ B = B_m \sin(kx - \omega t + \phi) \end{cases}$$

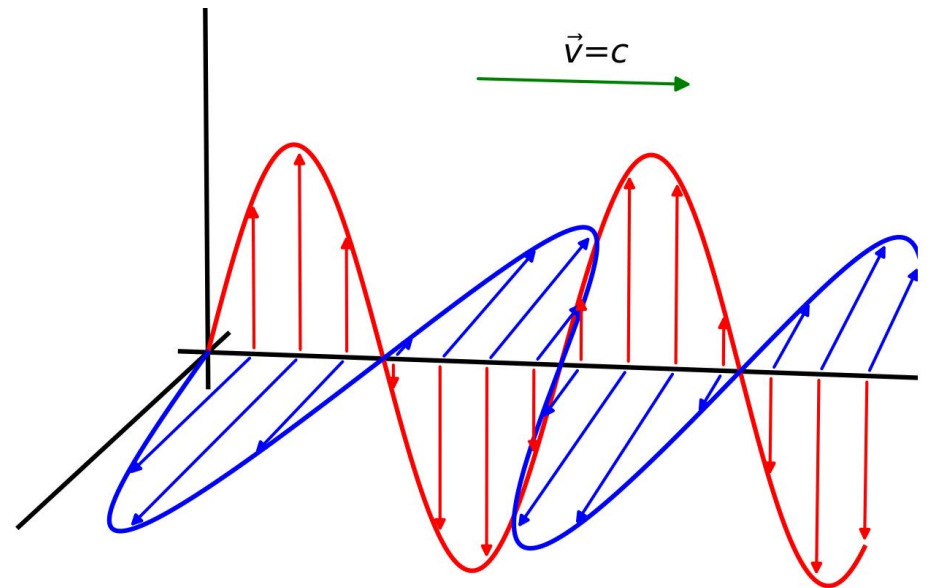


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$$\omega = 2\pi\nu = \frac{2\pi}{T}$$

$$k = \frac{2\pi}{\lambda}$$



Monochromatic plane wave

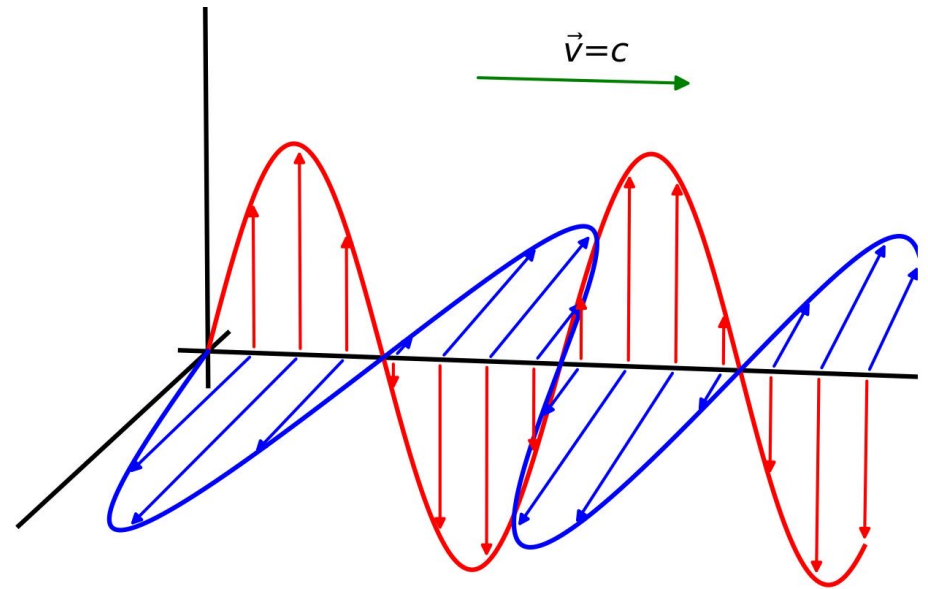
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$\omega = 2\pi\nu = \frac{2\pi}{T}$

$k = \frac{2\pi}{\lambda}$

(time-averaged) Flux density:

$$F = \frac{1}{2} \sqrt{\frac{\epsilon_0}{\mu_0}} |E_m|^2$$

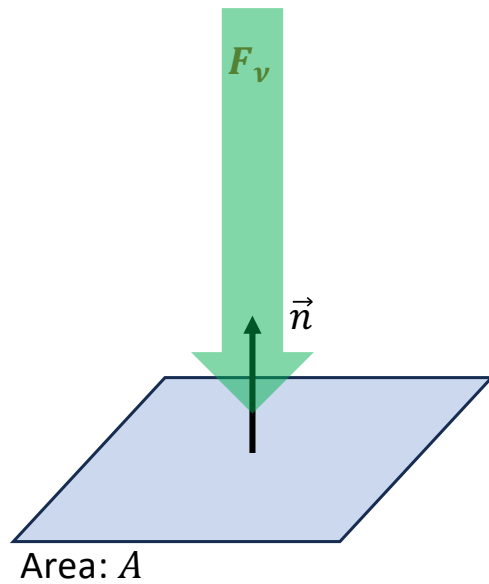


Flux density as a vector field

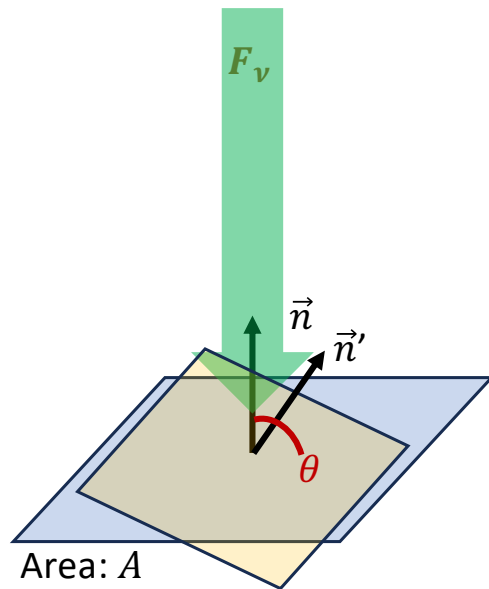
Normal incidence

Energy passing a CCD pixel that has area A in a unit of time

$$F_v A$$



Flux density as a vector field



Normal incidence

Energy passing a CCD pixel that has area A in a unit of time

$$F_v A$$

Inclined CCD pixel

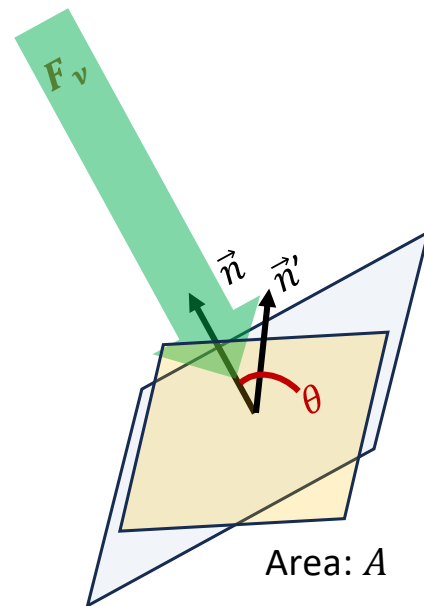
Energy passing a CCD pixel that has area A in a unit of time

$$F_v A (\vec{n} \cdot \vec{n}') = F_v A \cos \theta$$

$$\vec{n} \cdot \vec{n}' = \cos \theta$$

Flux density as a vector field

Inclined incident photon streams



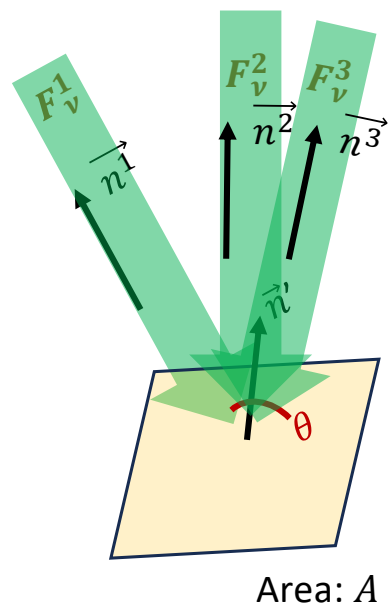
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Flux density as a vector field

Inclined, discrete incident photon streams

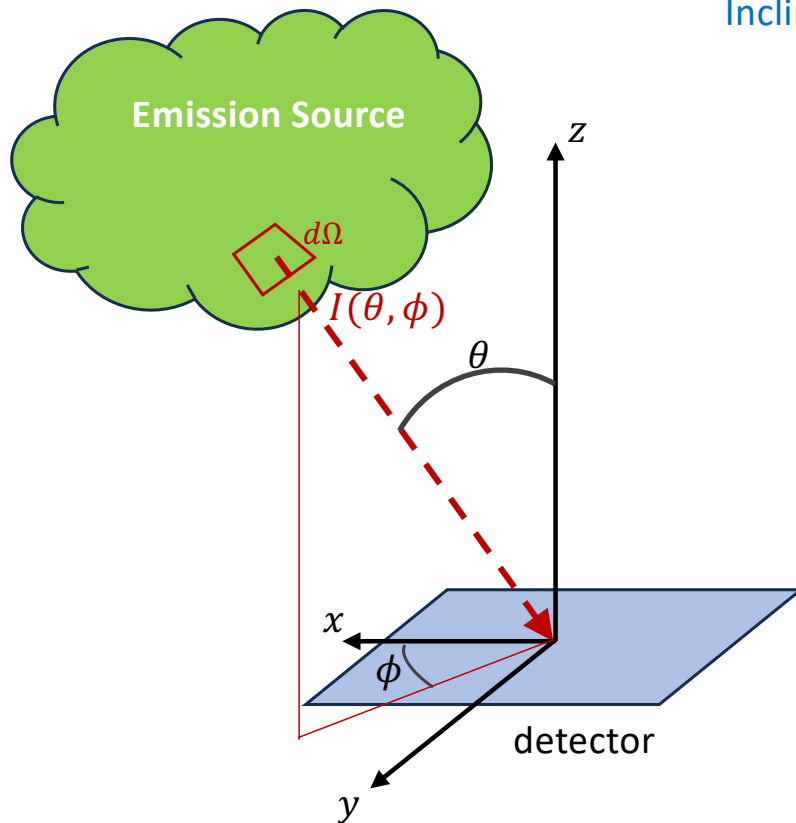


Energy passing a CCD pixel that has area A in a unit of time

$$\sum_i F_v^i A (\vec{n}^i \cdot \vec{n}') = \sum_i F_v^i A \cos \theta^i$$

$$\vec{n}^i \cdot \vec{n}' = \cos \theta^i$$

Continuous incident light



Inclined, discrete incident photon streams

Energy passing a CCD pixel that has area A in a unit of time

$$\sum_i F_v^i A (\vec{n}^i \cdot \vec{n}') = \sum_i F_v^i A \cos \theta^i$$

infinitesimal solid angle: $d\Omega \equiv \sin \theta d\theta d\phi$

Intensity $I(\theta, \phi)$: the amount of energy through a unit of area A in a unit of time t from a unit solid angle $d\Omega$ around the direction (θ, ϕ) [i.e., flux density per unit solid angle]

SI Unit: $\text{Joul s}^{-1} \text{m}^{-2} \text{Hz}^{-1} \text{Sr}^{-1}$

Flux density normal to our CCD pixel:

$$\int I_v(\theta, \phi) \cos \theta d\Omega$$

Flux density (incoming energy per unit time, frequency, and area) received by a CCD pixel:

$$F_v = \frac{1}{2} \int I_v(\theta, \phi) \cos \theta d\Omega$$

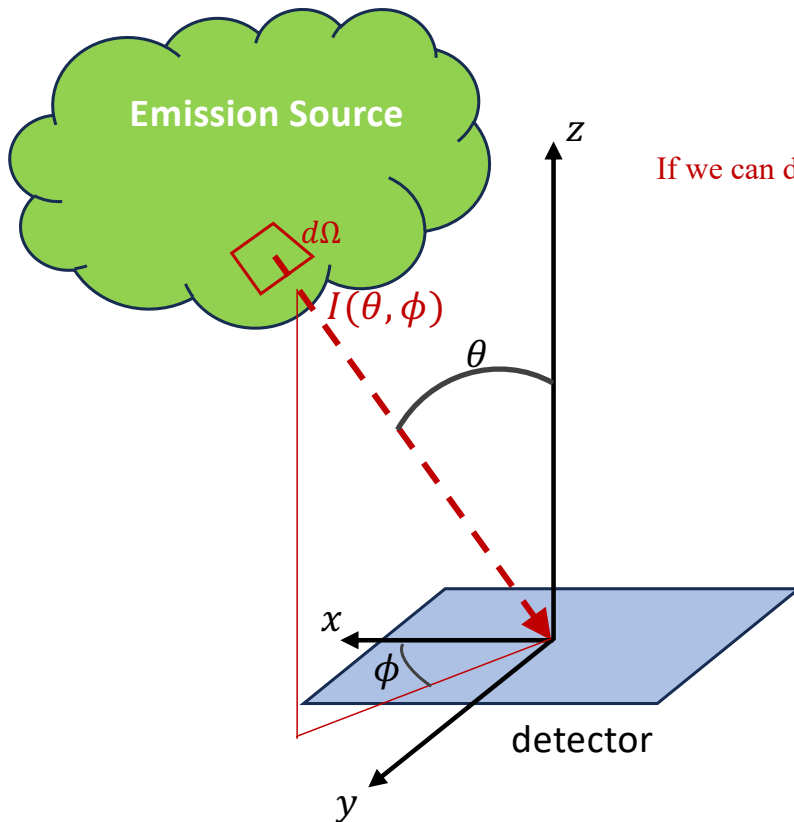
If we can detect one of the two polarizations

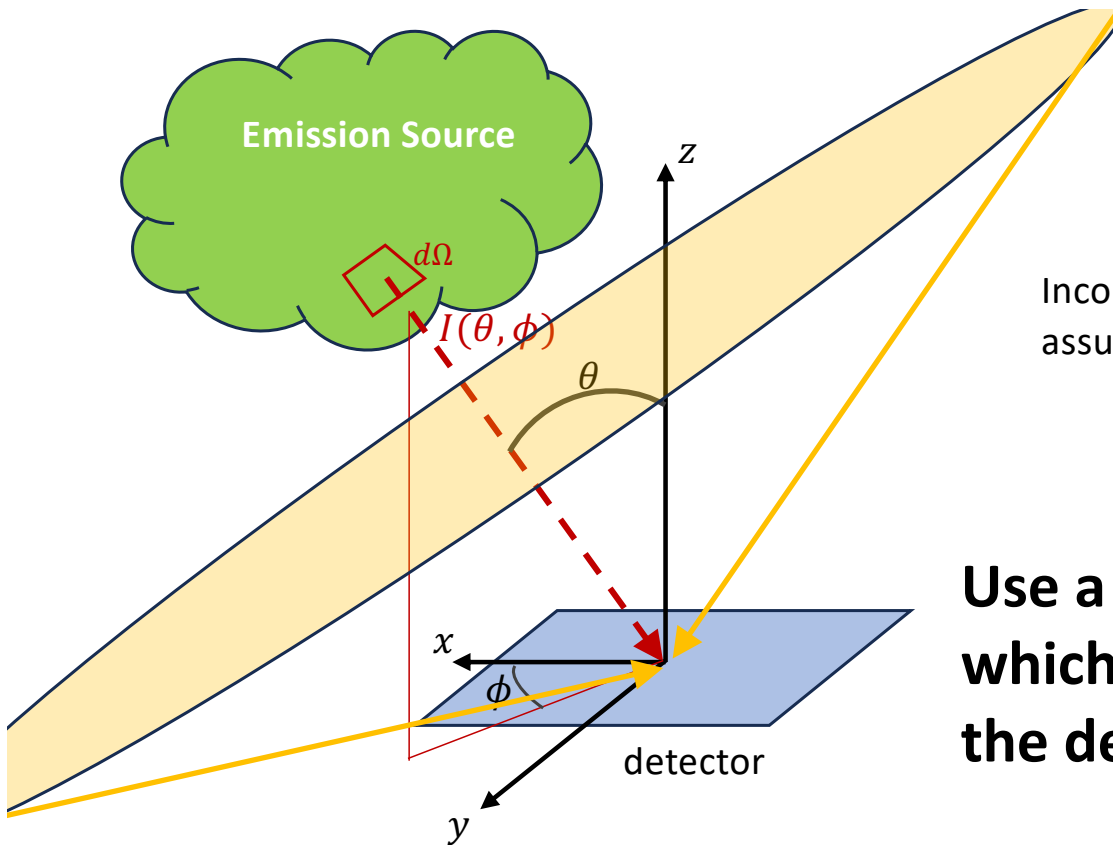
Incoming energy per unit time and frequency (i.e., power if assuming monochromatic light) received by a CCD pixel:

$$P_v = \frac{1}{2} A \int I_v(\theta, \phi) \cos \theta d\Omega$$

surface area of the detector

We receive more power in a unit of time (i.e., our device is more sensitive to weak signals) when the effective surface area of the detector is larger.



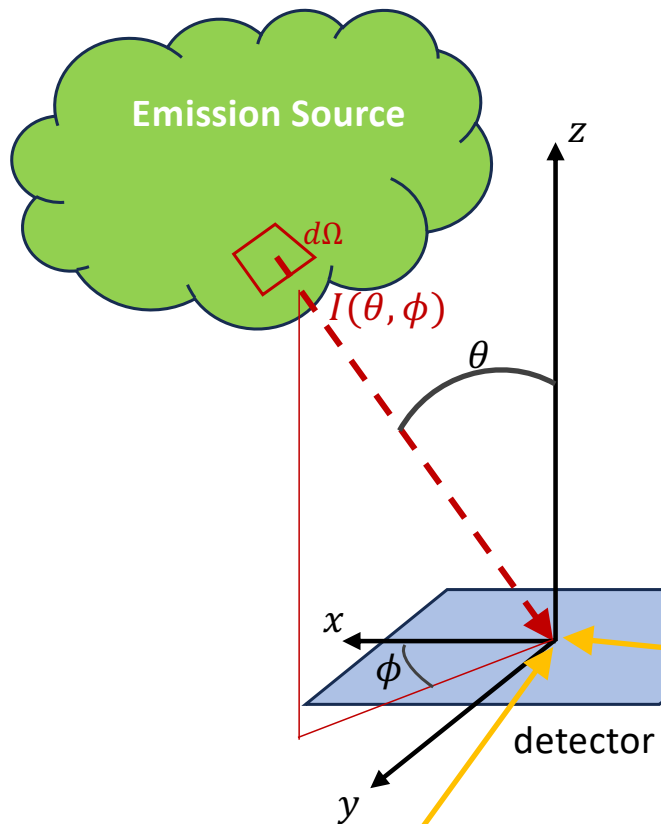


Incoming energy per unit time and frequency (i.e., power if assuming monochromatic light) received by a CCD pixel:

$$P_\nu = \frac{1}{2} A_e \int I_\nu(\theta, \phi) \cos \theta d\Omega$$

Use a lense to collect light from a larger area, which increases the effective surface area of the detector

We receive more power in a unit of time (i.e., our device is more sensitive to weak signals) when the effective surface area of the detector is larger.

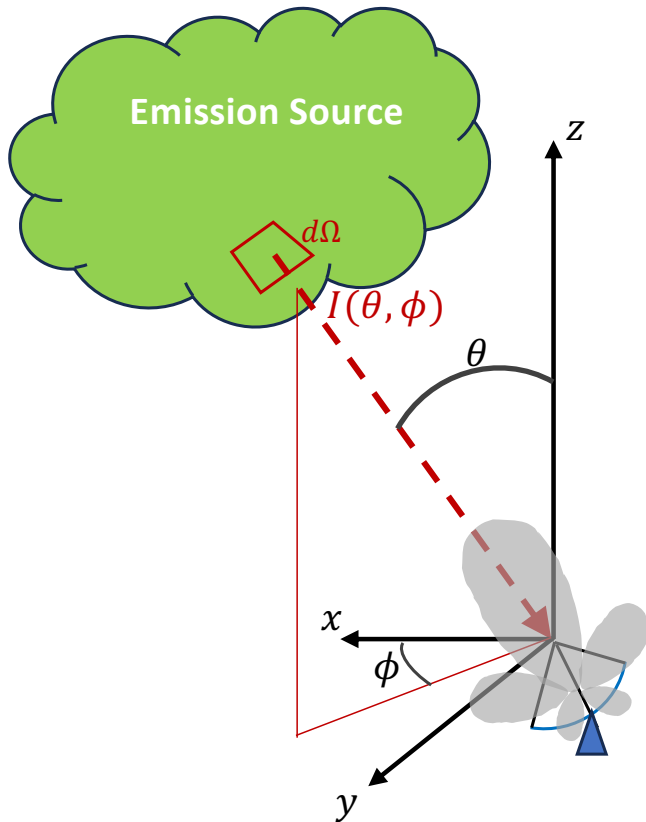


Incoming energy per unit time and frequency (i.e., power if assuming monochromatic light) received by a CCD pixel:

$$P_\nu = \frac{1}{2} A_e \int I_\nu(\theta, \phi) \cos \theta d\Omega$$

Use a reflector to collect light from a larger area, which increases the effective surface area of the detector

We receive more power in a unit of time (i.e., our device is more sensitive to weak signals) when the effective surface area of the detector is larger.



Incoming energy per unit time and frequency (i.e., power if assuming monochromatic light) received by a CCD pixel:

$$P_v = \frac{1}{2} A_e \int I_v(\theta, \phi) \cos \theta \underbrace{P(\theta, \phi)}_{\text{Response function of the observing device.}} d\Omega$$

Response function of the observing device.
Normalization: peak = 1.0 (i.e., 100% of incoming intensity can be detected.)

The optics determines the response function.
With a telescope, we:

- (1) Receive light from smaller solid angle(s)
(usually, provides better angular resolution)
- (2) Have a bigger collecting area

We receive more power in a unit of time (i.e., our device is more sensitive to weak signals) when the effective surface area of the detector is larger.

Example: 2D Gaussian beam

with Full width at half maximum (FWHM) θ_a and θ_b in the major and minor axes

(the relation between Gaussian standard deviation σ and FWHM θ

$$\theta = 2\sqrt{2\ln 2}\sigma \sim 2.35\sigma)$$

$$P = e^{\left(\frac{-\xi^2}{2\sigma_a^2} + \frac{-\eta^2}{2\sigma_b^2}\right)} = e^{\left[4\ln 2 \left(\frac{-\xi^2}{(2\sqrt{\ln 2}\sigma_a)^2} + \frac{-\eta^2}{(2\sqrt{\ln 2}\sigma_b)^2}\right)\right]} = e^{\left[4\ln 2 \left(\frac{-\xi^2}{(\theta_a)^2} + \frac{-\eta^2}{(\theta_b)^2}\right)\right]}$$

$$\Omega_B = \int e^{\left[4\ln 2 \left(\frac{-\xi^2}{(\theta_a)^2} + \frac{-\eta^2}{(\theta_b)^2}\right)\right]} d\xi d\eta = \frac{\pi\theta_a\theta_b}{4\ln 2}$$

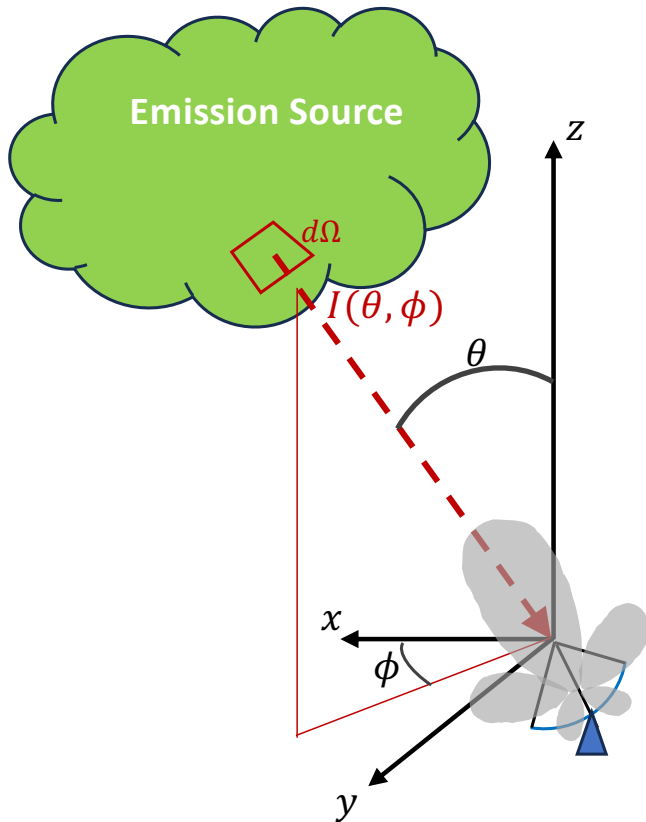
$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Beam solid angle:

$$\Omega_B = \int_0^{4\pi} P(\theta, \phi) d\Omega = \int_0^{4\pi} P(\theta, \phi) \sin \theta d\theta d\phi$$

Primary beam solid angle:

$$\Omega_{PB} = \int_{\text{Primary beam}} P(\theta, \phi) \sin \theta d\theta d\phi$$



Incoming energy per unit time and frequency (i.e., power if assuming monochromatic light) received by a CCD pixel:

$$P_v = A_e \int I_v(\theta, \phi) \cos \theta P(\theta, \phi) d\Omega$$

Response function of the observing device.

Normalization: peak = 1.0 (i.e., 100% of incoming intensity can be detected.)

$$\frac{P_v}{A_e} = \int I_v(\theta, \phi) \cos \theta P(\theta, \phi) d\Omega \sim \int I_v(\theta, \phi) P(\theta, \phi) d\Omega$$

$\theta \rightarrow 0 \Rightarrow \cos \theta \rightarrow 1$

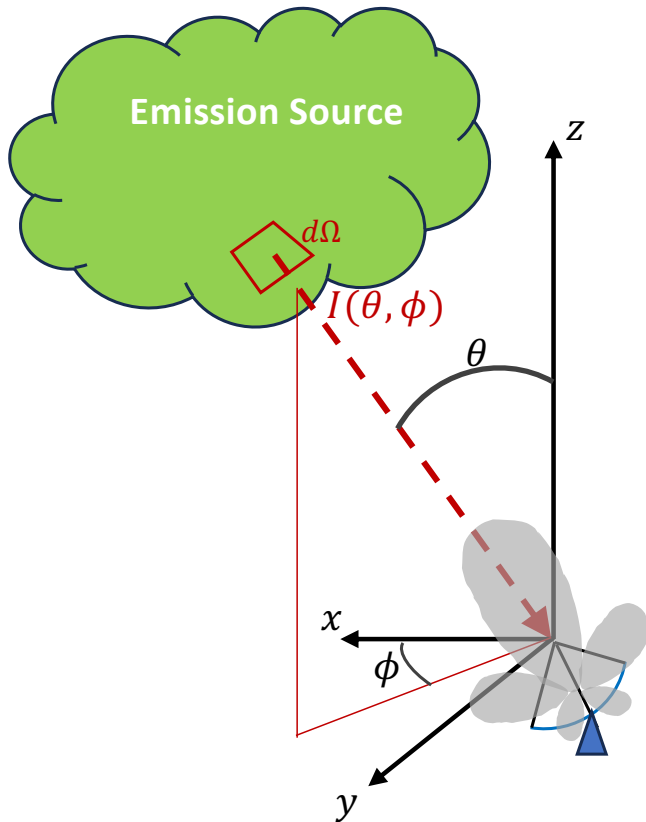
If $I_v(\theta, \phi)$ is approximately constant over the beam

$$\frac{P_v}{A_e} \sim I_v \int P(\theta, \phi) d\Omega = I_v \Omega_B = 1 \text{ Jy}$$

Solid angle in Sr unit

Flux density

Intensity in Jy Sr^{-1} unit



Incoming energy per unit time and frequency (i.e., power if assuming monochromatic light) received by a CCD pixel:

$$P_\nu = A_e \int I_\nu(\theta, \phi) \cos \theta P(\theta, \phi) d\Omega \equiv k T_A$$

Boltzman
n
constant

Antenna
temperature

Response function of the observing device.

Normalization: peak = 1.0 (i.e., 100% of incoming intensity can be detected.)

$$\frac{P_\nu}{A_e} = \int I_\nu(\theta, \phi) \cos \theta P(\theta, \phi) d\Omega \sim \int I_\nu(\theta, \phi) P(\theta, \phi) d\Omega$$

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If $I_\nu(\theta, \phi)$ is approximately constant over the beam

$$\frac{P_\nu}{A_e} = \sim I_\nu \int P(\theta, \phi) d\Omega = I_\nu \Omega_B = I_\nu^{Jy/beam} \times 1 = 1 Jy$$

Flux density

Intensity in Jy Sr⁻¹ unit

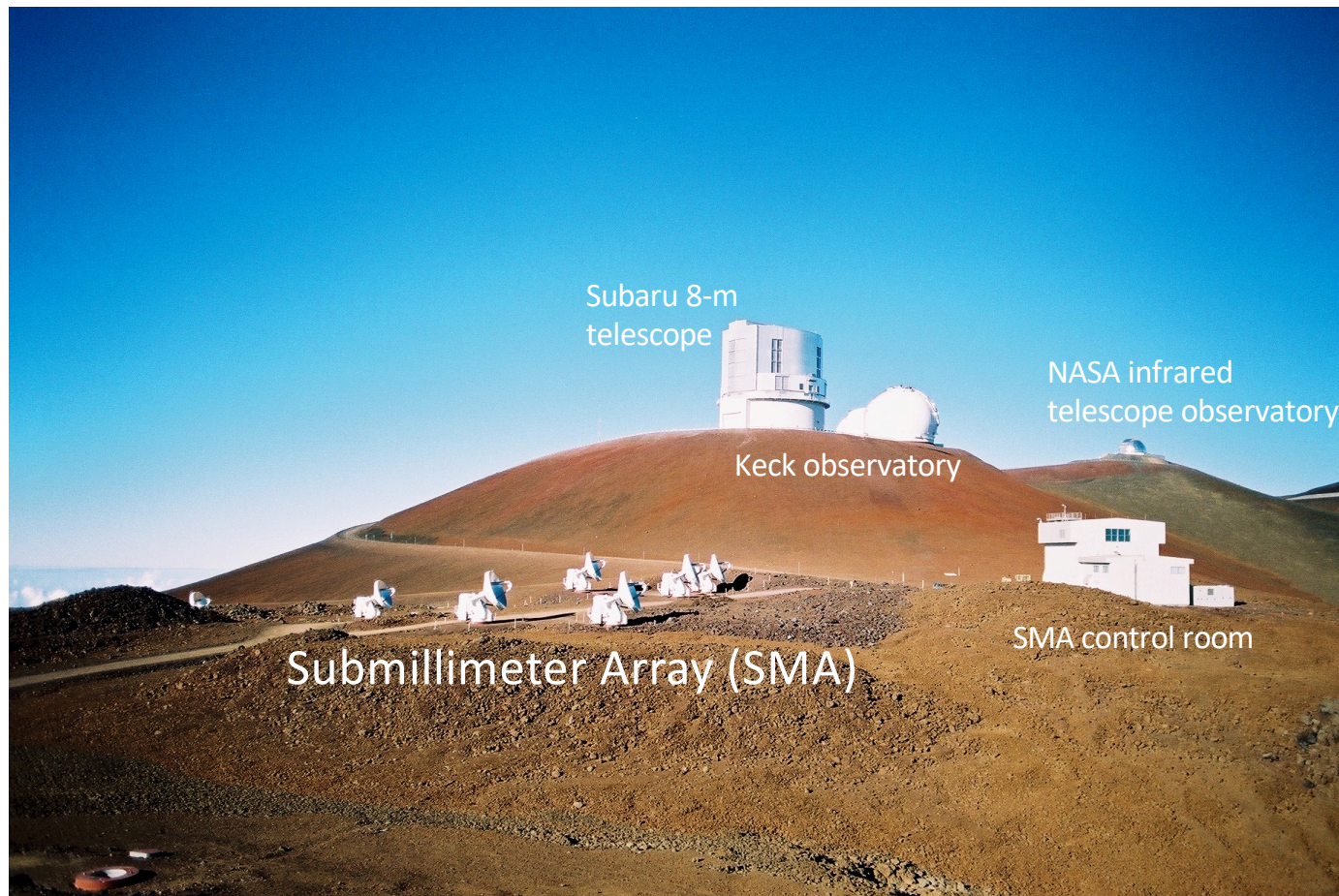
changing unit

Intensity in Jy beam⁻¹ unit

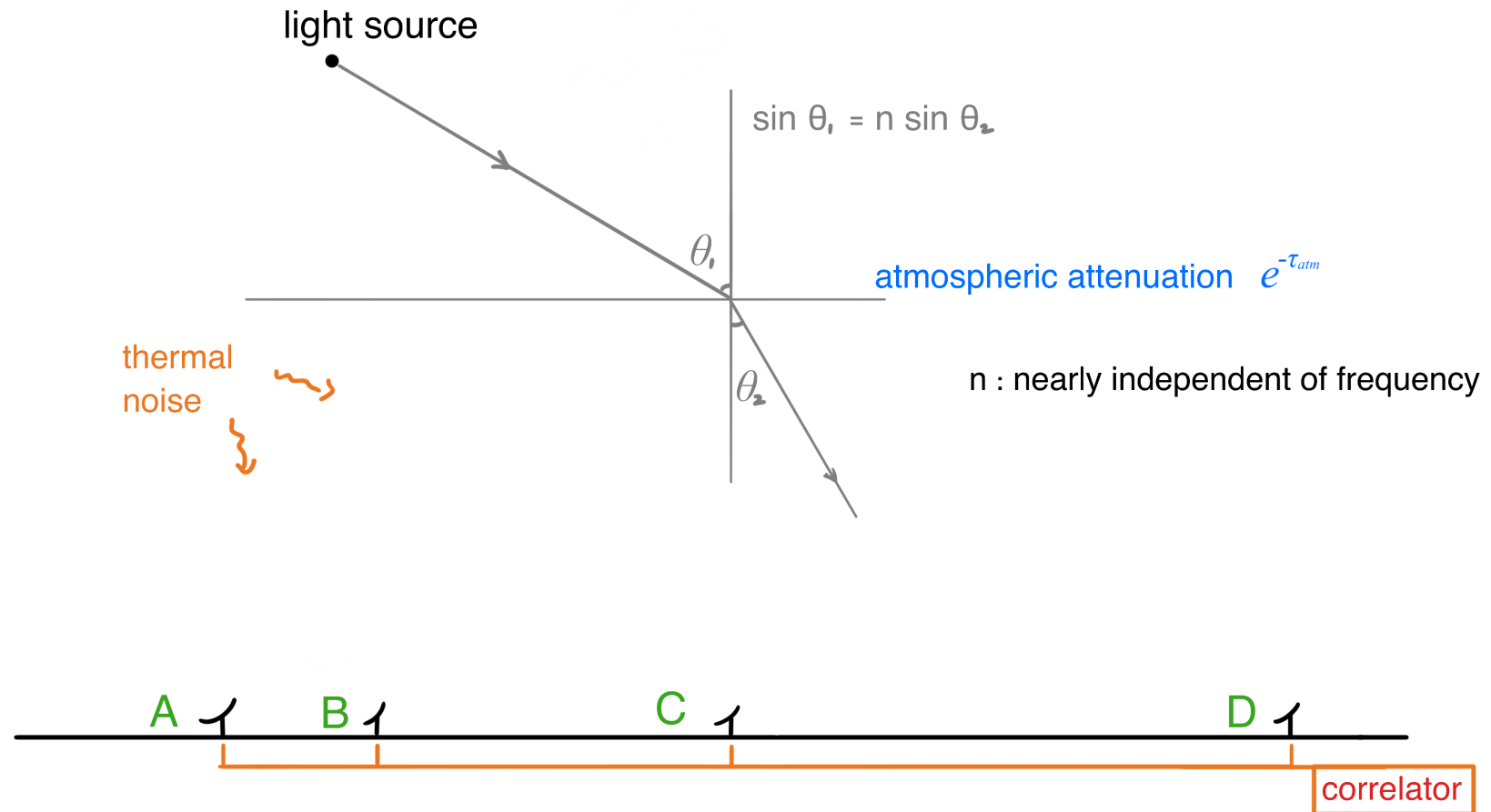
Solid angle in Sr unit

Solid angle in beam unit

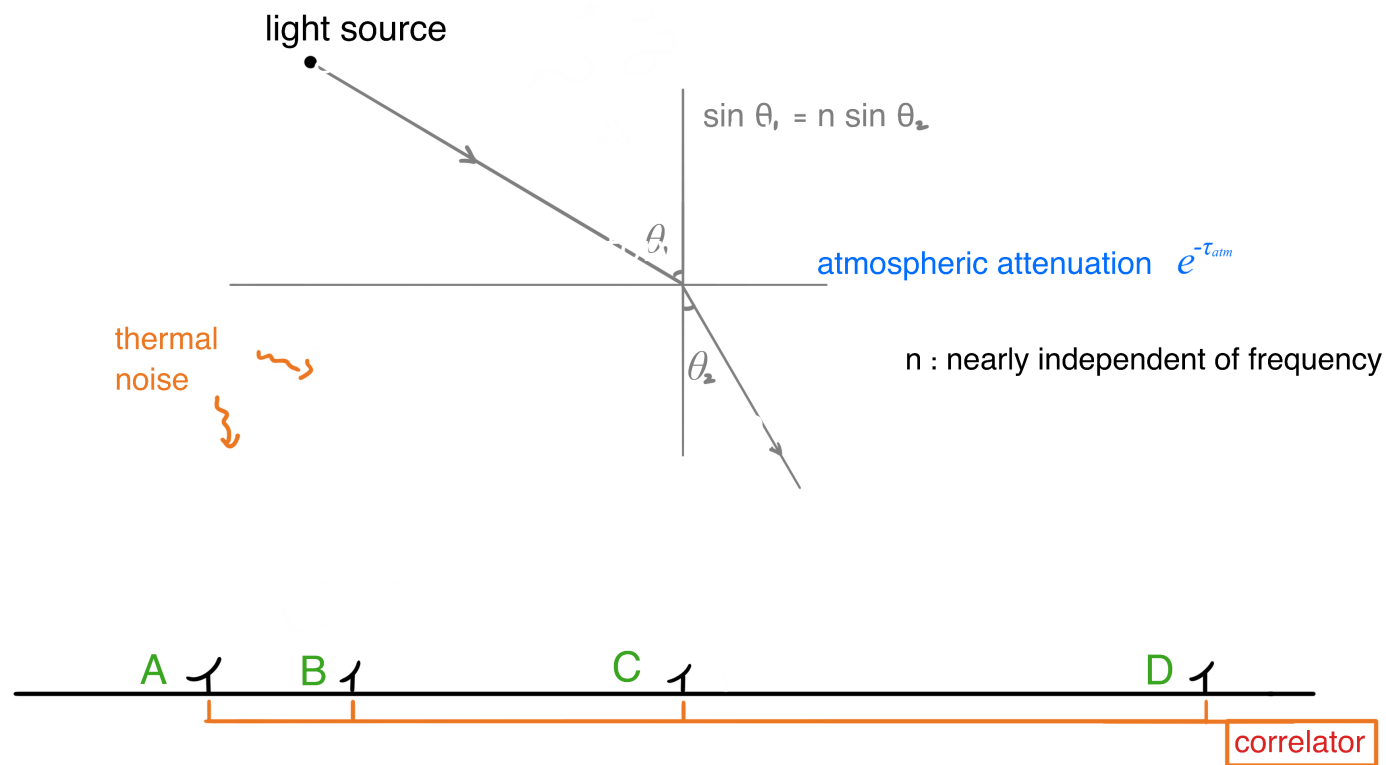
How a radio or (sub)millimeter interferometric observatory look like



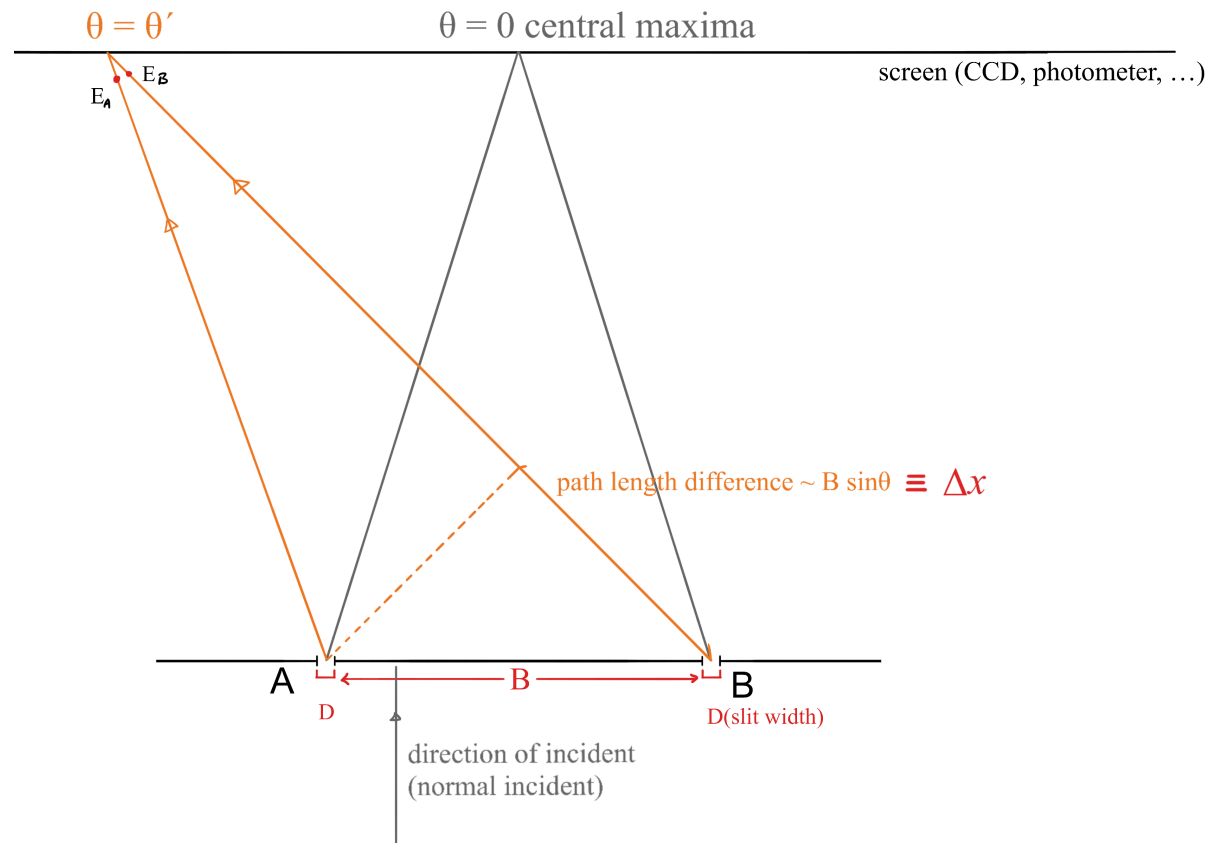
How a radio or (sub)millimeter interferometric observatory look like



干涉儀直接測量的物理量為Fourier transformed的intensity distribution

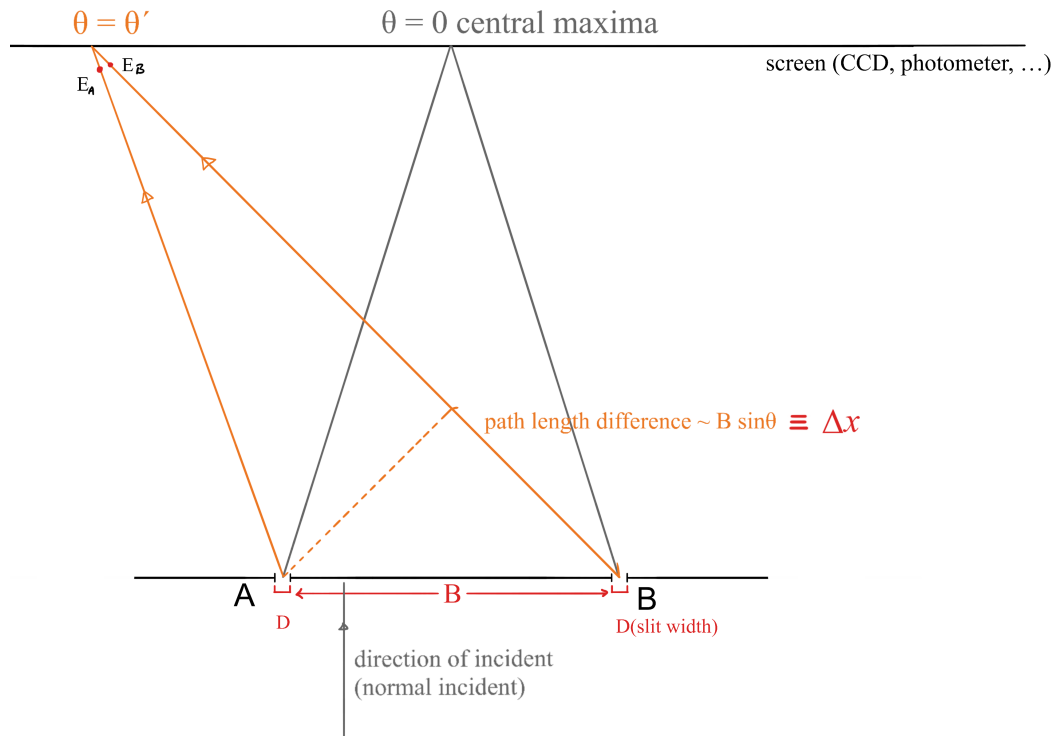


測量及成像之數學原理與光柵雷同。從最簡單情形，一維雙狹縫之成像來介紹。



1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
2. Energy flux density $\propto E_0^2$

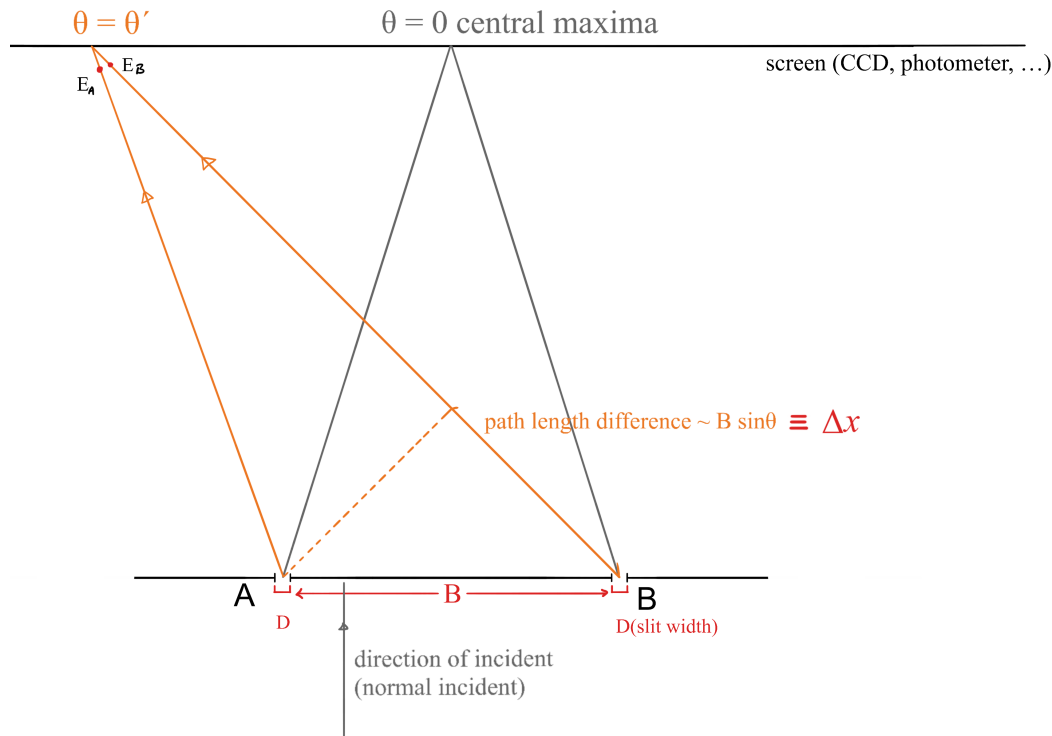
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Number of incoming photons in a unit of time and a unit area
(at the angle of emergence $\theta = \theta'$)

$$\begin{aligned}
 &\propto [E_A + E_B]^2 \\
 &= \tilde{P}(\theta) [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \\
 &= \tilde{P}(\theta) \{ [\cos(kx - \omega t + \phi_0)]^2 \\
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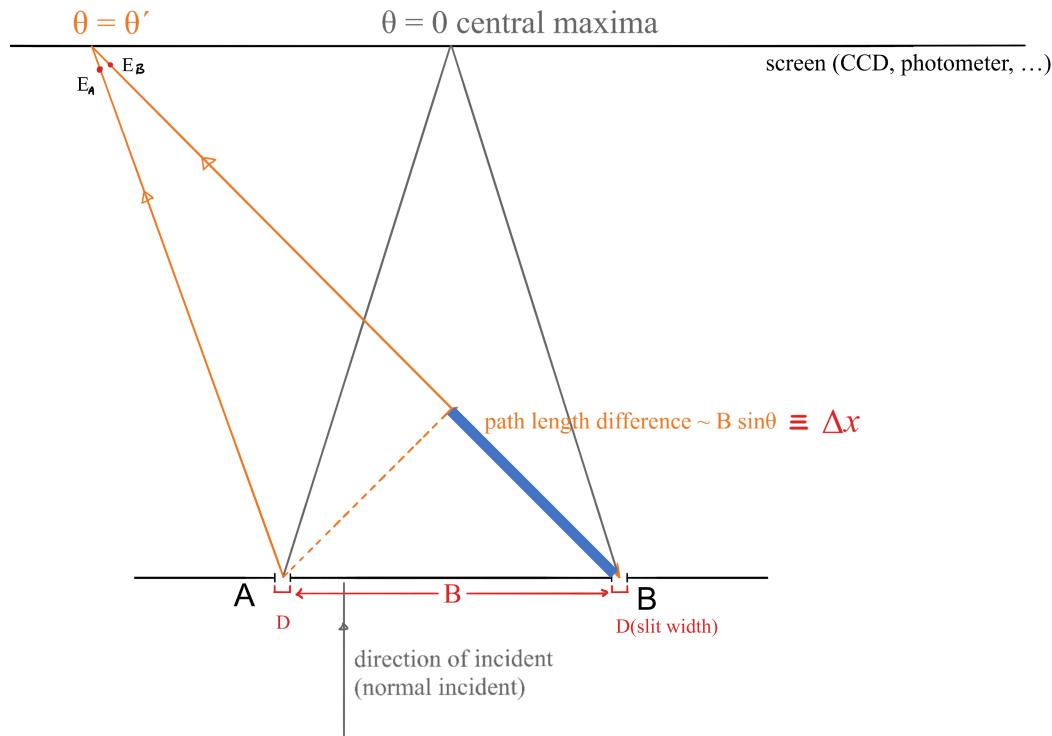
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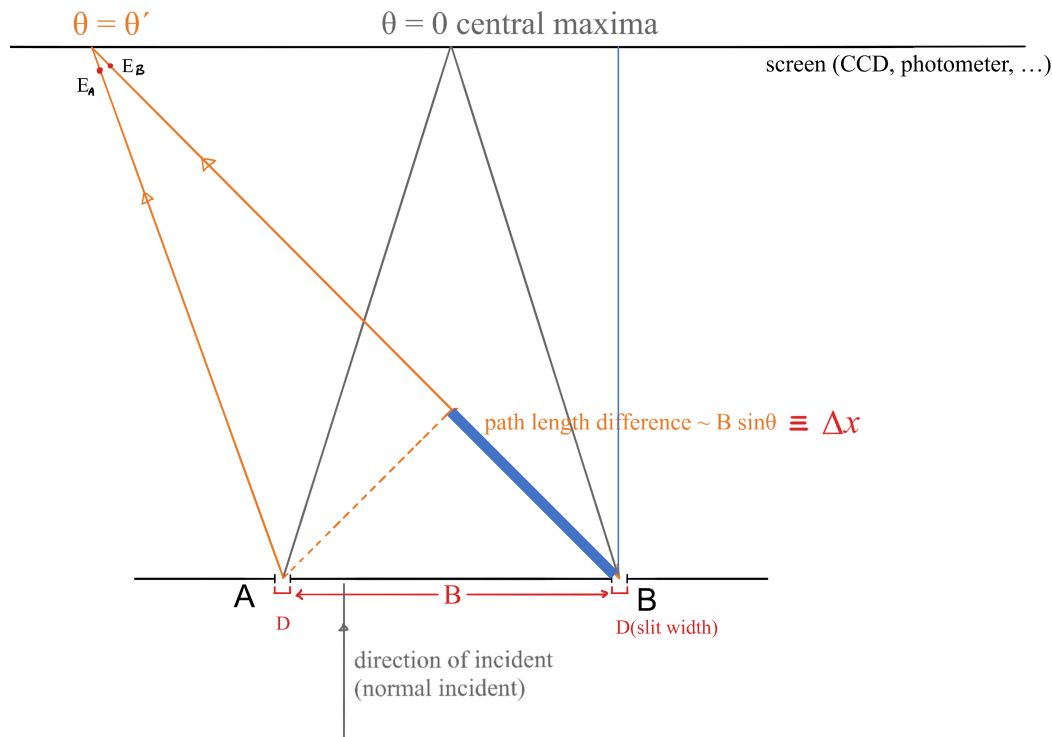
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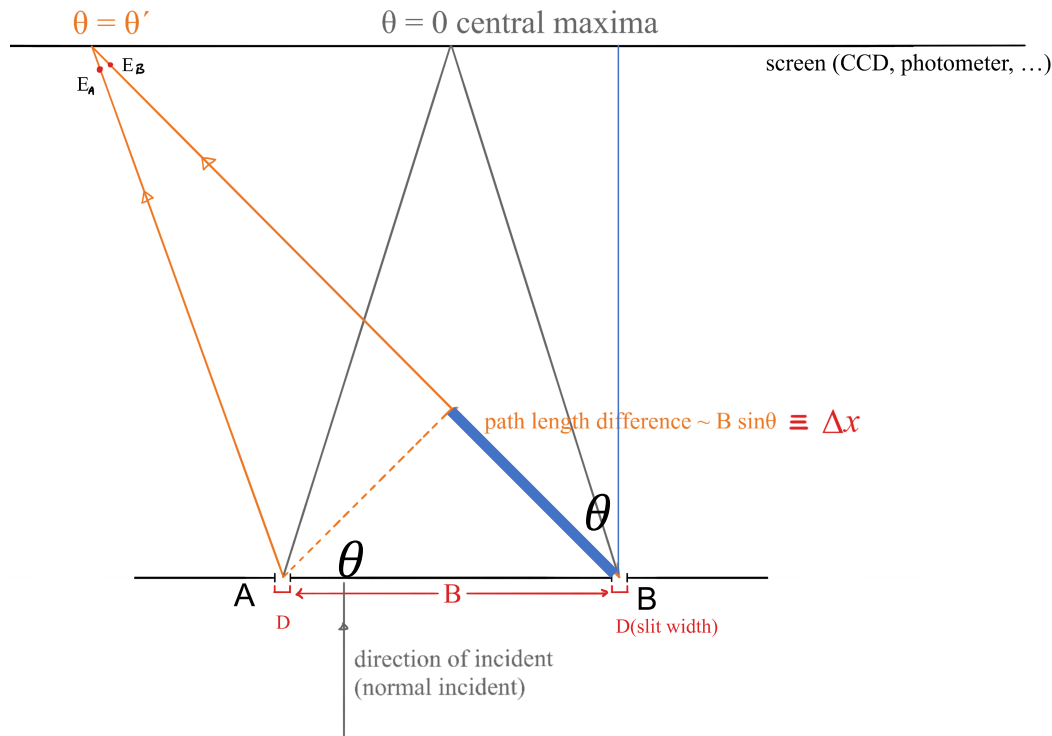
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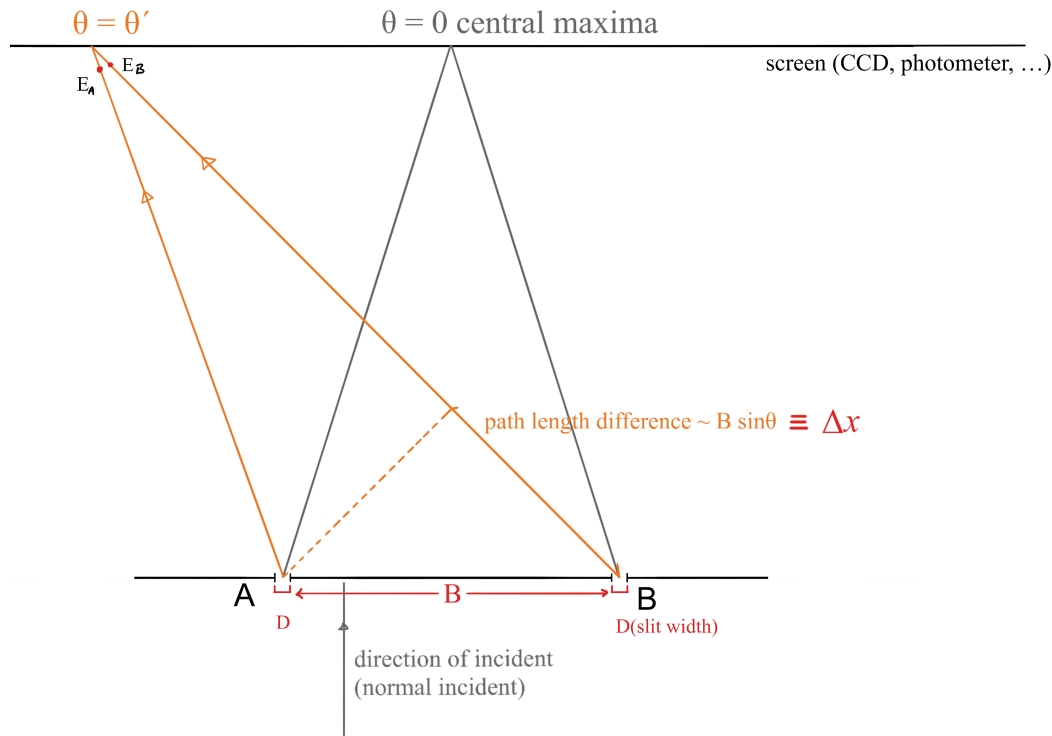
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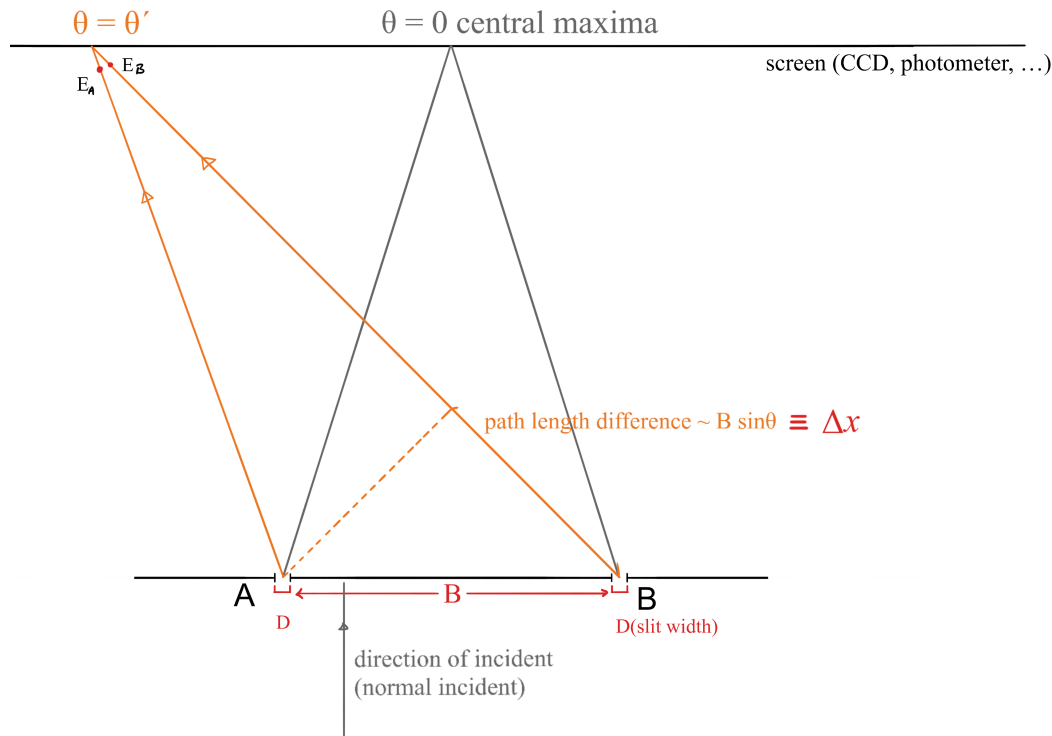
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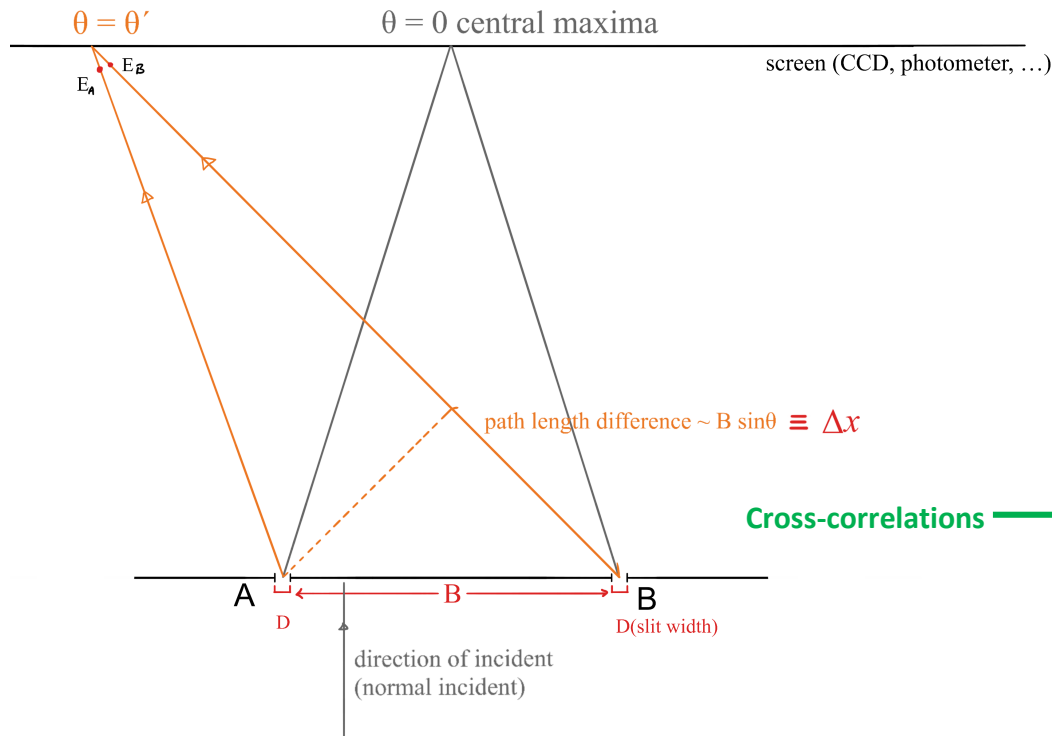


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$$\langle \cdot \rangle \equiv \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \cdot dt$$

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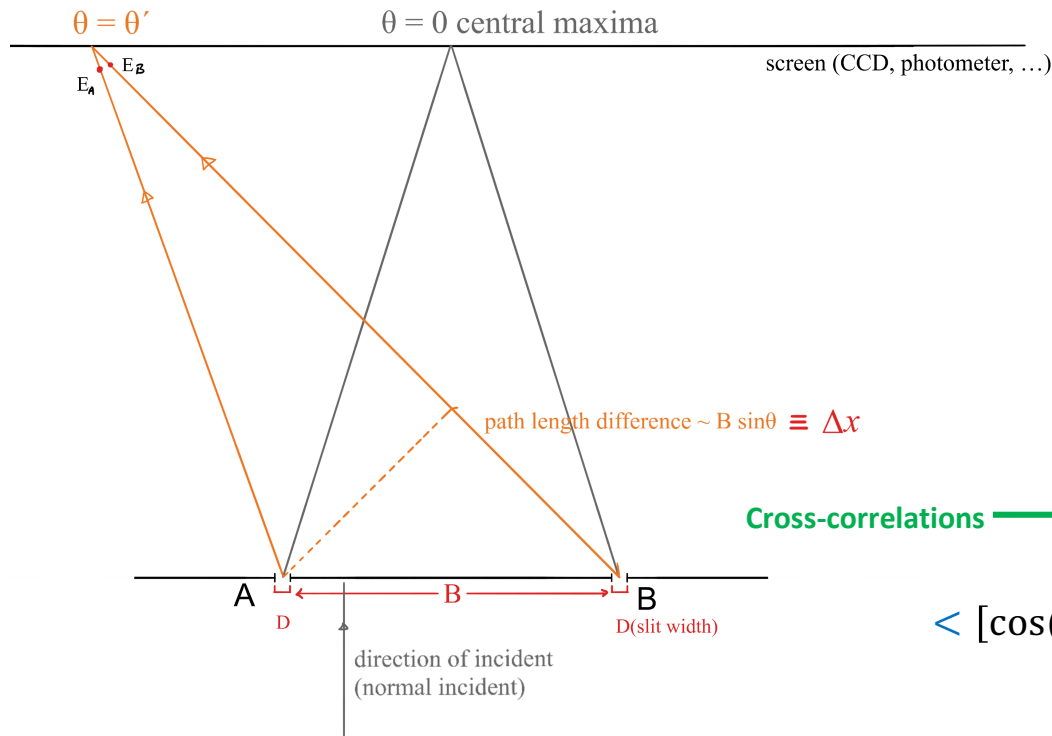


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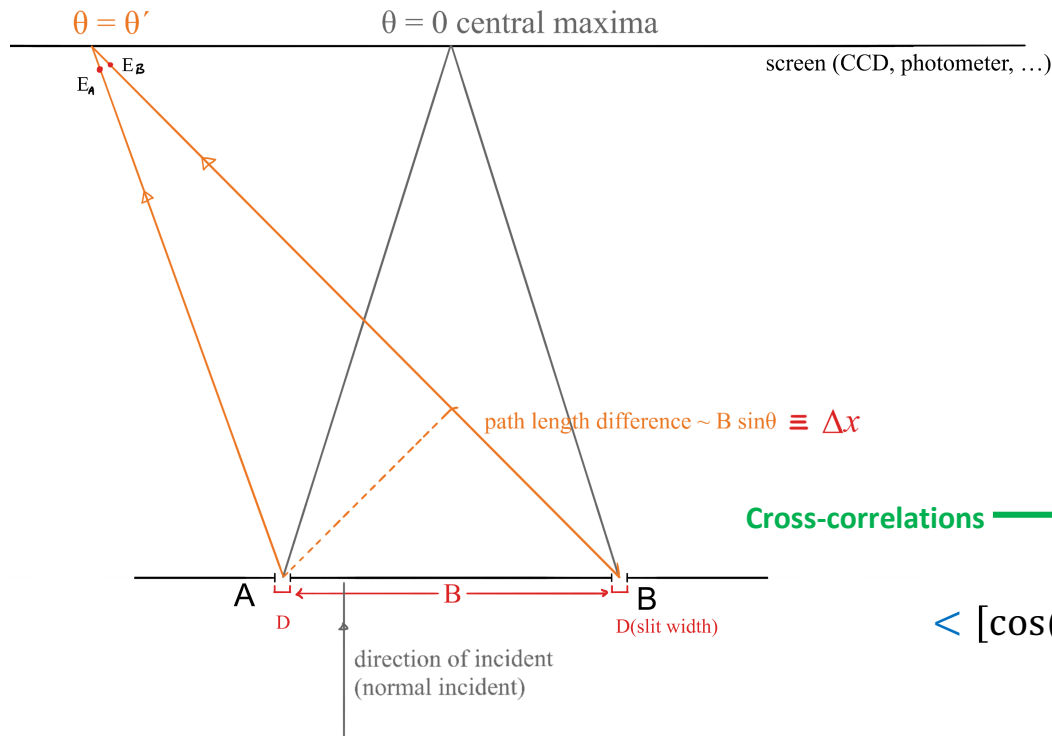


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$$\begin{aligned}
 \langle [\cos(kx - \omega t + \phi_0)]^2 \rangle &\equiv \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} [\cos(\underbrace{kx - \omega t + \phi_0}_{\beta})]^2 dt \\
 &= \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \frac{1}{2} [(\cos^2 \beta + \sin^2 \beta) - (\cos^2 \beta - \sin^2 \beta)] dt
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
2. Energy flux density $\propto E_0^2$

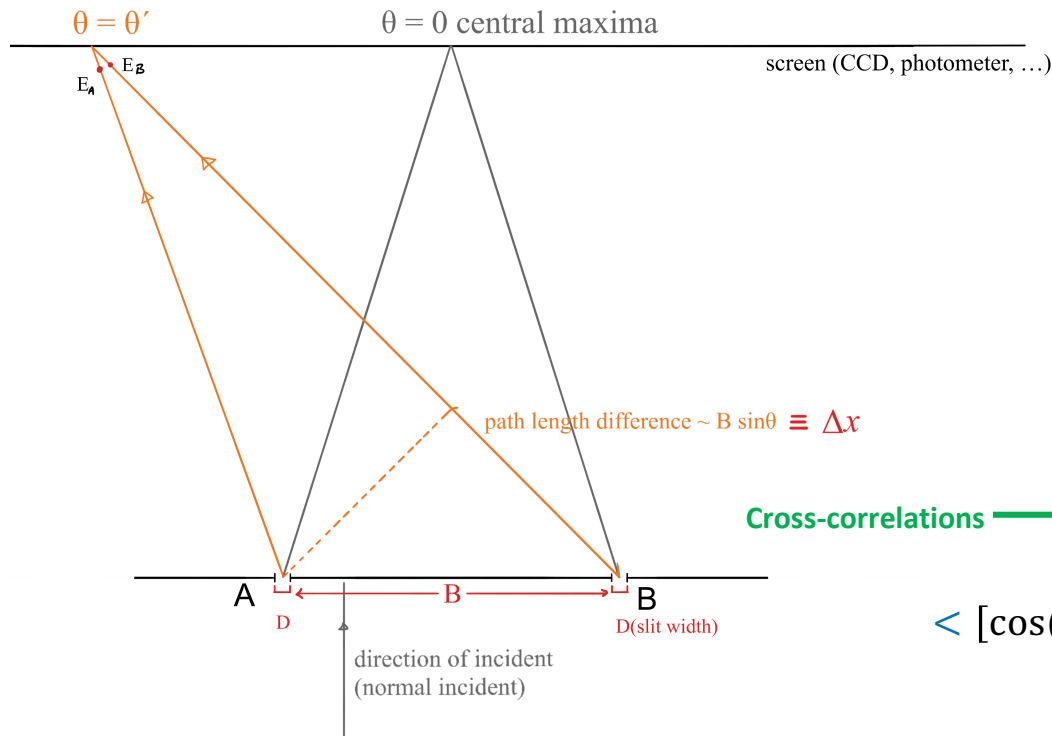


Number of incoming photons in a unit of time and a unit area
(at the angle of emergence $\theta = \theta'$)

$$\begin{aligned}
 &\propto \langle [E_A + E_B]^2 \rangle \\
 &\propto \langle [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \\
 &= \langle [\cos(kx - \omega t + \phi_0)]^2 \rangle \\
 &\quad + \langle [\cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \quad \left. \vphantom{\langle [\cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle} \right\} \text{Auto-correlations} \\
 &\quad + \langle 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \quad \text{Cross-correlations}
 \end{aligned}$$

$$\begin{aligned}
 \langle [\cos(kx - \omega t + \phi_0)]^2 \rangle &\equiv \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} [\cos(\underbrace{kx - \omega t + \phi_0}_{\beta})]^2 dt \\
 &= \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \frac{1}{2} [(\underbrace{\cos^2 \beta + \sin^2 \beta}_1) - (\underbrace{\cos^2 \beta - \sin^2 \beta}_{\cos 2\beta})] dt
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
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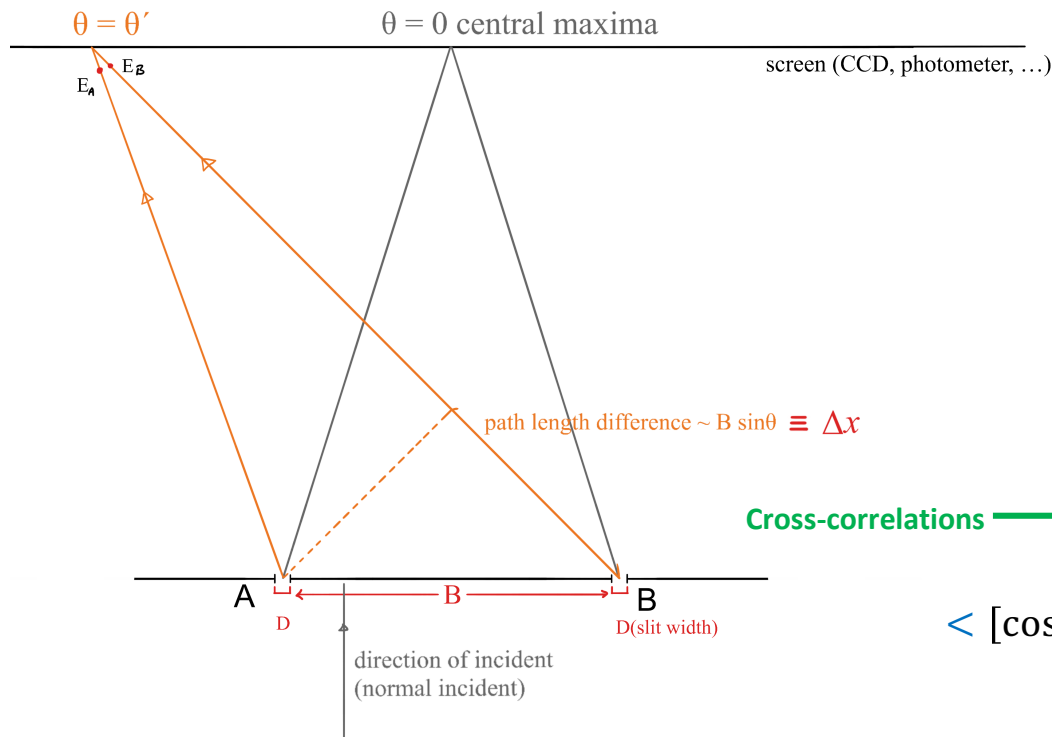


Number of incoming photons in a unit of time and a unit area
(at the angle of emergence $\theta = \theta'$)

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 &\propto \langle [E_A + E_B]^2 \rangle \\
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 \end{aligned}$$

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 \end{aligned}$$

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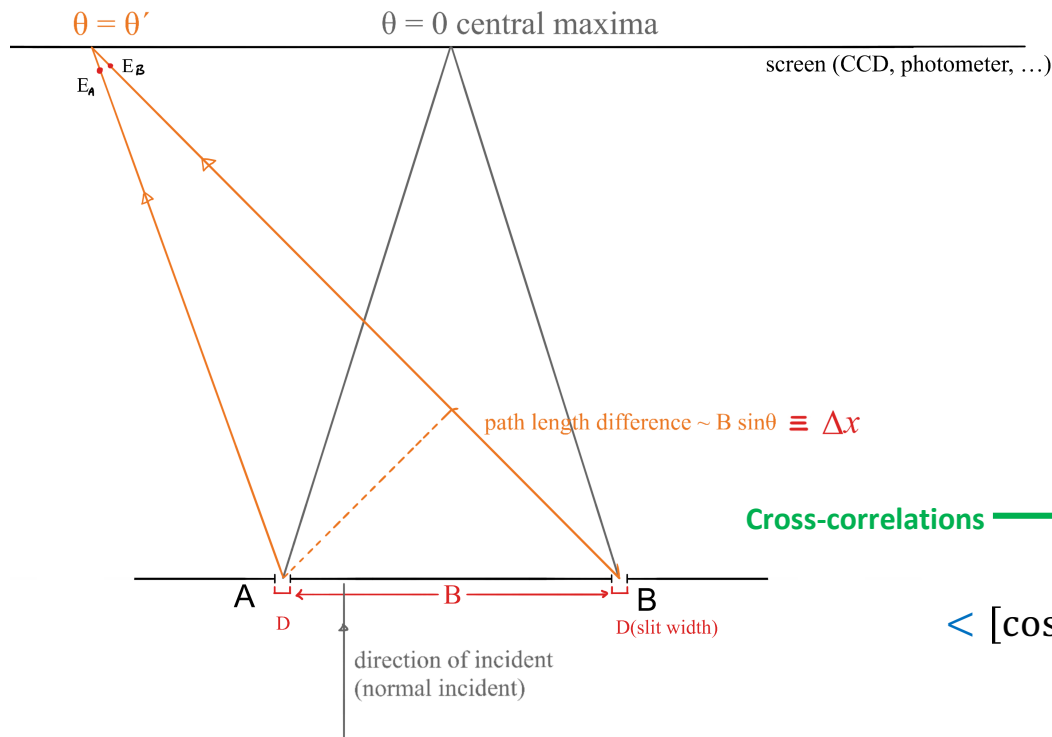


Number of incoming photons in a unit of time and a unit area
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 &\quad + \langle 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \quad \leftarrow \text{Cross-correlations}
 \end{aligned}$$

$$\begin{aligned}
 \langle [\cos(kx - \omega t + \phi_0)]^2 \rangle &\equiv \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \underbrace{[\cos(kx - \omega t + \phi_0)]^2}_{\beta} dt \\
 &= \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \frac{1}{2} \underbrace{[(\cos^2 \beta + \sin^2 \beta)]}_{1} dt = \frac{1}{\Delta t} \frac{1}{2} \left(\frac{1}{2}\Delta t - \frac{-1}{2}\Delta t \right) \\
 &= \frac{1}{2}
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
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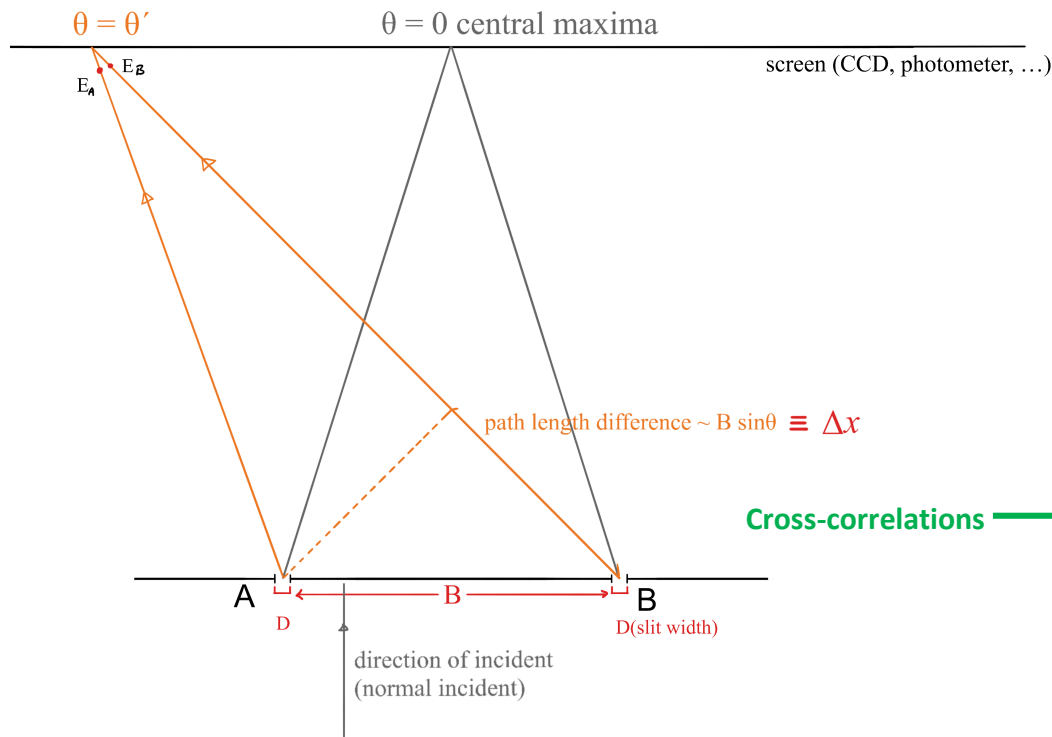


Number of incoming photons in a unit of time and a unit area
(at the angle of emergence $\theta = \theta'$)

$$\begin{aligned}
 &\propto \langle [E_A + E_B]^2 \rangle \\
 &\propto \langle [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \\
 &= \left. \begin{aligned} &\cdot \frac{1}{2} \\ &+ \frac{1}{2} \end{aligned} \right\} \text{Auto-correlations} \\
 &\quad \text{Cross-correlations} \longrightarrow + \langle 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle
 \end{aligned}$$

$$\begin{aligned}
 \langle [\cos(kx - \omega t + \phi_0)]^2 \rangle &\equiv \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} [\cos(\underbrace{kx - \omega t + \phi_0}_{\beta})]^2 dt \\
 &= \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \frac{1}{2} [(\underbrace{\cos^2 \beta + \sin^2 \beta}_{1})] dt = \frac{1}{\Delta t} \frac{1}{2} \left(\frac{1}{2}\Delta t - \frac{-1}{2}\Delta t \right) \\
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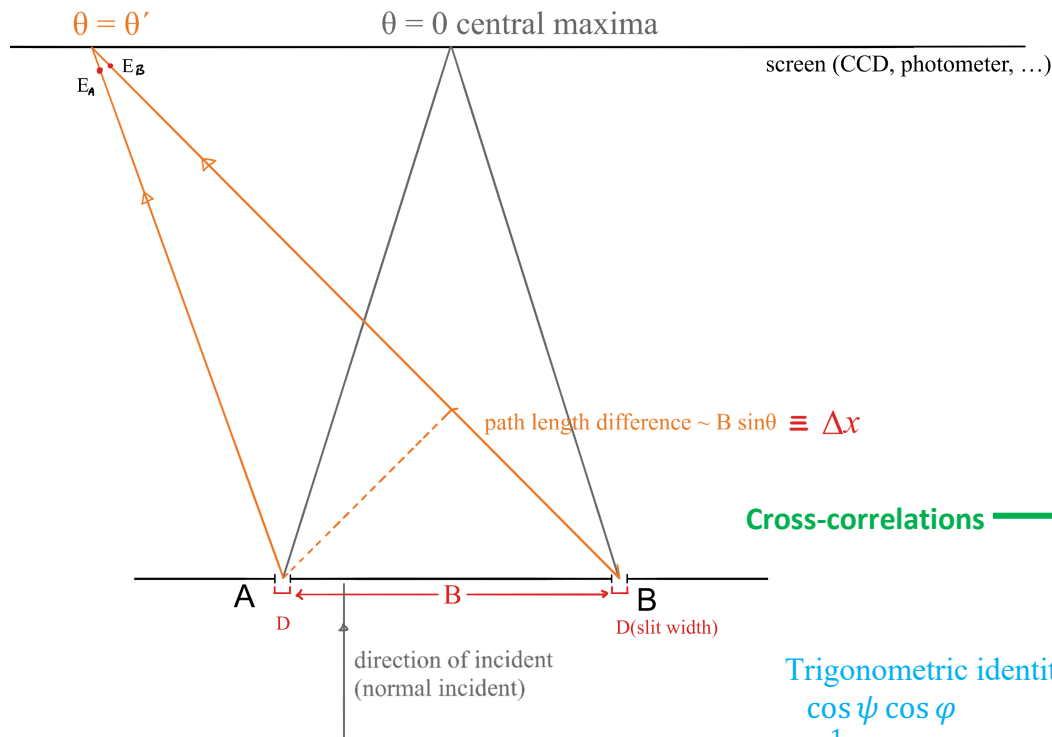


Number of incoming photons in a unit of time and a unit area
(at the angle of emergence $\theta = \theta'$)

$$\begin{aligned}
 &\propto \langle [E_A + E_B]^2 \rangle \\
 &= \langle [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \\
 &= \left. \begin{aligned} &\frac{1}{2} \\ &+ \frac{1}{2} \end{aligned} \right\} \text{Auto-correlations} \\
 &\quad + \langle 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle
 \end{aligned}$$

Cross-correlations

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
2. Energy flux density $\propto E_0^2$



Number of incoming photons in a unit of time and a unit area (at the angle of emergence $\theta = \theta'$)

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 &\propto \langle [E_A + E_B]^2 \rangle \\
 &\propto \langle [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \\
 &= \left. \begin{aligned} &\frac{1}{2} \\ &+ \frac{1}{2} \end{aligned} \right\} \text{Auto-correlations} \\
 &\quad + \langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle
 \end{aligned}$$

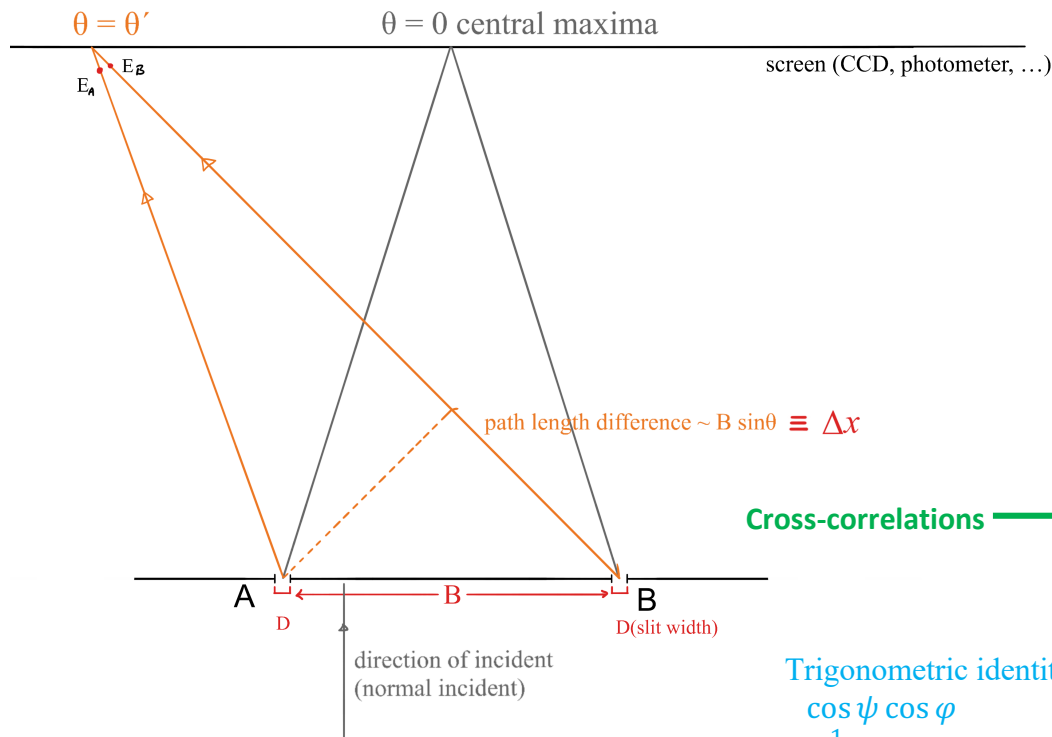
ψ φ

Cross-correlations

Trigonometric identity:

$$\begin{aligned}
 &\cos \psi \cos \varphi \\
 &= \frac{1}{2} [\cos(\psi - \varphi) + \cos(\psi + \varphi)]
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
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Number of incoming photons in a unit of time and a unit area (at the angle of emergence $\theta = \theta'$)

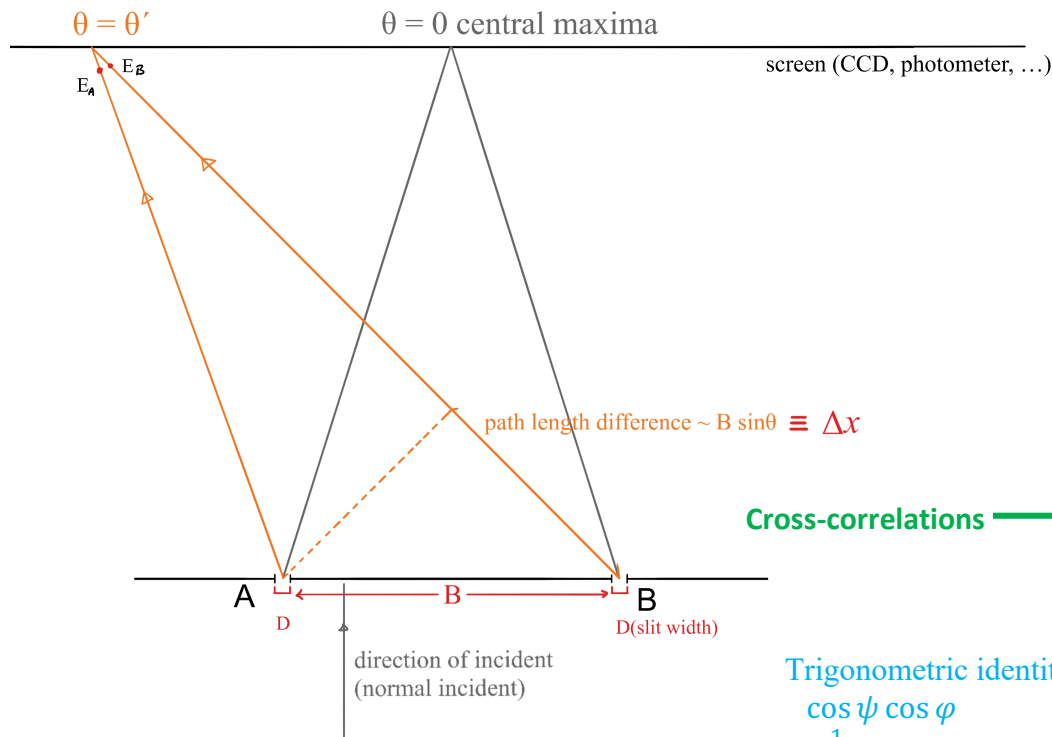
$$\begin{aligned}
 &\propto \langle [E_A + E_B]^2 \rangle \\
 &= \langle [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \\
 &= \frac{1}{2} \left[\text{Auto-correlations} \right] \\
 &\quad + \langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \\
 &\quad \underbrace{\qquad \qquad \qquad \phi \qquad \qquad \qquad \psi \qquad \qquad \qquad}_{\cos(k\Delta x) + \cos(2kx - 2\omega t + 2\phi_0 + k\Delta x)}
 \end{aligned}$$

Cross-correlations

Trigonometric identity:

$$\begin{aligned}
 &\cos \psi \cos \phi \\
 &= \frac{1}{2} [\cos(\psi - \phi) + \cos(\psi + \phi)]
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1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
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 &= \frac{1}{2} \left[\text{Auto-correlations} \right] \\
 &\quad + \langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle
 \end{aligned}$$

Cross-correlations

Trigonometric identity:

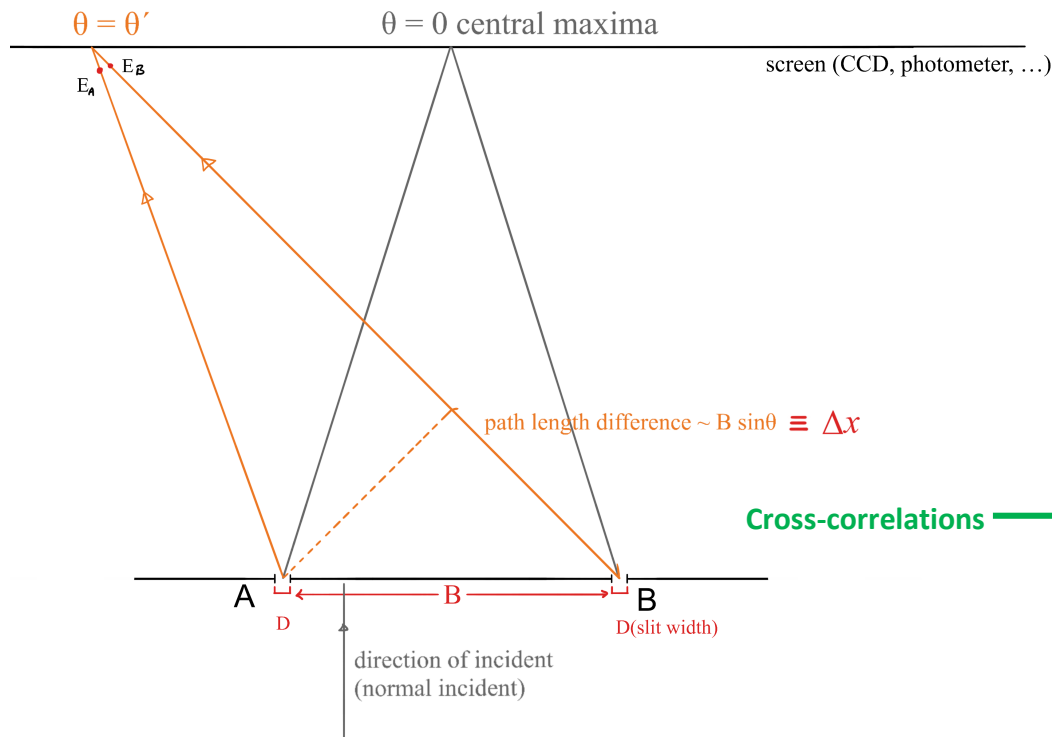
$$\begin{aligned}
 &\cos \psi \cos \varphi \\
 &= \frac{1}{2} [\cos(\psi - \varphi) + \cos(\psi + \varphi)]
 \end{aligned}$$

$$\cos(k\Delta x) + \cos(2kx - 2\omega t + 2\phi_0 + k\Delta x)$$

$$\text{Minima: } \cos(k\Delta x) = -1$$

$$\Rightarrow k\Delta x = \frac{2\pi}{\lambda} B \sin \theta = m\pi, m = \pm 1, 3, 5, \dots$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
2. Energy flux density $\propto E_0^2$



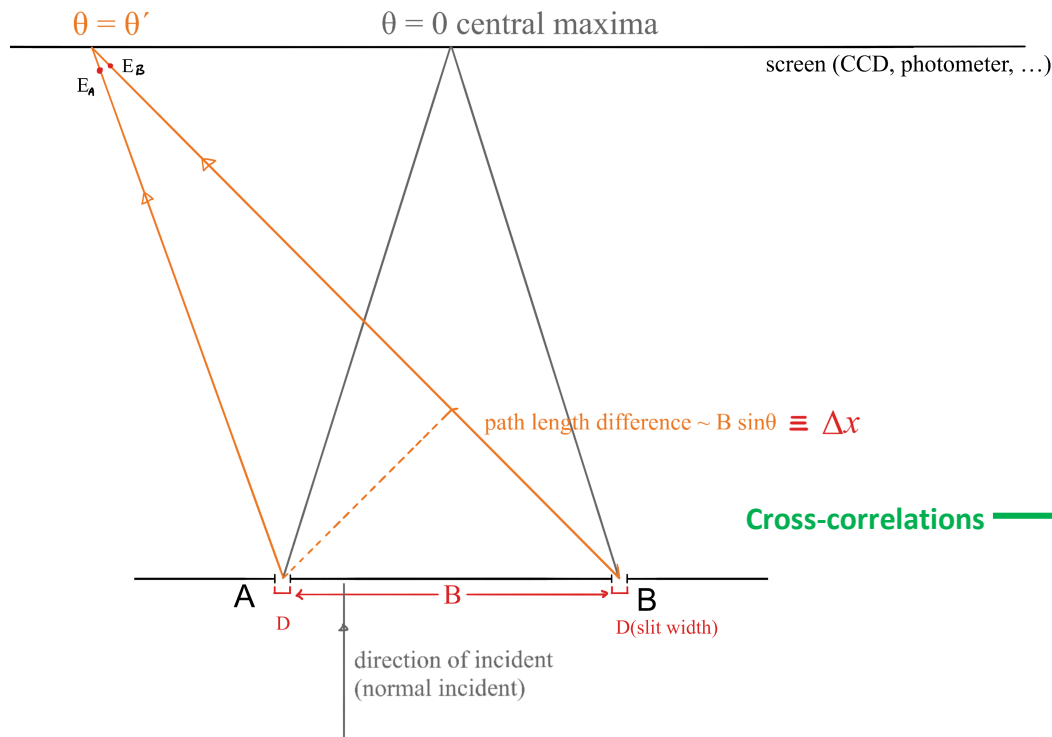
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 &= \left[\frac{1}{2} + \frac{1}{2} \right] \text{Auto-correlations} + \langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \\
 &\quad \underbrace{\hspace{10em}}_{\phi} \quad \underbrace{\hspace{10em}}_{\psi}
 \end{aligned}$$

Cross-correlations $\longrightarrow$

$$\begin{aligned}
 \langle [E_A + E_B]^2 \rangle &= \tilde{P}(\theta) \left[1 + \cos \left(\frac{2\pi}{\lambda} B \sin \theta \right) \right] \\
 &\sim \tilde{P}(\theta) \left[1 + \cos \left(2\pi \frac{B}{\lambda} \theta \right) \right]
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
2. Energy flux density $\propto E_0^2$

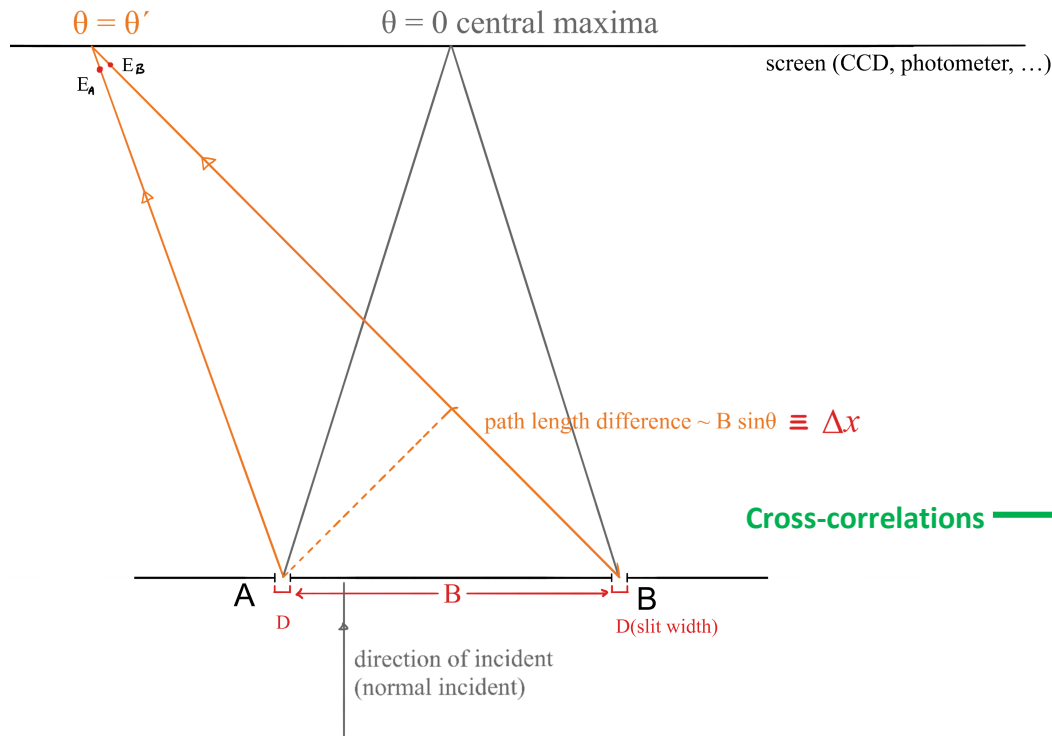


Number of incoming photons in a unit of time and a unit area (at the angle of emergence $\theta = \theta'$)

$$\begin{aligned}
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 &= \left[\frac{1}{2} + \frac{1}{2} \right] \text{Auto-correlations} + \underbrace{\langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle}_{\text{Cross-correlations}}
 \end{aligned}$$

$$\begin{aligned}
 \langle [E_A + E_B]^2 \rangle &= \tilde{P}(\theta) \left[1 + \cos \left(\frac{2\pi}{\lambda} B \sin \theta \right) \right] \\
 &\sim \underbrace{\tilde{P}(\theta)}_{\text{Auto-correlations}} \left[\underbrace{1}_{\text{Auto-correlations}} + \underbrace{\cos \left(2\pi \frac{B}{\lambda} \theta \right)}_{\text{Cross-correlations}} \right]
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
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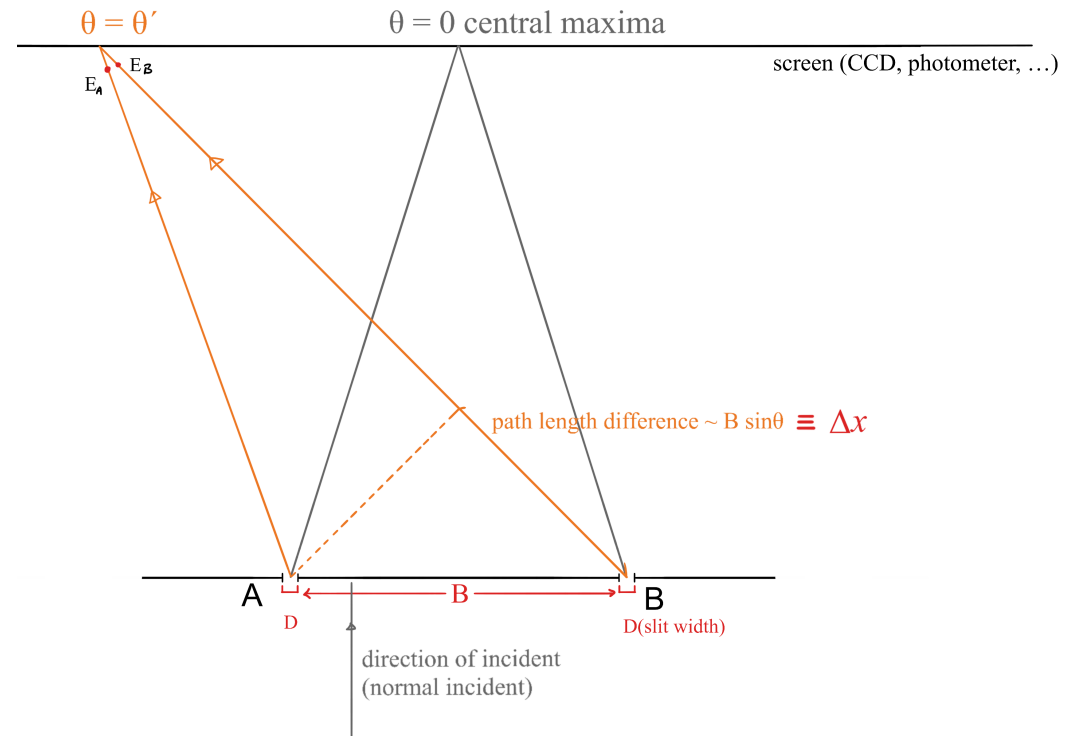
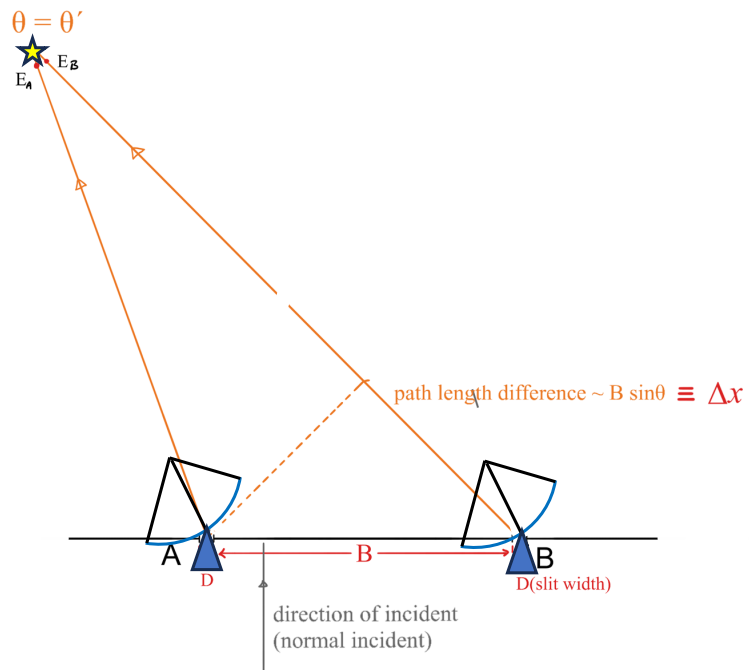
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 &\quad + \underbrace{\langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle}_{\phi \quad \psi}
 \end{aligned}$$

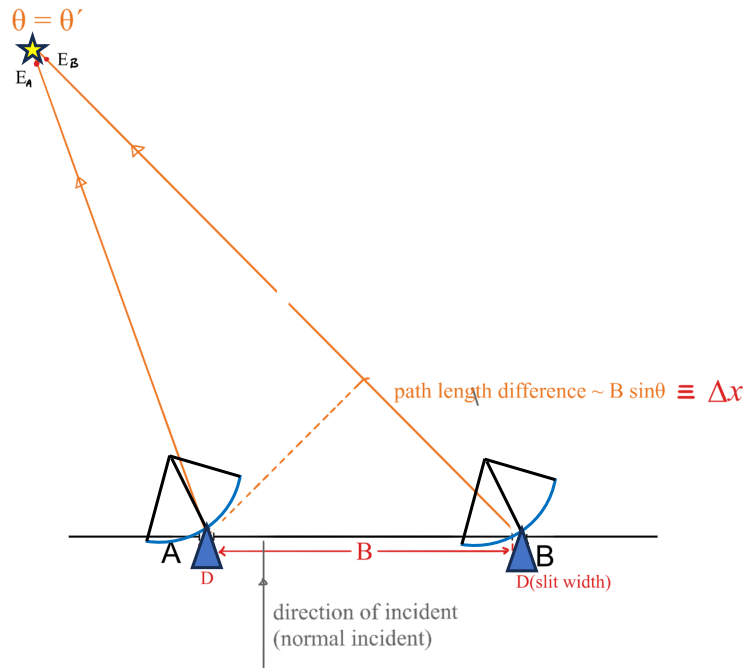
Cross-correlations

$$\begin{aligned}
 \langle [E_A + E_B]^2 \rangle &= \tilde{P}(\theta) \left[1 + \cos \left(\frac{2\pi}{\lambda} B \sin \theta \right) \right] \\
 &\sim \tilde{P}(\theta) \left[1 + \cos \left(2\pi \frac{B}{\lambda} \theta \right) \right] \sim \tilde{P}(\theta) [1 + \cos(2\pi u \theta)]
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
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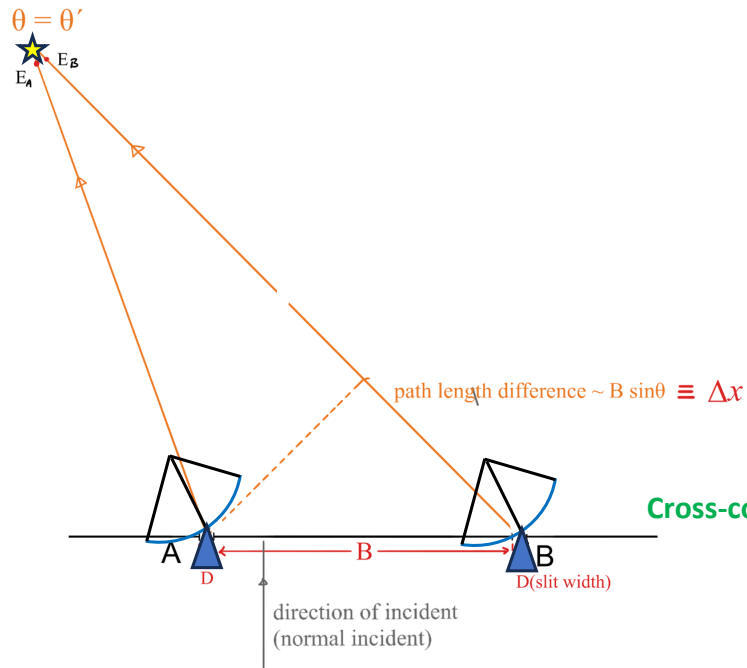
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Number of incoming photons in a unit of time and a unit area
(at the angle of emergence $\theta = \theta'$)

$$\begin{aligned}
 &\propto [E_A + E_B]^2 \\
 &= \tilde{P}(\theta) [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \\
 &= \tilde{P}(\theta) \{ [\cos(kx - \omega t + \phi_0)]^2 \\
 &\quad + [\cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \\
 &\quad + 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \}
 \end{aligned}$$

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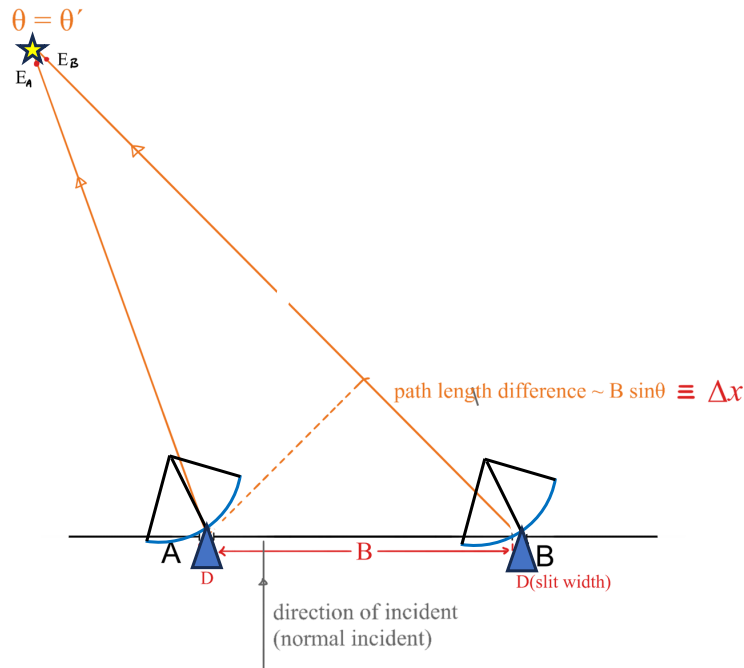
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 &= \langle [\cos(kx - \omega t + \phi_0)]^2 \rangle \\
 &\quad + \langle [\cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \quad \left. \vphantom{\langle [\cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle} \right\} \text{Auto-correlations} \\
 &\quad + \langle 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle
 \end{aligned}$$

Cross-correlations $\longrightarrow$

$$\langle \cdot \rangle \equiv \lim_{\Delta t \rightarrow \infty} \frac{1}{\Delta t} \int_{-\frac{1}{2}\Delta t}^{\frac{1}{2}\Delta t} \cdot dt$$

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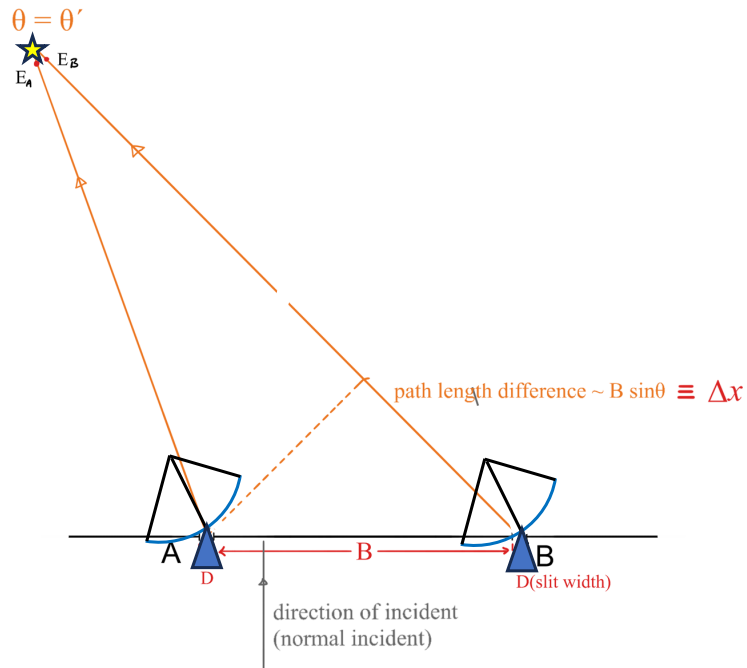


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 &= \left[\frac{1}{2} + \frac{1}{2} \right] \underbrace{\langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle}_{\text{Cross-correlations}} \\
 &\quad \text{Auto-correlations}
 \end{aligned}$$

$$\begin{aligned}
 \langle [E_A + E_B]^2 \rangle &= \tilde{P}(\theta) \left[1 + \cos\left(\frac{2\pi}{\lambda} B \sin \theta\right) \right] \\
 &\sim \tilde{P}(\theta) \left[1 + \cos\left(2\pi \frac{B}{\lambda} \theta\right) \right]
 \end{aligned}$$

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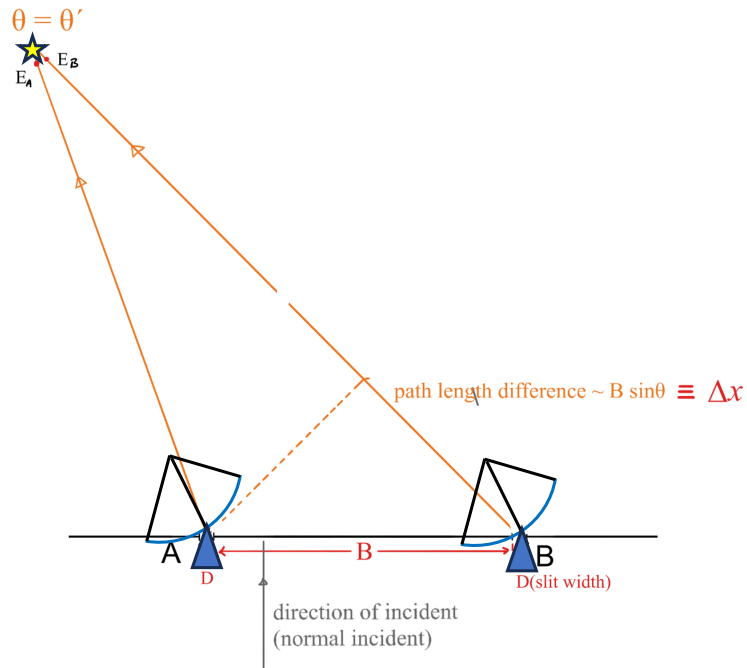


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 &\quad \text{Auto-correlations}
 \end{aligned}$$

$$\langle [E_A + E_B]^2 \rangle \sim \underbrace{\tilde{P}(\theta)}_{\text{Auto-correlations}} [1 + \cos(2\pi u \theta)]$$

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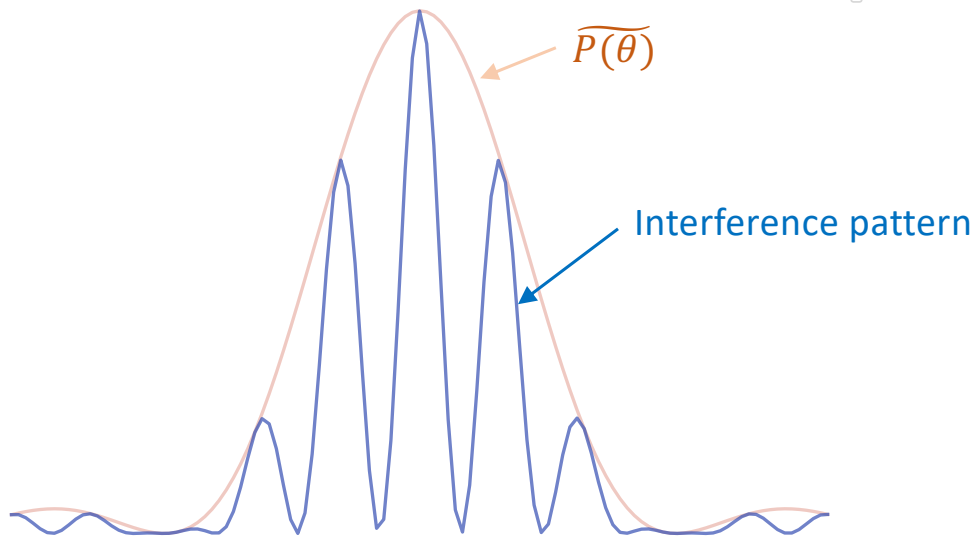


Number of incoming photons in a unit of time and a unit area
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 &= \left[\frac{1}{2} + \frac{1}{2} \right] \underbrace{\langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle}_{\text{Cross-correlations}} \\
 &\quad \text{Auto-correlations}
 \end{aligned}$$

$$\langle [E_A + E_B]^2 \rangle \sim \tilde{P}(\theta) [1 + \underbrace{\cos(2\pi u \theta)}_{\text{Cross-correlations}}]$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-



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 &= \langle [\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0)]^2 \rangle \\
 &= \left. \begin{aligned} &+ \frac{1}{2} \\ &+ \frac{1}{2} \end{aligned} \right\} \text{Auto-correlations} \\
 &+ \underbrace{\langle 2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle}_{\text{Cross-correlations}}
 \end{aligned}$$

$$\langle [E_A + E_B]^2 \rangle \sim \underbrace{\tilde{P}(\theta)}_{\text{Primary beam: response function of each reflector}} [1 + \cos(2\pi u \theta)]$$

Primary beam: response function of
each reflector

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta) [1 + \cos(2\pi u \theta)]$

Sine correlation: $\propto \langle [E_A + E'_B]^2 \rangle$

$$= \langle \left[\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \right]^2 \rangle$$

$$= \left. \begin{array}{l} \frac{1}{2} \\ + \frac{1}{2} \end{array} \right\} \text{Auto-} \\ \text{correlations}$$

$$+ \langle 2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \rangle$$

$$2 \cos(kx - \omega t + \phi_0) \left[\cos(k(x + \Delta x) - \omega t + \phi_0) \underbrace{\cos \frac{\pi}{2}}_0 - \sin(k(x + \Delta x) - \omega t + \phi_0) \underbrace{\sin \frac{\pi}{2}}_1 \right]$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta) [1 + \cos(2\pi u \theta)]$

Sine correlation: $\propto \langle [E_A + E'_B]^2 \rangle$

$$\begin{aligned}
 &= \langle \left[\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \right]^2 \rangle \\
 &= \left. \begin{array}{l} \frac{1}{2} \\ + \frac{1}{2} \end{array} \right\} \text{Auto-correlations} \\
 &+ \langle \underbrace{2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2})}_{-2 \cos(kx - \omega t + \phi_0) \sin(k(x + \Delta x) - \omega t + \phi_0)} \rangle
 \end{aligned}$$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta) [1 + \cos(2\pi u \theta)]$

Sine correlation: $\propto \langle [E_A + E'_B]^2 \rangle$

$$\begin{aligned}
 &= \langle \left[\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \right]^2 \rangle \\
 &= \left. \begin{array}{l} \frac{1}{2} \\ + \frac{1}{2} \end{array} \right\} \text{Auto-correlations} \\
 &+ \langle \underbrace{2\cos(kx - \omega t + \phi_0) \cos\left(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}\right)}_{-2\cos(kx - \omega t + \phi_0) \sin(k(x + \Delta x) - \omega t + \phi_0)} \rangle \\
 &\quad - [\sin(2kx - 2\omega t + 2\phi_0 + k\Delta x) - \sin(k\Delta x)]
 \end{aligned}$$

Trigonometric identity:
 $\cos \psi \sin \varphi$
 $= \frac{1}{2} [\sin(\psi + \varphi) - \sin(\psi - \varphi)]$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u \theta)]$

Sine correlation: $\propto \langle [E_A + E'_B]^2 \rangle$

$$\begin{aligned}
 &= \langle \left[\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \right]^2 \rangle \\
 &= \left. \begin{array}{l} \frac{1}{2} \\ + \frac{1}{2} \end{array} \right\} \text{Auto-correlations} \\
 &+ \langle \underbrace{2\cos(kx - \omega t + \phi_0) \cos\left(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}\right)}_{-2\cos(kx - \omega t + \phi_0) \sin(k(x + \Delta x) - \omega t + \phi_0)} \rangle \\
 &\quad \underbrace{-[\sin(2kx - 2\omega t + 2\phi_0 + k\Delta x) - \sin(k\Delta x)]}_{\text{Zero in the } \diamond \text{ bracket}}
 \end{aligned}$$

Trigonometric identity:
 $\cos \psi \sin \varphi$
 $= \frac{1}{2} [\sin(\psi + \varphi) - \sin(\psi - \varphi)]$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u \theta)]$

Sine correlation: $\propto \langle [E_A + E'_B]^2 \rangle$

$$\begin{aligned}
 &= \langle \left[\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \right]^2 \rangle \\
 &= \left. \begin{array}{l} \frac{1}{2} \\ + \frac{1}{2} \end{array} \right\} \text{Auto-correlations} \\
 &+ \langle \underbrace{2\cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2})}_{-2\cos(kx - \omega t + \phi_0) \sin(k(x + \Delta x) - \omega t + \phi_0)} \rangle \\
 &\quad \underbrace{\hspace{10em}}_{[\sin(k\Delta x)]}
 \end{aligned}$$

Trigonometric identity:
 $\cos \psi \sin \varphi$
 $= \frac{1}{2} [\sin(\psi + \varphi) - \sin(\psi - \varphi)]$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta) [1 + \cos(2\pi u \theta)]$

Sine correlation: $\propto \langle [E_A + E'_B]^2 \rangle$

$$\begin{aligned}
 &= \langle \left[\cos(kx - \omega t + \phi_0) + \cos(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}) \right]^2 \rangle \\
 &= \left. \begin{array}{l} \frac{1}{2} \\ + \frac{1}{2} \end{array} \right\} \text{Auto-correlations} \\
 &+ \langle \underbrace{2\cos(kx - \omega t + \phi_0) \cos\left(k(x + \Delta x) - \omega t + \phi_0 + \frac{\pi}{2}\right)}_{-2\cos(kx - \omega t + \phi_0) \sin(k(x + \Delta x) - \omega t + \phi_0)} \rangle \\
 &\qquad \qquad \qquad \sin(k\Delta x) = \sin(2\pi u \theta)
 \end{aligned}$$

Trigonometric identity:
 $\cos \psi \sin \varphi$
 $= \frac{1}{2} [\sin(\psi + \varphi) - \sin(\psi - \varphi)]$

1. EM-wave at long-distance limit: plane wave $E = E_0 \cos(kx - \omega t + \phi_0)$ $k = \frac{2\pi}{\lambda}$
 2. Energy flux density $\propto E_0^2$
-

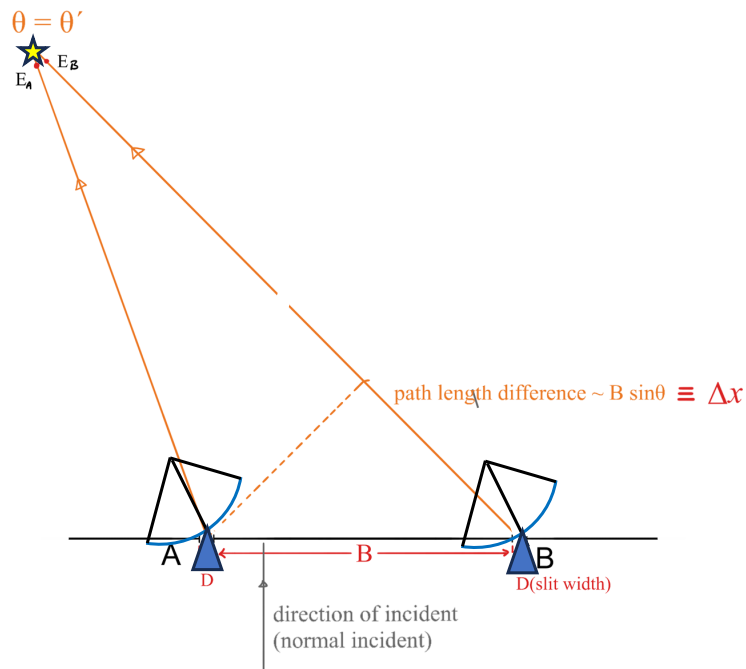
Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta) [1 + \cos(2\pi u \theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta) [1 + \sin(2\pi u \theta)]$

Cosine correlation and sine correlation are the direct measurement of a two-element radio interferometer

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

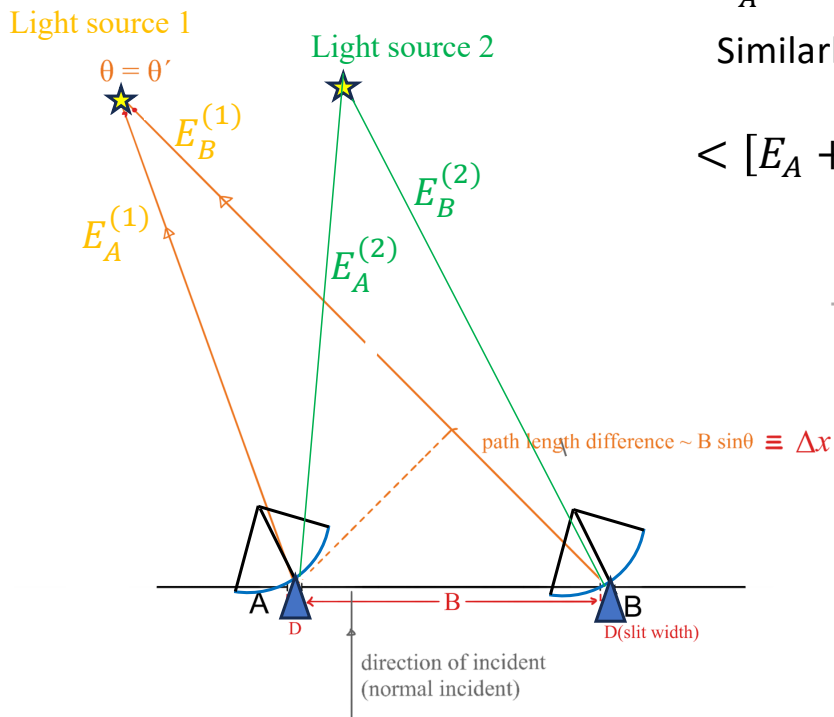


When there is one target source

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there are two incoherent sources



E_A : electric field received at antenna A = $E_A^{(1)} + E_A^{(2)}$,

Similarly, $E_B = E_B^{(1)} + E_B^{(2)}$

$$\begin{aligned} \langle [E_A + E_B]^2 \rangle &= \langle E_A^{(1)2} \rangle + \langle E_B^{(1)2} \rangle + \langle 2E_A^{(1)} E_B^{(1)} \rangle \\ &\quad + \langle E_A^{(2)2} \rangle + \langle E_B^{(2)2} \rangle + \langle 2E_A^{(2)} E_B^{(2)} \rangle \\ &\quad + \langle 2E_A^{(1)} E_A^{(2)} \rangle + \langle 2E_A^{(1)} E_B^{(2)} \rangle + \langle 2E_B^{(1)} E_A^{(2)} \rangle + \langle 2E_B^{(1)} E_B^{(2)} \rangle \end{aligned}$$

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

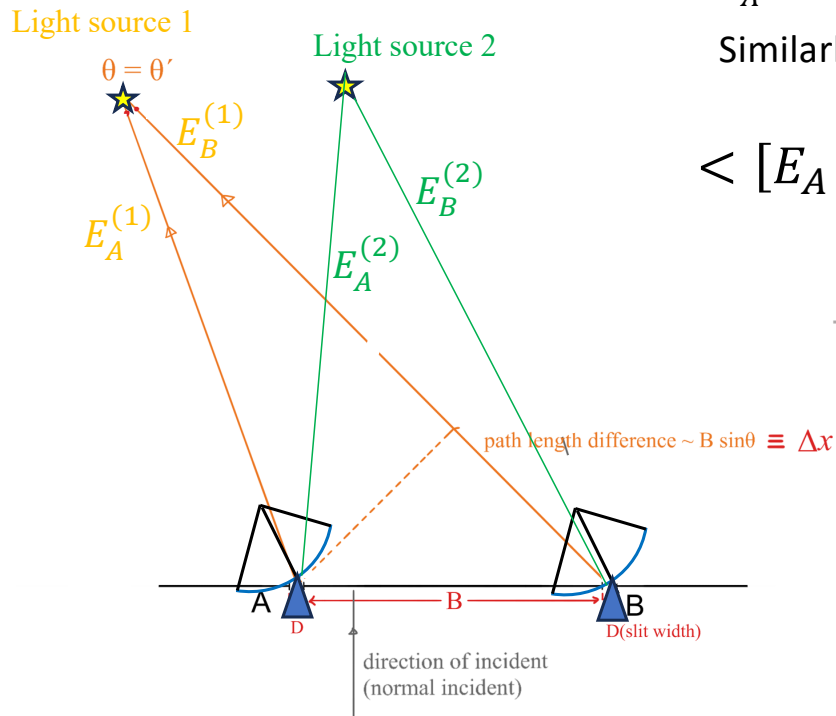
Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there are two incoherent sources

E_A : electric field received at antenna A = $E_A^{(1)} + E_A^{(2)}$,

Similarly, $E_B = E_B^{(1)} + E_B^{(2)}$

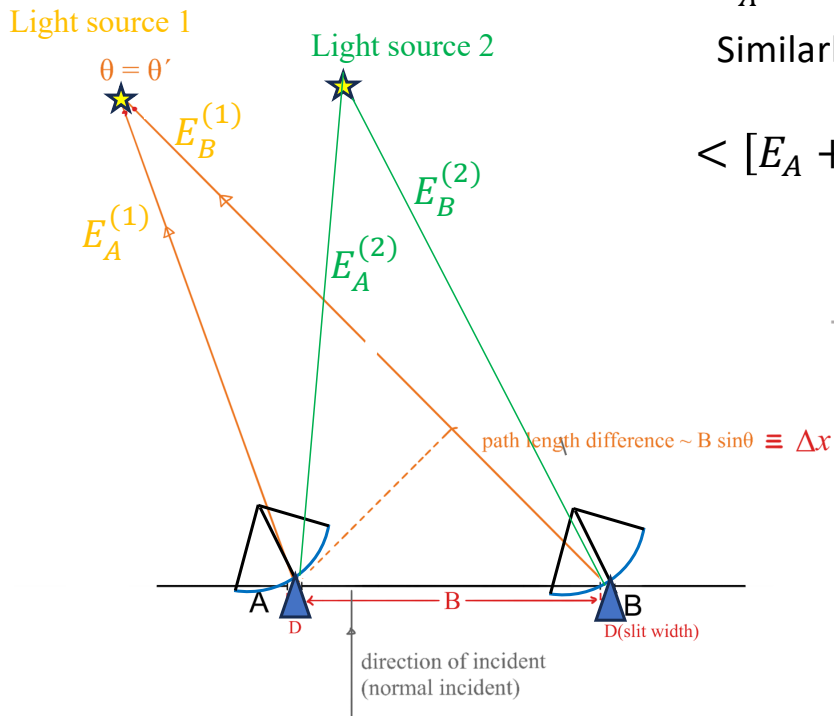
$$\begin{aligned} \langle [E_A + E_B]^2 \rangle &= \langle E_A^{(1)2} \rangle + \langle E_B^{(1)2} \rangle + \langle 2E_A^{(1)}E_B^{(1)} \rangle \\ &\quad + \langle E_A^{(2)2} \rangle + \langle E_B^{(2)2} \rangle + \langle 2E_A^{(2)}E_B^{(2)} \rangle \\ &\quad + \langle 2E_A^{(1)}E_A^{(2)} \rangle + \langle 2E_A^{(1)}E_B^{(2)} \rangle + \langle 2E_B^{(1)}E_A^{(2)} \rangle + \langle 2E_B^{(1)}E_B^{(2)} \rangle \end{aligned}$$



Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there are two incoherent sources



E_A : electric field received at antenna A = $E_A^{(1)} + E_A^{(2)}$,

Similarly, $E_B = E_B^{(1)} + E_B^{(2)}$

$$\begin{aligned} \langle [E_A + E_B]^2 \rangle &= \langle E_A^{(1)2} \rangle + \langle E_B^{(1)2} \rangle + \langle 2E_A^{(1)}E_B^{(1)} \rangle \\ &\quad + \langle E_A^{(2)2} \rangle + \langle E_B^{(2)2} \rangle + \langle 2E_A^{(2)}E_B^{(2)} \rangle \\ &\quad + \langle 2E_A^{(1)}E_A^{(2)} \rangle + \langle 2E_A^{(1)}E_B^{(2)} \rangle + \langle 2E_B^{(1)}E_A^{(2)} \rangle + \langle 2E_B^{(1)}E_B^{(2)} \rangle \end{aligned}$$

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

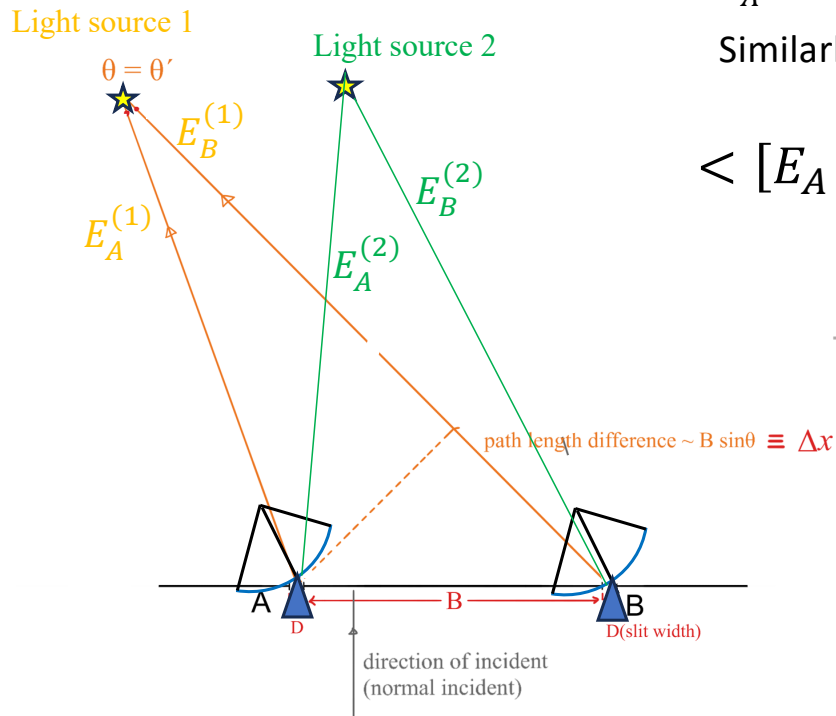
Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there are two incoherent sources

E_A : electric field received at antenna A = $E_A^{(1)} + E_A^{(2)}$,

Similarly, $E_B = E_B^{(1)} + E_B^{(2)}$

$$\begin{aligned} \langle [E_A + E_B]^2 \rangle &= \langle E_A^{(1)2} \rangle + \langle E_B^{(1)2} \rangle + \langle 2E_A^{(1)}E_B^{(1)} \rangle \\ &\quad + \langle E_A^{(2)2} \rangle + \langle E_B^{(2)2} \rangle + \langle 2E_A^{(2)}E_B^{(2)} \rangle \\ &\quad + \underbrace{\langle 2E_A^{(1)}E_A^{(2)} \rangle + \langle 2E_A^{(1)}E_B^{(2)} \rangle + \langle 2E_B^{(1)}E_A^{(2)} \rangle + \langle 2E_B^{(1)}E_B^{(2)} \rangle}_{\text{Definition of incoherent sources}} \end{aligned}$$



Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

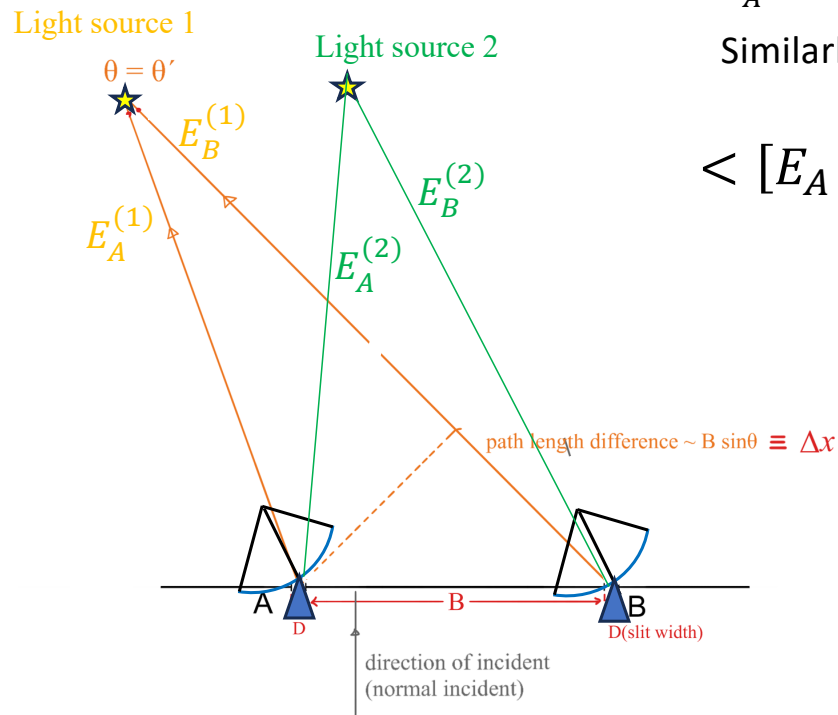
When there are two incoherent sources

E_A : electric field received at antenna A = $E_A^{(1)} + E_A^{(2)}$,

Similarly, $E_B = E_B^{(1)} + E_B^{(2)}$

$$\begin{aligned} \langle [E_A + E_B]^2 \rangle &= \langle E_A^{(1)2} \rangle + \langle E_B^{(1)2} \rangle + \langle 2E_A^{(1)}E_B^{(1)} \rangle \\ &+ \langle E_A^{(2)2} \rangle + \langle E_B^{(2)2} \rangle + \langle 2E_A^{(2)}E_B^{(2)} \rangle \end{aligned}$$

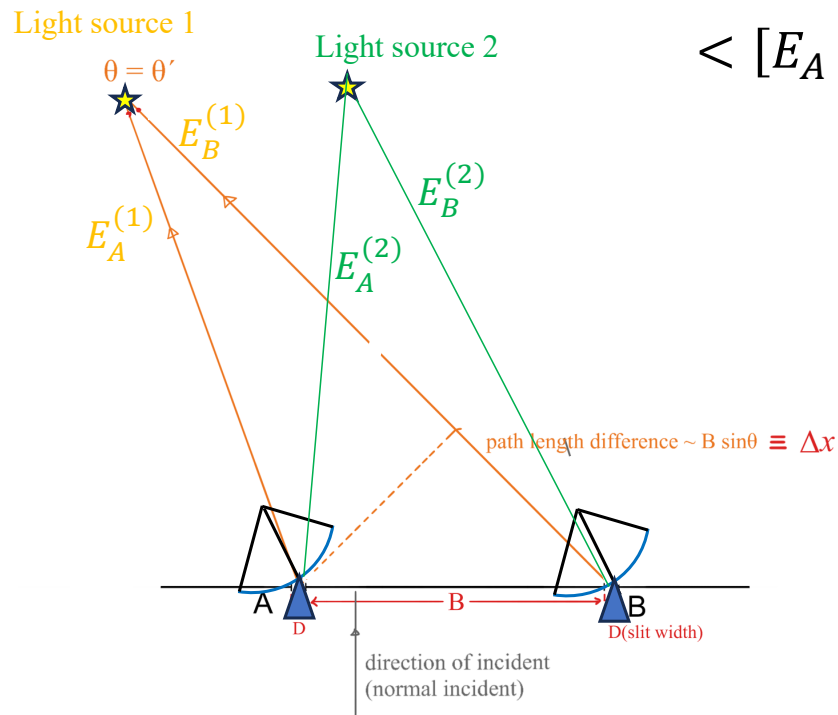
Correlation of multiple incoherent sources can be evaluated by summing over the contribution of individual sources.



Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there are more incoherent sources



$$\langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} \propto \sum_{i=0}^{N-1} \underbrace{P(\theta_i')}_{\text{A coefficient to describe how bright is source } i} [A_i \cos(2\pi u \theta_i')]$$

A coefficient to describe how bright is source i
Dimension: flux density

Looking at cross correlations at this moment
(i.e., neglecting auto-correlation terms)

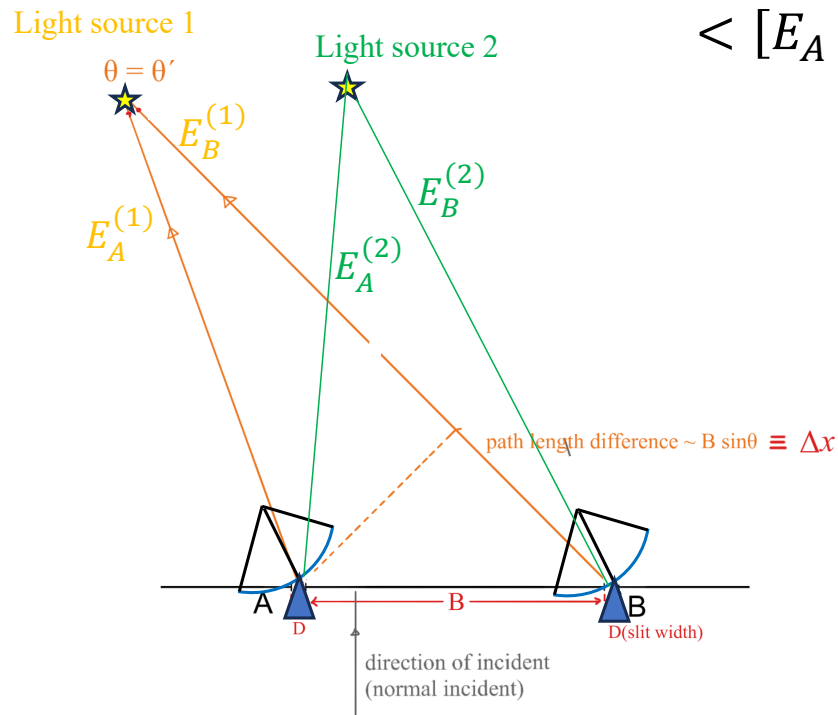
Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there is a continuous distribution of incoherent sources

$$\langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} \propto \int \tilde{P}(\theta) \underbrace{A(\theta)}_{\text{coefficient}} \cos(2\pi u\theta) d\theta$$

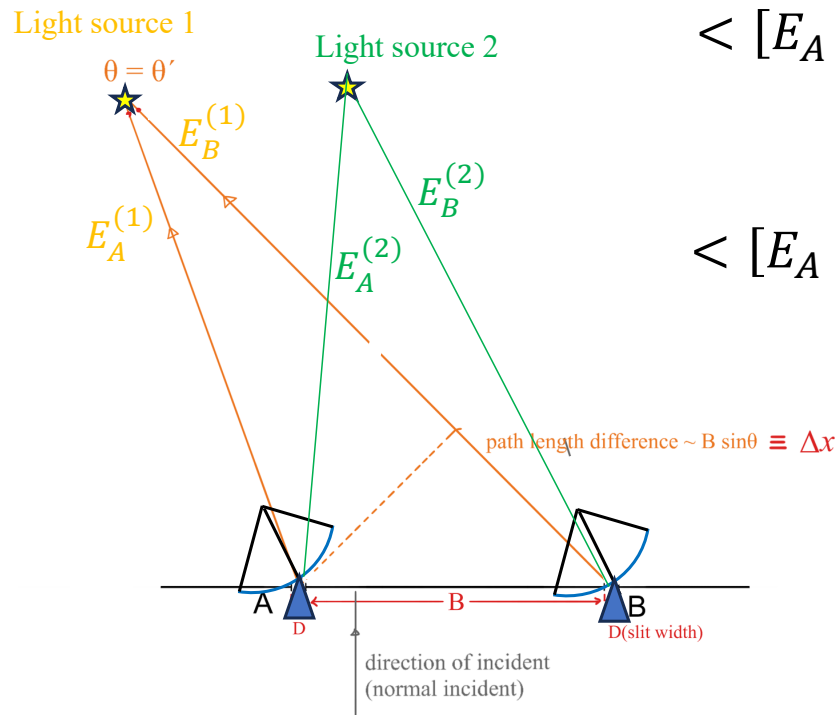
A coefficient to describe the intensity of the target source at incident angle θ
Dimension: intensity



Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there is a continuous distribution of incoherent sources



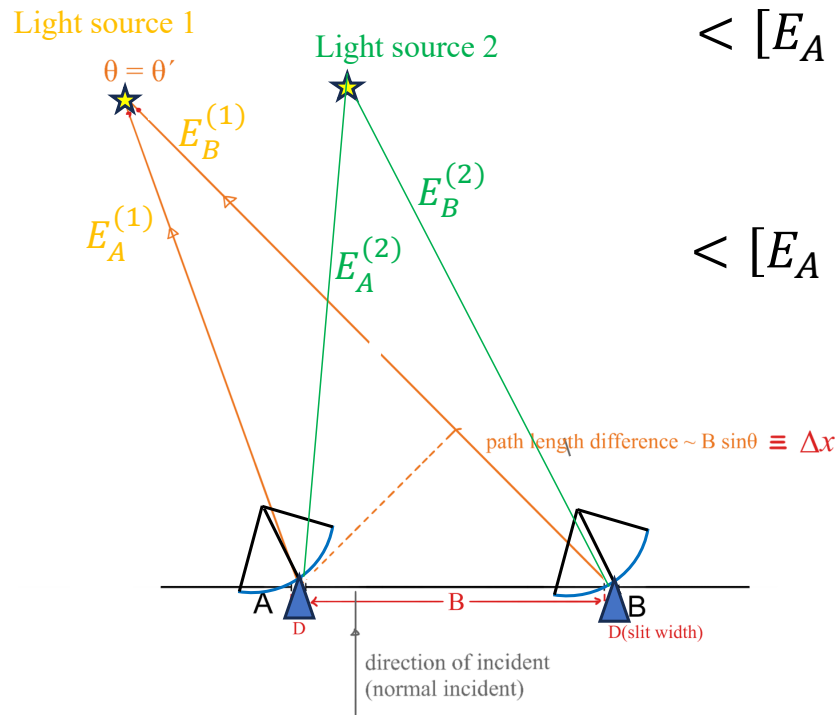
$$\langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} \propto \int \tilde{P}(\theta) \underbrace{A(\theta)}_{\text{A coefficient to describe the intensity of the target source at incident angle } \theta} \cos(2\pi u\theta) d\theta$$

$$\langle [E_A + E_B]^2 \rangle_{\sin}^{\text{cross}} \propto \int \tilde{P}(\theta) A(\theta) \sin(2\pi u\theta) d\theta$$

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

When there is a continuous distribution of incoherent sources



$$\langle [E_A + E_B]^2 \rangle_{cos}^{cross} \propto \int \tilde{P}(\theta) \underbrace{A(\theta)}_{\text{A coefficient to describe the intensity of the target source at incident angle } \theta} \cos(2\pi u\theta) d\theta$$

$$\langle [E_A + E_B]^2 \rangle_{sin}^{cross} \propto \int \tilde{P}(\theta) A(\theta) \sin(2\pi u\theta) d\theta$$

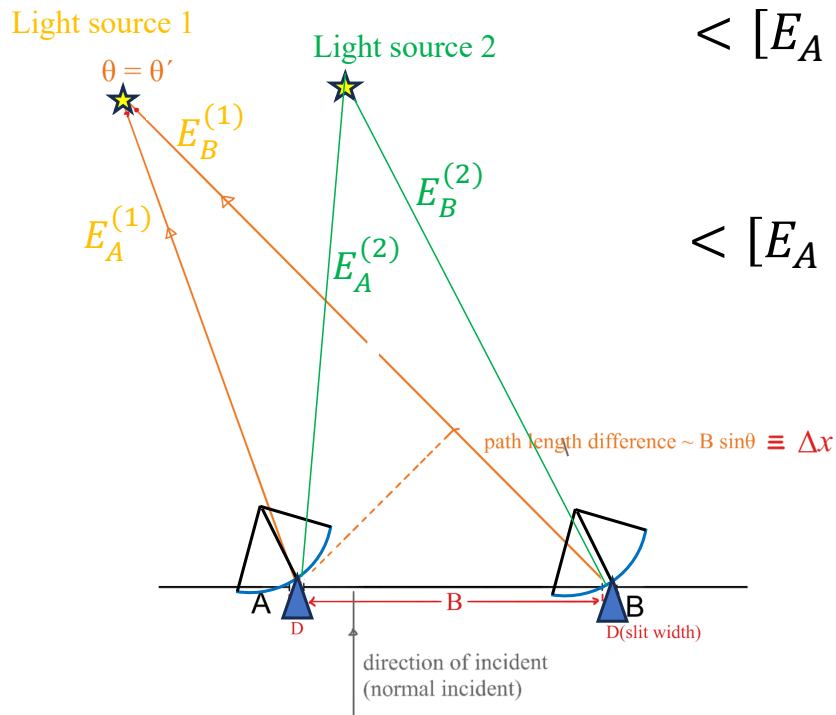
$$\text{Complex visibility } V_{AB} = \langle [E_A + E_B]^2 \rangle_{cos}^{cross} + i \langle [E_A + E_B]^2 \rangle_{sin}^{cross}$$

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

Complex visibility $V_{AB} = \langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} + i \langle [E_A + E_B]^2 \rangle_{\sin}^{\text{cross}}$

Intensity distribution of a point-like source: $A(\theta) = \delta(\theta - \theta')$



$$\begin{aligned} \langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} &= \int \tilde{P}(\theta) A(\theta) \cos(2\pi u\theta) d\theta \\ &= \tilde{P}(\theta') \cos(2\pi u\theta') \\ \langle [E_A + E_B]^2 \rangle_{\sin}^{\text{cross}} &= \tilde{P}(\theta') \sin(2\pi u\theta') \end{aligned}$$

Complex visibility $V_{AB} = \tilde{P}(\theta') \cos(2\pi u\theta') + i \tilde{P}(\theta') \sin(2\pi u\theta')$

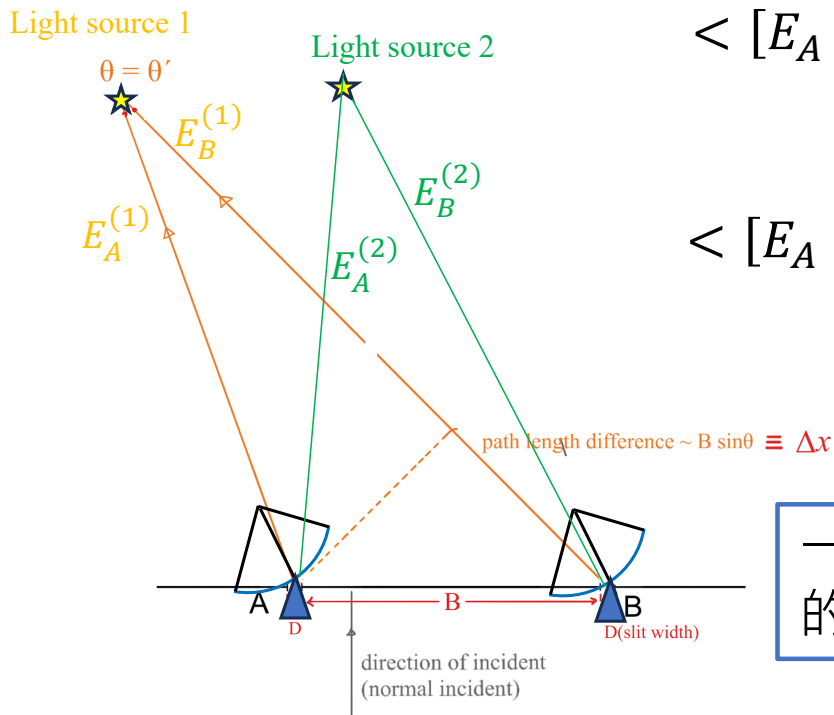
$$= \underbrace{\tilde{P}(\theta')}_{\text{Visibility amplitude}} e^{i \underbrace{2\pi u\theta'}_{\text{Visibility phase}}}$$

Cosine correlation: $\langle [E_A + E_B]^2 \rangle_{\cos} \sim \tilde{P}(\theta)[1 + \cos(2\pi u\theta)]$

Sine correlation: $\langle [E_A + E_B]^2 \rangle_{\sin} \sim \tilde{P}(\theta)[1 + \sin(2\pi u\theta)]$

Complex visibility $V_{AB} = \langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} + i \langle [E_A + E_B]^2 \rangle_{\sin}^{\text{cross}}$

Intensity distribution of a point-like source: $A(\theta) = \delta(\theta - \theta')$



$$\begin{aligned} \langle [E_A + E_B]^2 \rangle_{\cos}^{\text{cross}} &= \int \tilde{P}(\theta) A(\theta) \cos(2\pi u\theta) d\theta \\ &= \tilde{P}(\theta') \cos(2\pi u\theta') \\ \langle [E_A + E_B]^2 \rangle_{\sin}^{\text{cross}} &= \tilde{P}(\theta') \sin(2\pi u\theta') \end{aligned}$$

Complex visibility $V_{AB} = \tilde{P}(\theta') \cos(2\pi u\theta')$

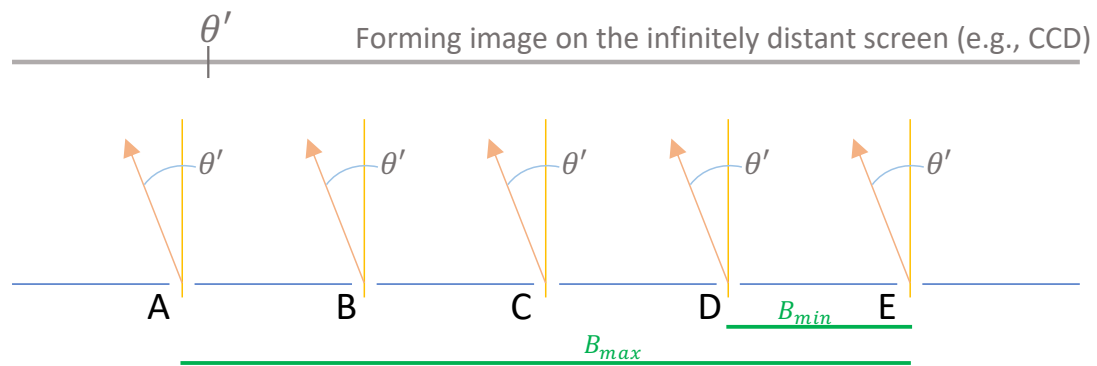
$+ i \tilde{P}(\theta') \sin(2\pi u\theta')$

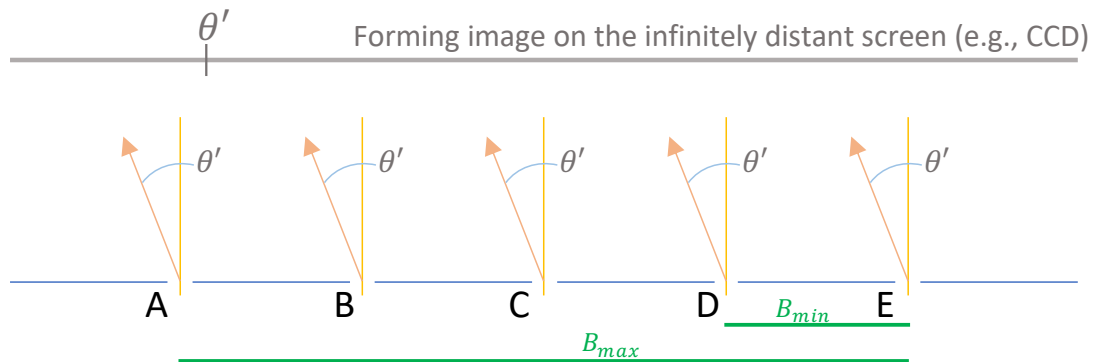
$= \tilde{P}(\theta') e^{i 2\pi u\theta'}$

Visibility phase

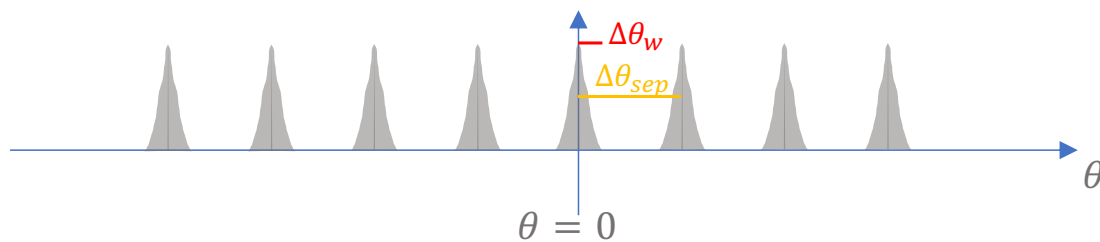
Visibility amplitude

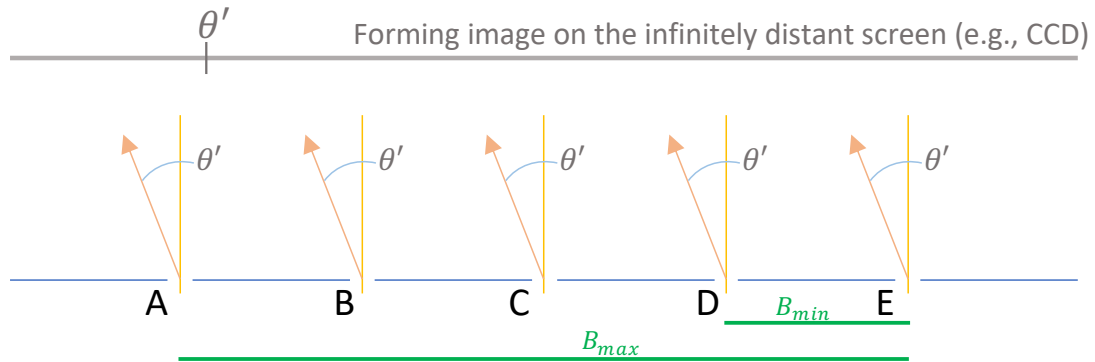
一組基線(baseline)
的測量





Diffraction pattern
(proportional to the number of incoming photons
in a unit of time and a unit of area)



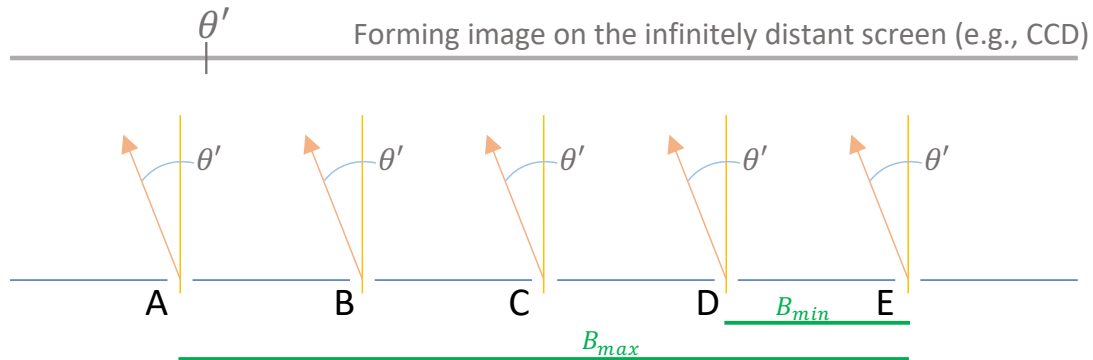


Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

$$\langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle$$

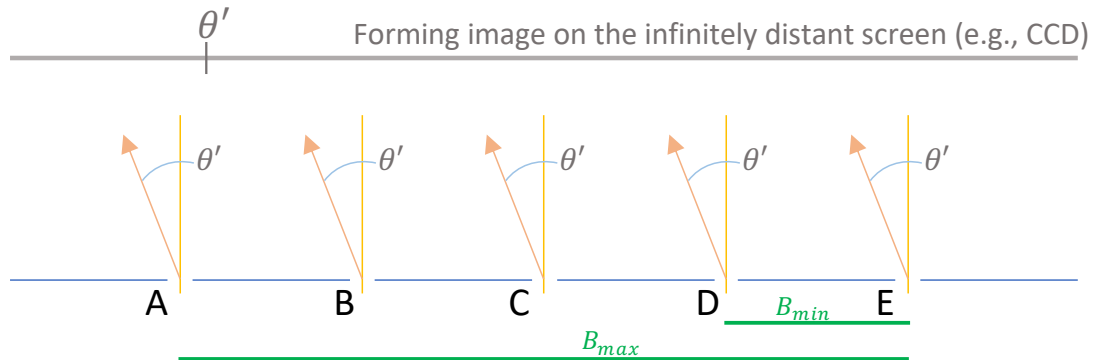


Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle = & \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ & + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ & + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ & + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ & + \langle 2E_D E_E \rangle \end{aligned}$$



Evaluation of diffraction pattern

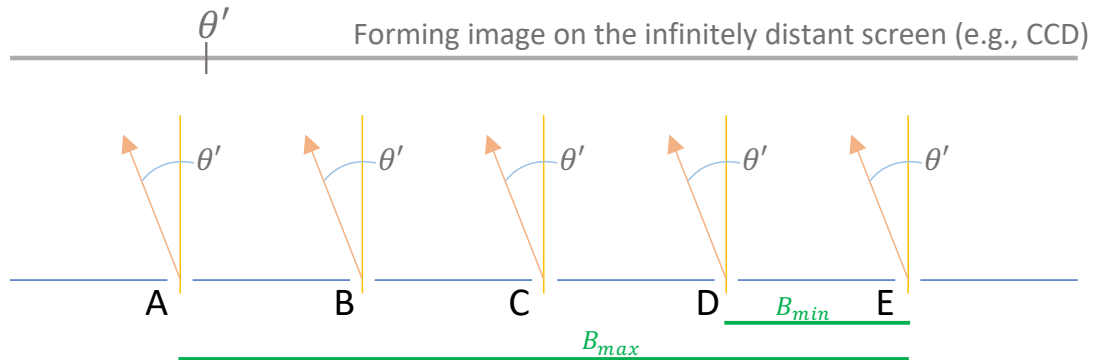
(proportional to the number of incoming photons in a unit of time and a unit of area)

$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

Auto-correlation

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ &\quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ &\quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ &\quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ &\quad + \langle 2E_D E_E \rangle \end{aligned}$$

Cross-
correlation



Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

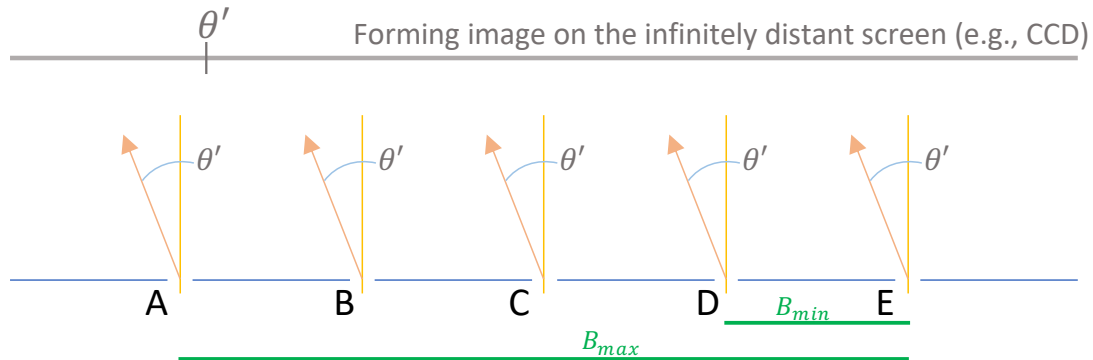
$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

Auto-correlation

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ &\quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ &\quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ &\quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ &\quad + \langle 2E_D E_E \rangle \end{aligned}$$

Cross-correlation

4 slit-pairs at shortest separation



Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

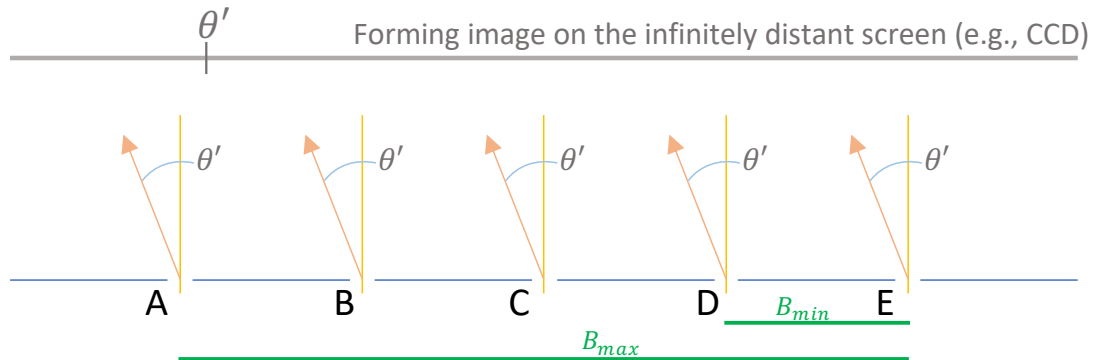
$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

Auto-correlation

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ &\quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ &\quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ &\quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ &\quad + \langle 2E_D E_E \rangle \end{aligned}$$

Cross-
correlation

3 slit-pairs at the 2nd shortest separation



Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

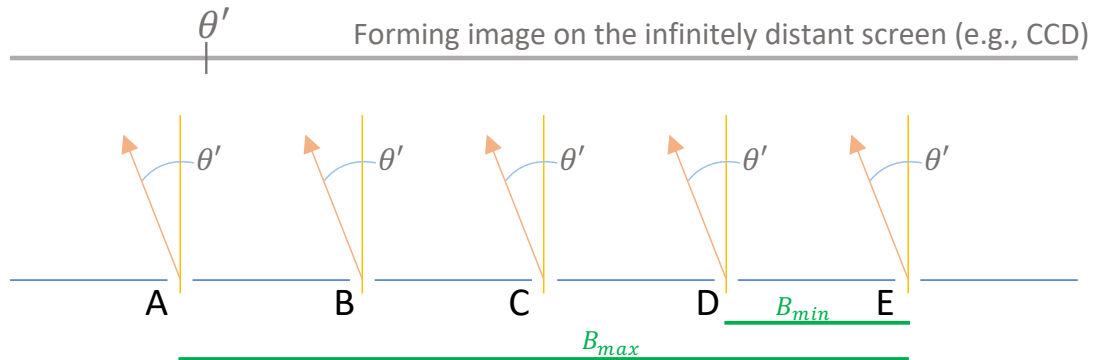
$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

Auto-correlation

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ &\quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ &\quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ &\quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ &\quad + \langle 2E_D E_E \rangle \end{aligned}$$

Cross-correlation

2 slit-pairs at the 3rd shortest separation



Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

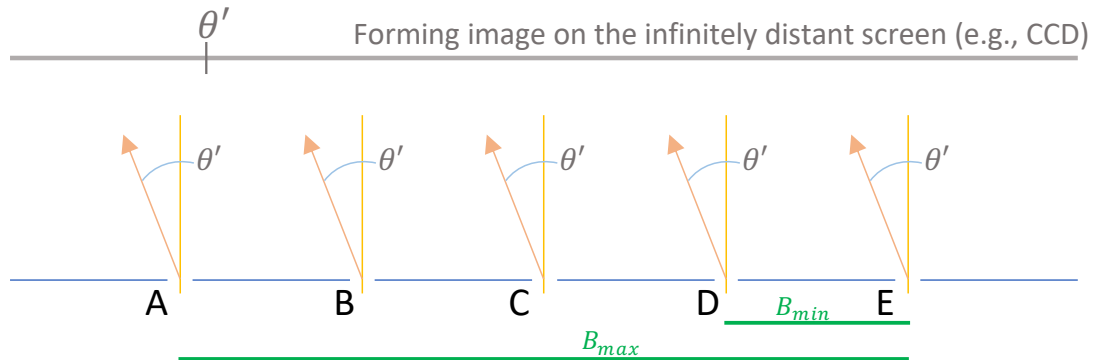
$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

Auto-correlation

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ &\quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ &\quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ &\quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ &\quad + \langle 2E_D E_E \rangle \end{aligned}$$

Cross-correlation

1 slit-pair at longest separation



Evaluation of diffraction pattern

(proportional to the number of incoming photons in a unit of time and a unit of area)

$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

Auto-correlation

$$\begin{aligned} \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ &\quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ &\quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ &\quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ &\quad + \langle 2E_D E_E \rangle \end{aligned}$$

Cross-correlation

1 slit-pair at longest separation

Form of each cross-term

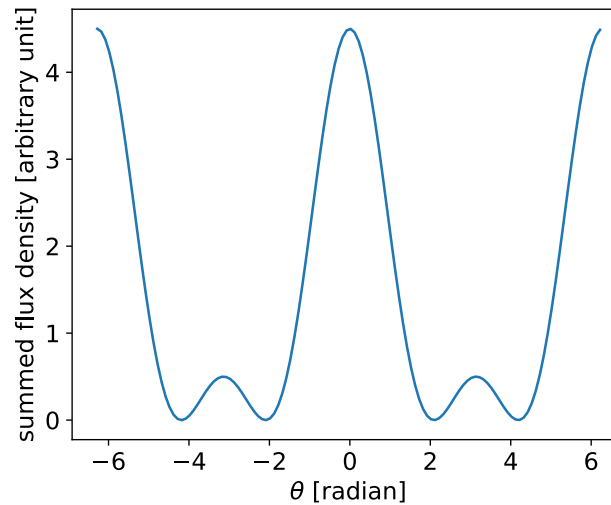
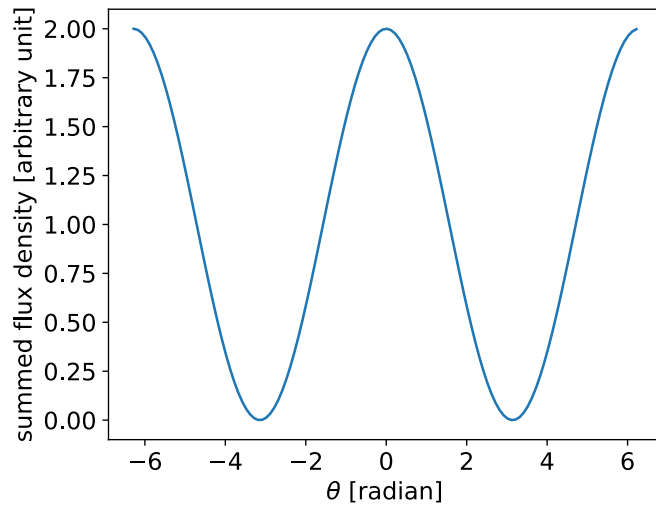
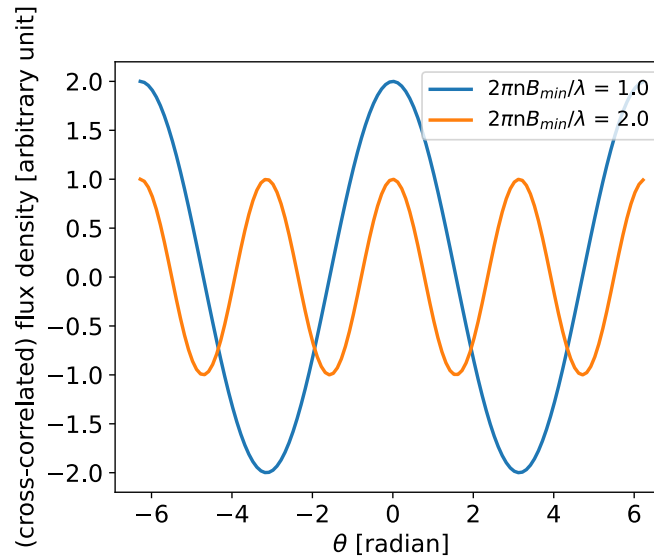
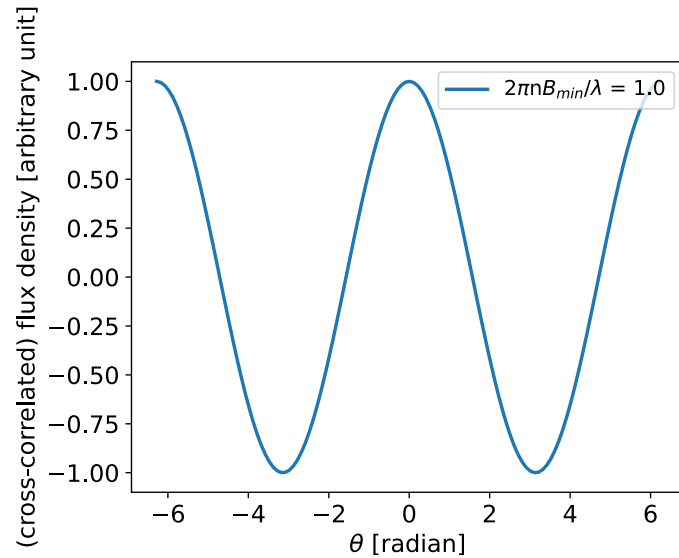
$$\begin{aligned} &\langle 2E_0^2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \\ &= \langle \cos(k\Delta x) \rangle + \langle \cos(2kx - 2\omega t + 2\phi_0 + k\Delta x) \rangle, \quad \Delta x = nB_{\min} \sin \theta' \\ &= \cos\left(\frac{2\pi}{\lambda} nB_{\min} \sin \theta'\right) \sim \cos\left(\frac{2\pi}{\lambda} nB_{\min} \theta'\right) \end{aligned}$$

Trigonometric identity:

$$\begin{aligned} &\cos \psi \cos \varphi \\ &= \frac{1}{2} [\cos(\psi - \varphi) + \cos(\psi + \varphi)] \end{aligned}$$

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Department of Physics

Assuming a diffraction grating with 2 and 3 slits



Flux density:

$$\cos(knB_{min}\theta) = \cos\left(2\pi \frac{nB_{min}}{\lambda} \theta\right),$$

$$n = 1, 2, 3, 4, 5, \dots, (N-1)$$

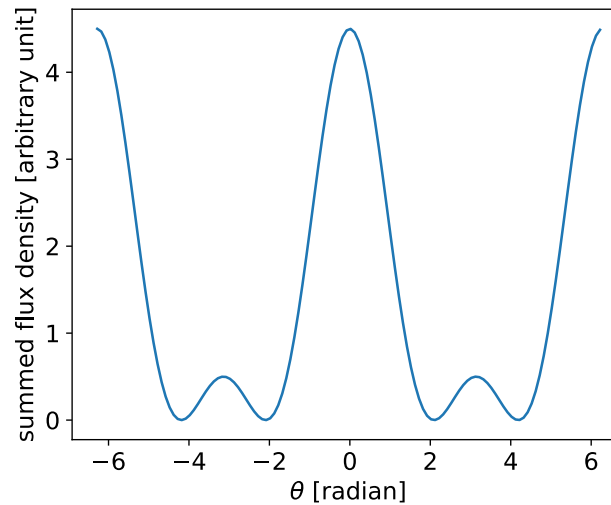
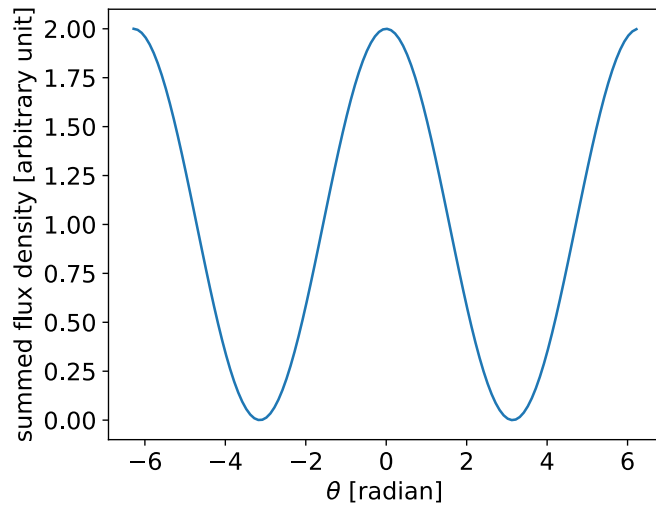
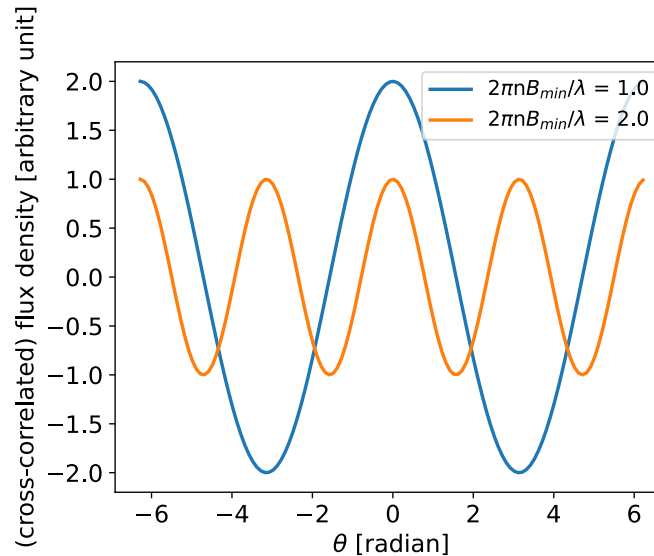
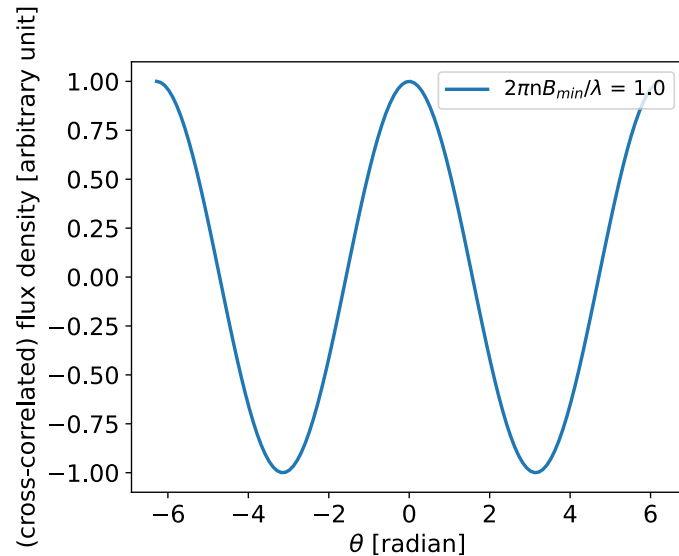
Summed flux density:

$$\sum_n (N - n) \cos(knB_{min}\theta) + \text{auto-corr.}$$

Properties:

1. For any n , $\theta = 0$ is always the location of the central maximum (i.e., constructive interference)

Assuming a diffraction grating with 2 and 3 slits



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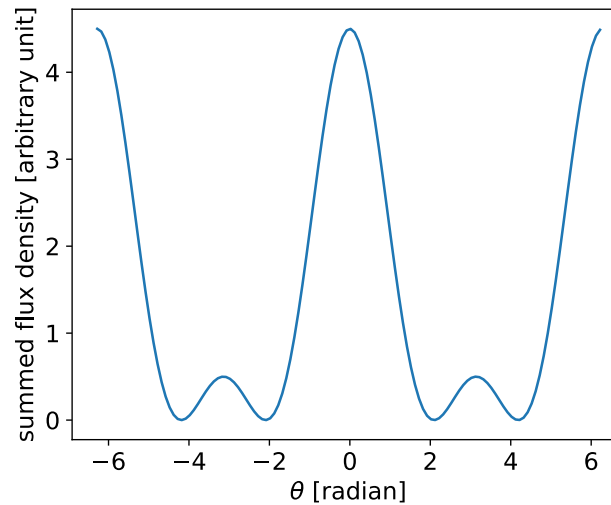
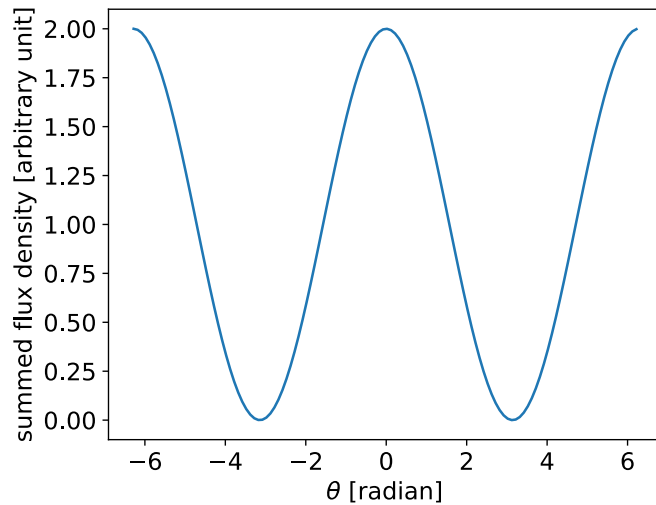
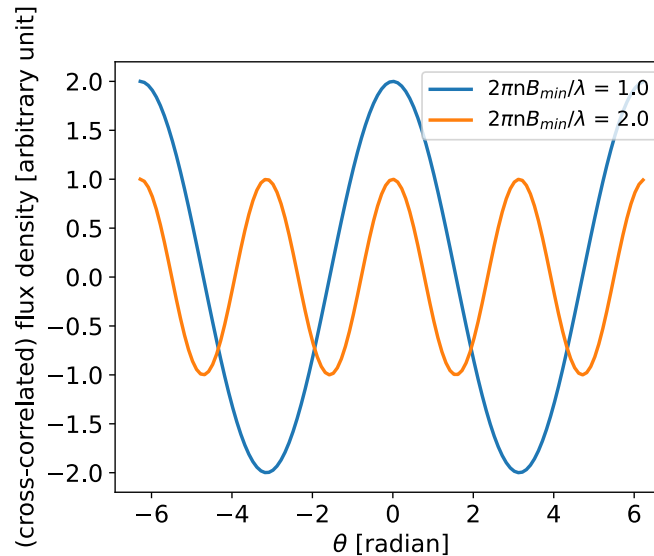
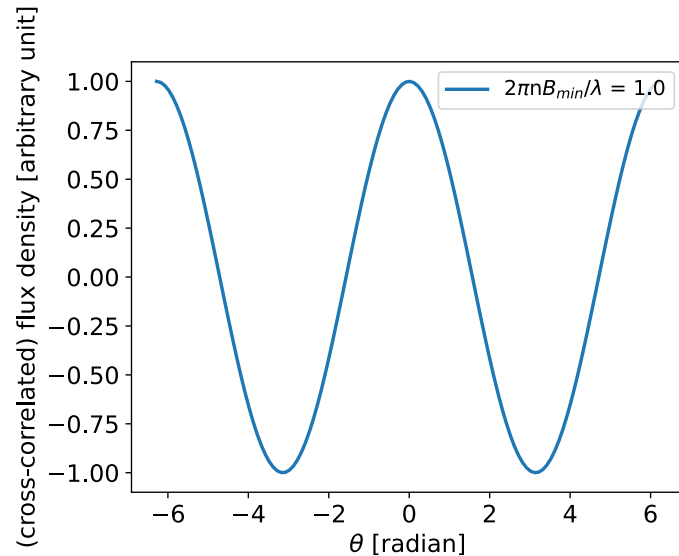
Summed flux density:

$$\sum_n (N - n) \cos(knB_{min}\theta) + \text{auto-corr.}$$

Properties:

1. For any n , $\theta = 0$ is always the location of the central maximum (i.e., constructive interference)
2. The locations of the maxima for the $n = 1$ slit-pairs are always the locations of the maxima for any other pairs of slits.

Assuming a diffraction grating with 2 and 3 slits



Flux density:

$$\cos(knB_{min}\theta) = \cos\left(2\pi \frac{nB_{min}}{\lambda} \theta\right),$$

$$n = 1, 2, 3, 4, 5, \dots, (N-1)$$

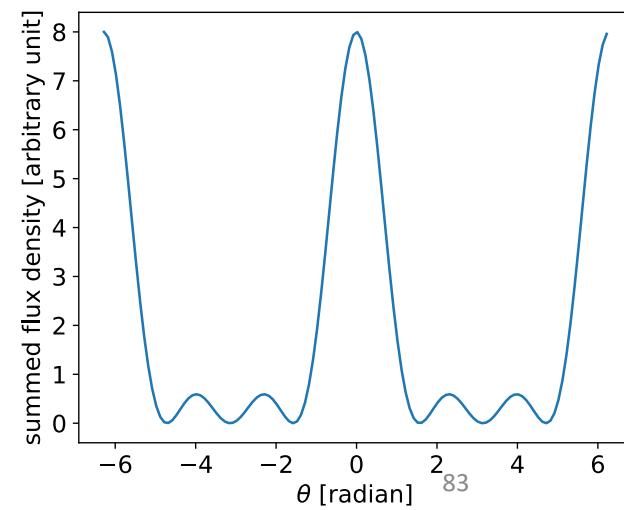
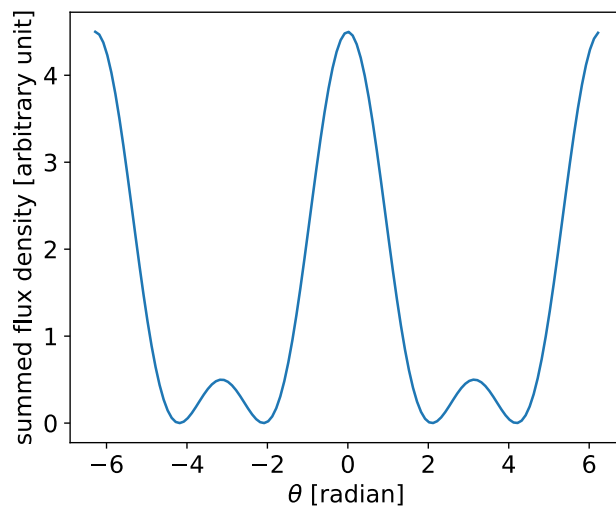
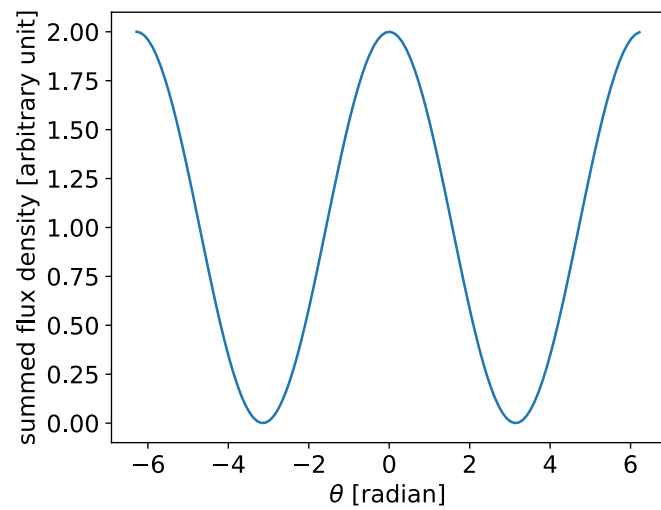
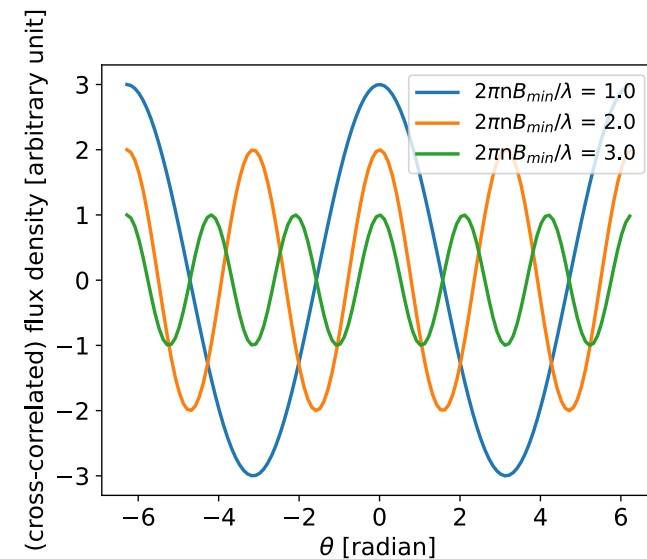
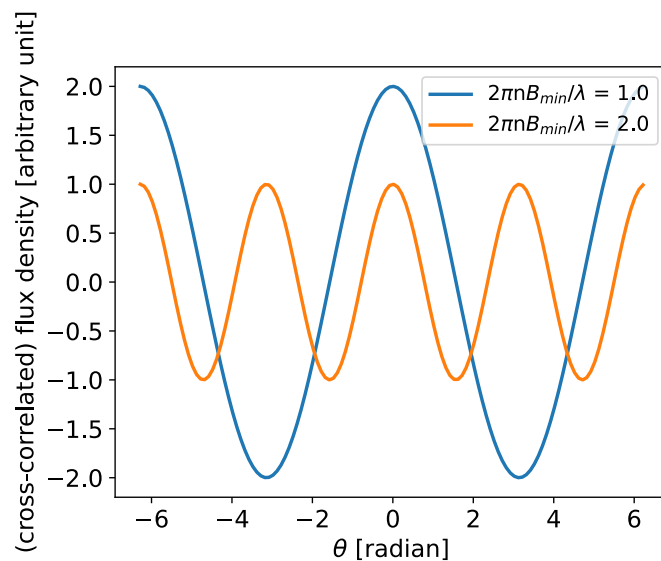
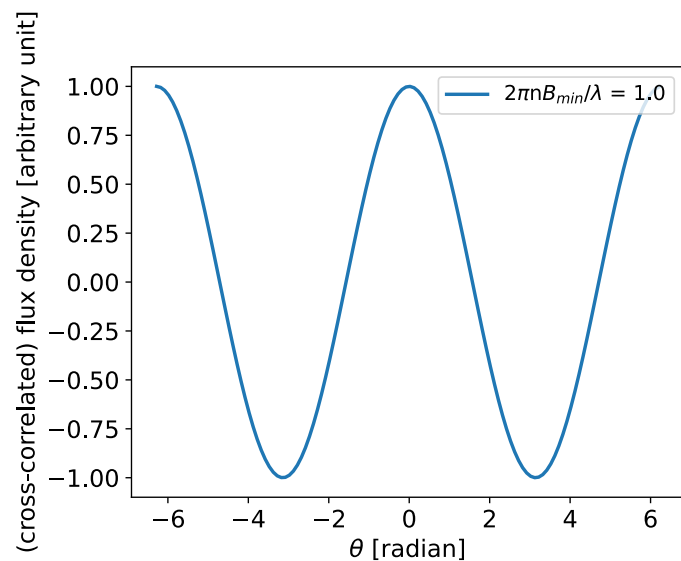
Summed flux density:

$$\sum_n (N - n) \cos(knB_{min}\theta) + \text{auto-corr.}$$

Properties:

1. For any n , $\theta = 0$ is always the location of the central maximum (i.e., constructive interference)
2. The locations of the maxima for the $n = 1$ slit-pairs are always the locations of the maxima for any other pairs of slits.
3. Between the maxima of the $n=1$ pairs of slits, the destructive interference makes the photon flux density approach 0. (i.e., the cross-correlation terms cancel each other).

Assuming a diffraction grating with 2, 3, and 4 slits



Assuming a diffraction grating with 2, 3, and 10 slits

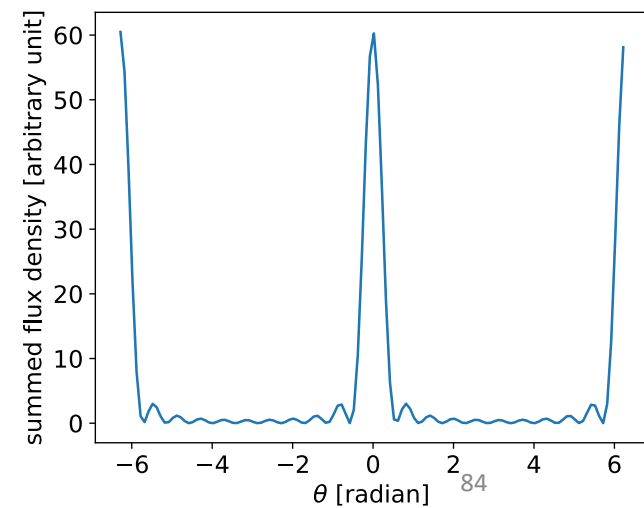
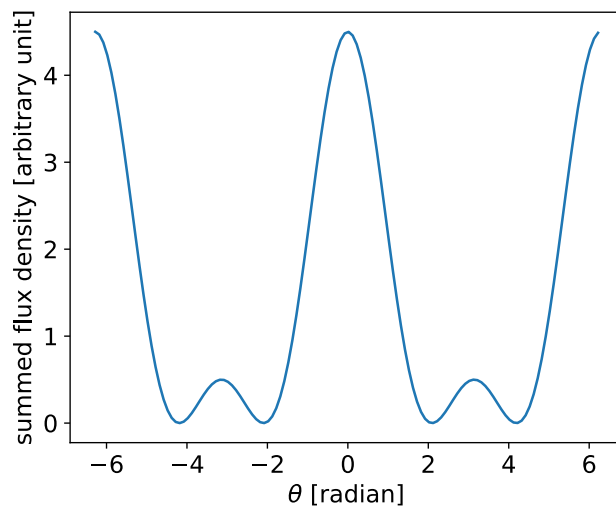
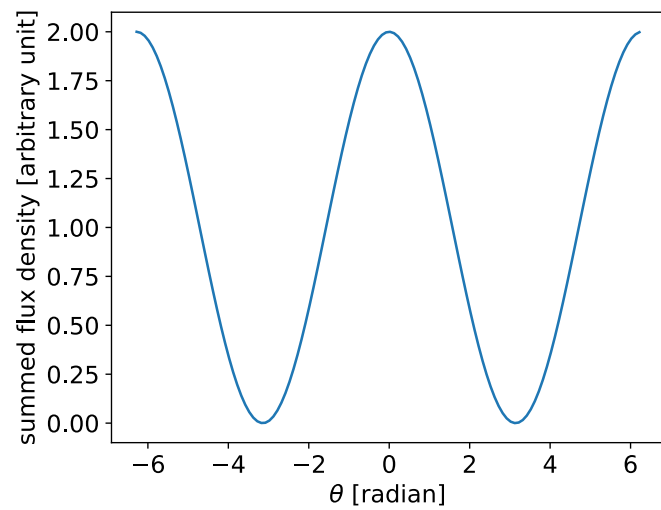
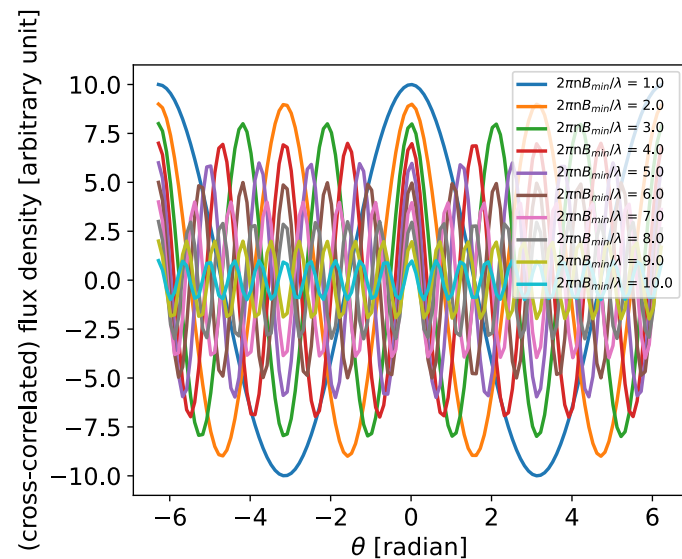
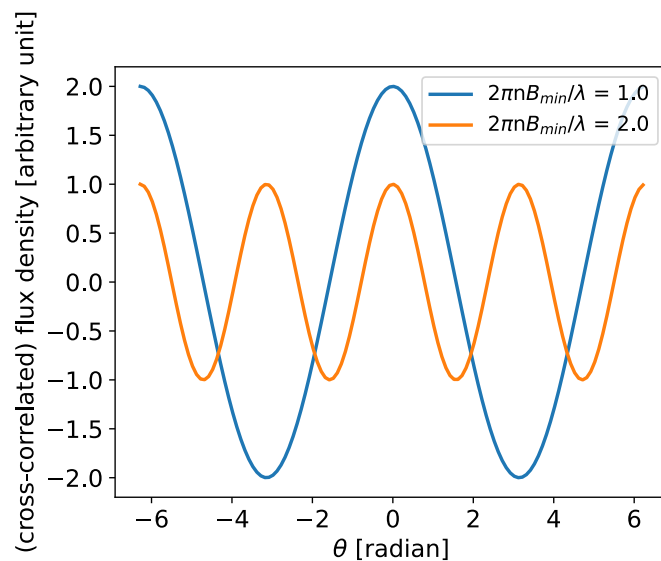
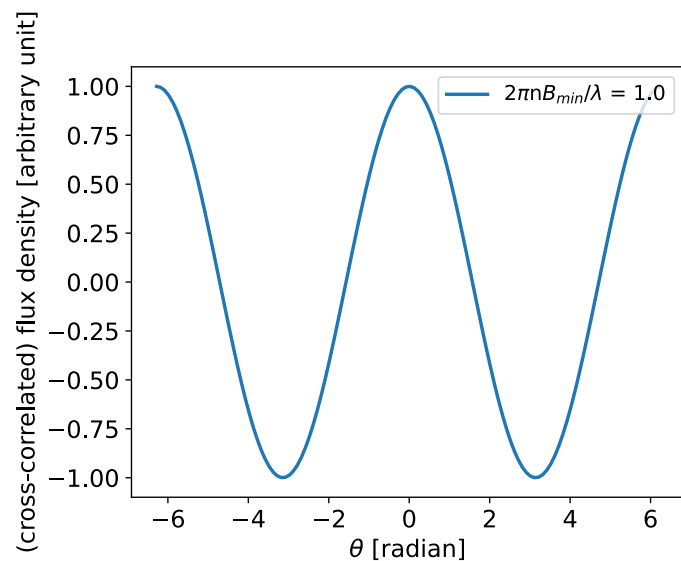
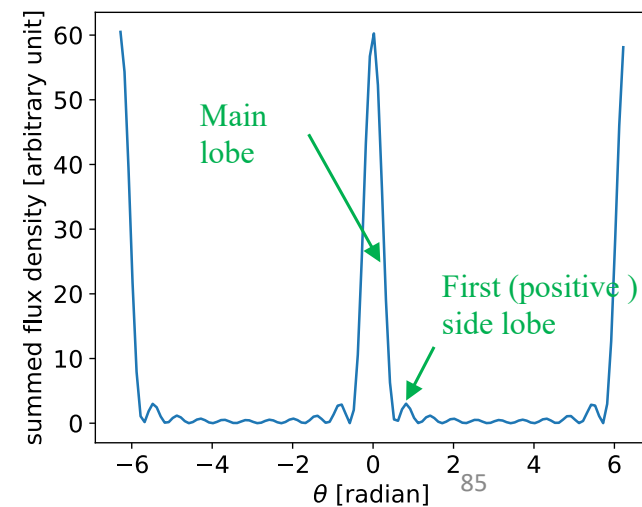
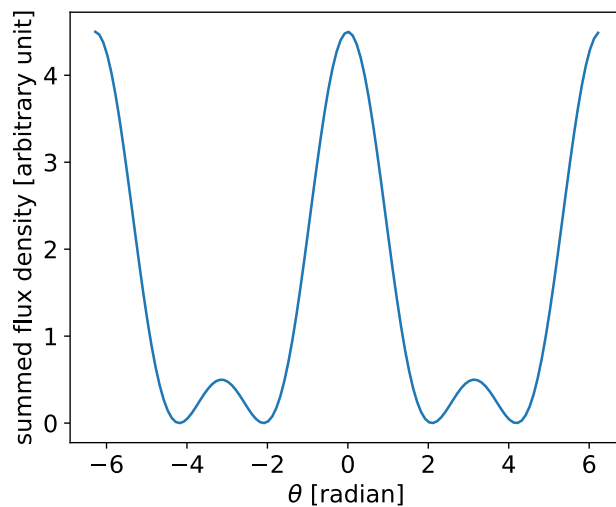
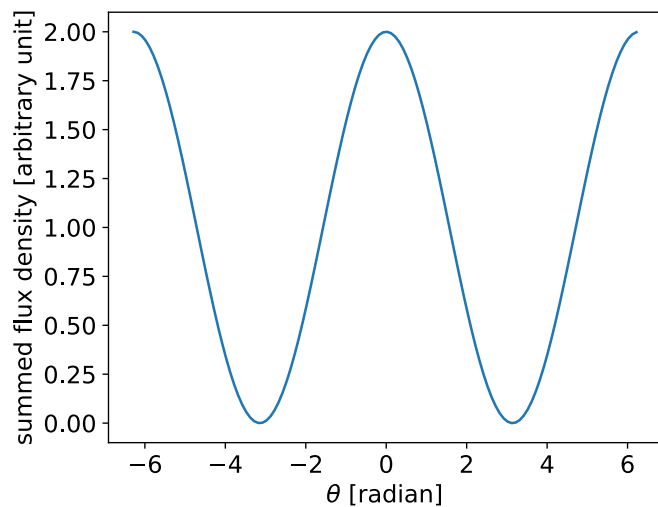
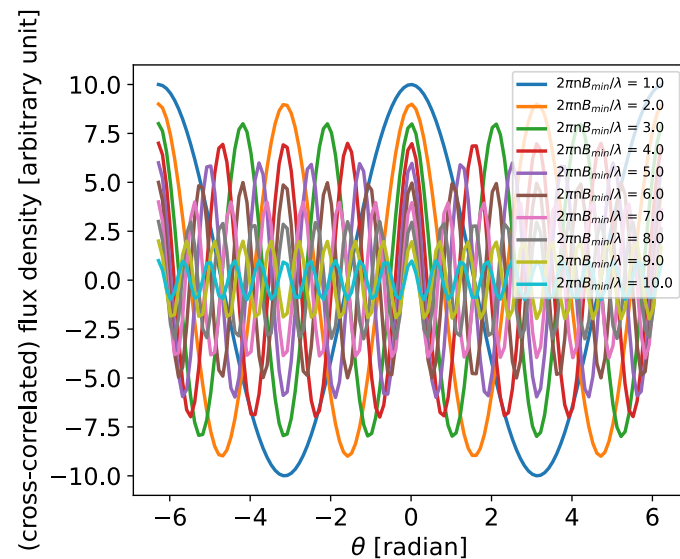
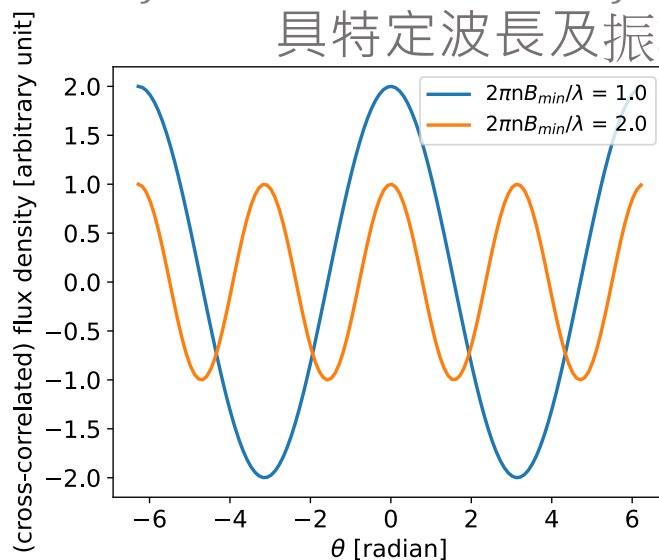
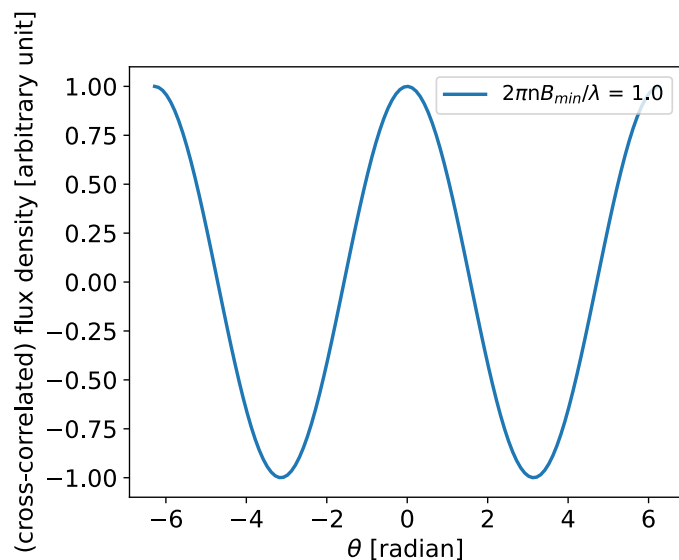
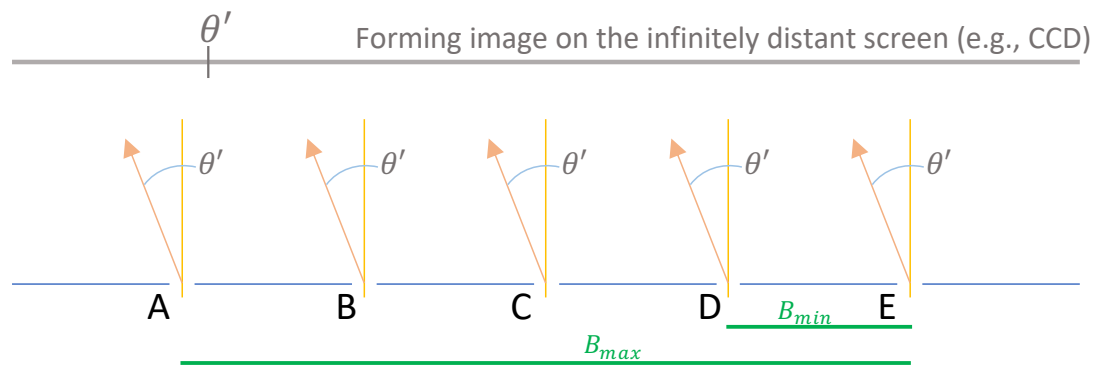
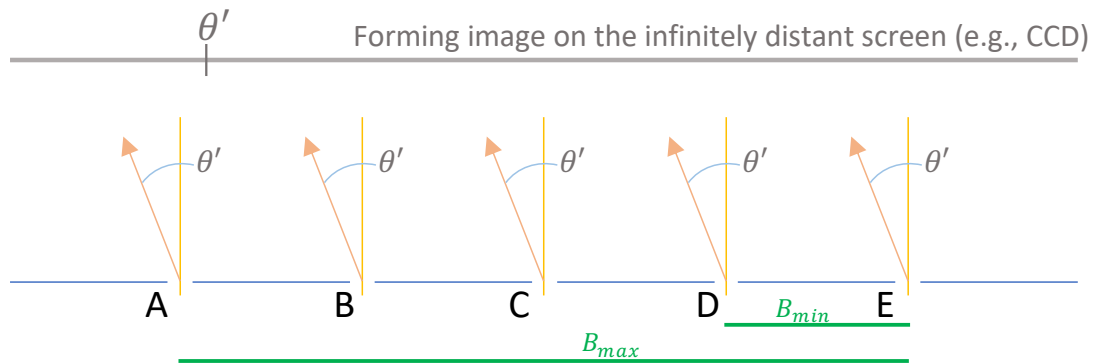


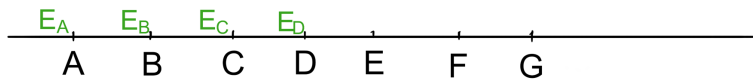
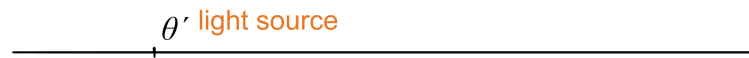
Image is formed by co-adding $\langle 2E_i E_j \rangle$ (什麼是 $\langle 2E_i E_j \rangle$? 空間方向cosine函式
具特定波長及振幅)

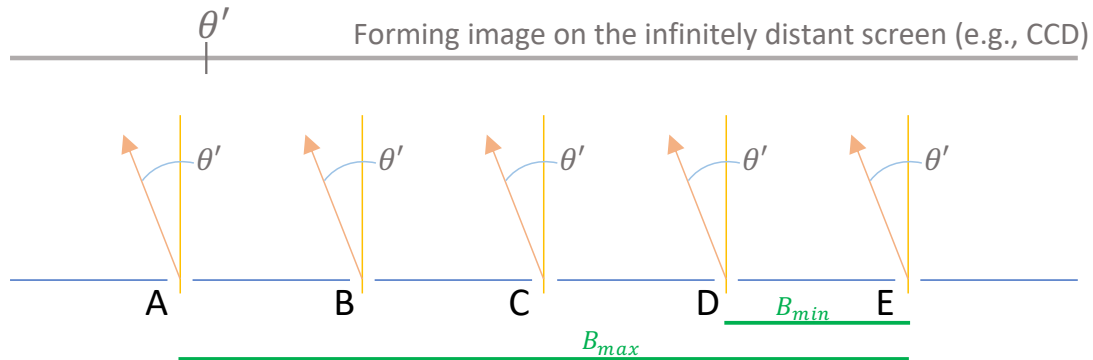






Interferometric array observation towards a infinitely distant, monochromatic source



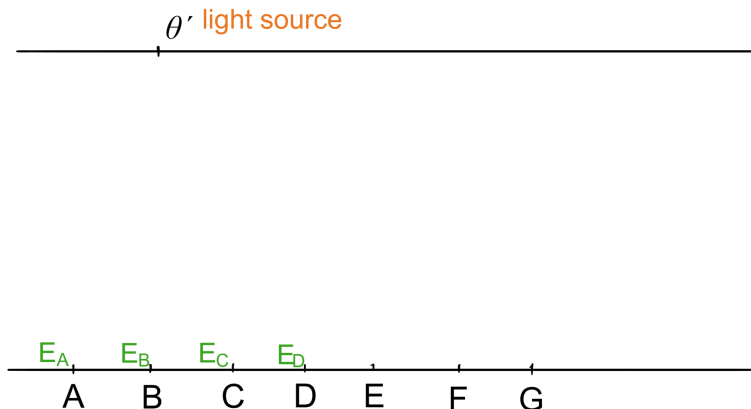


$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

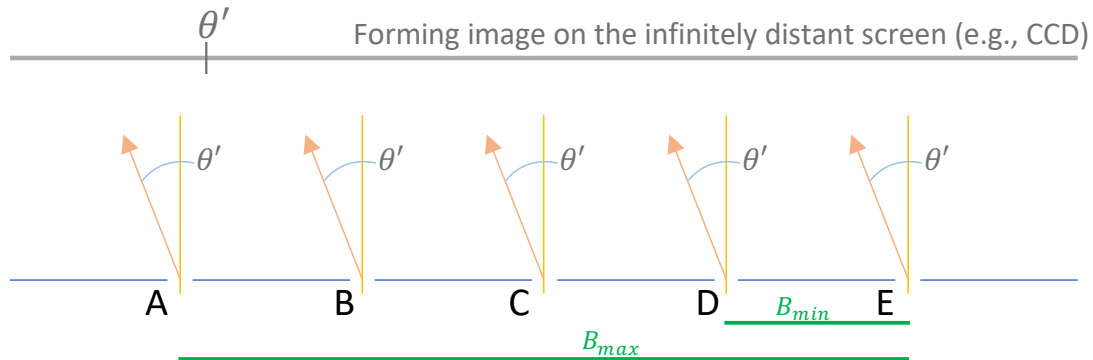
With correlators, we can evaluate each of the following terms:

$$\begin{aligned} & \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle \\ &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ & \quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ & \quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ & \quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ & \quad + \langle 2E_D E_E \rangle \end{aligned}$$

Interferometric array observation towards a infinitely distant, monochromatic source



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Department of Physics

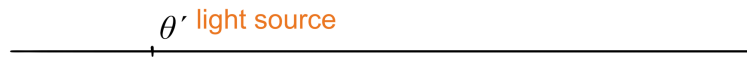


$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

With correlators, we can evaluate each of the following terms:

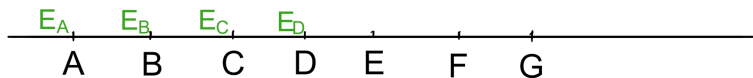
$$\begin{aligned} & \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle \\ &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ & \quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ & \quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ & \quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ & \quad + \langle 2E_D E_E \rangle \end{aligned}$$

Interferometric array observation towards a infinitely distant, monochromatic source

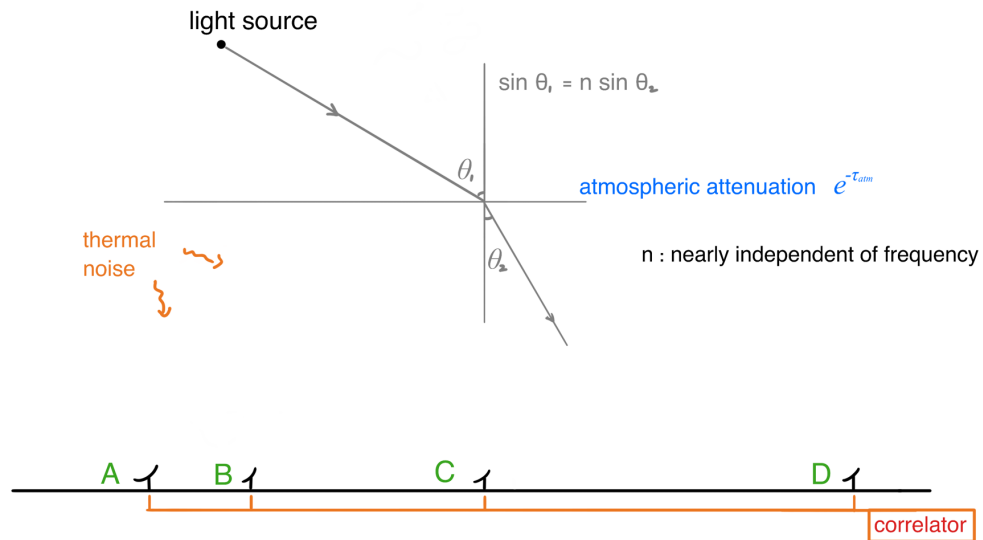


Form of each cross-term

$$\begin{aligned} & \langle 2E_0^2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \\ &= \langle \cos(k\Delta x) \rangle + \langle \cos(2kx - 2\omega t + 2\phi_0 + k\Delta x) \rangle, \quad \Delta x = nB_{min} \sin \theta' \\ &= \cos\left(\frac{2\pi}{\lambda} nB_{min} \sin \theta'\right) \sim \cos\left(\frac{2\pi}{\lambda} nB_{min} \theta'\right) \end{aligned}$$



1D linear radio interferometer



$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

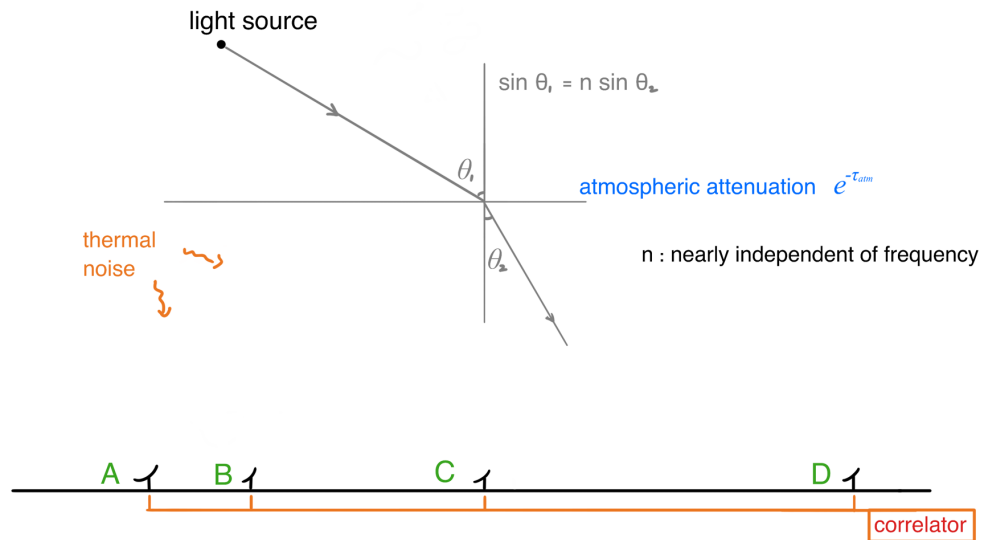
With correlators, we can evaluate each of the following terms:

$$\begin{aligned} & \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle \\ &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ & \quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ & \quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ & \quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ & \quad + \langle 2E_D E_E \rangle \end{aligned}$$

General expression for the cross (cosine) correlation of antennae i and j:

$$\begin{aligned} & \langle 2E_0^2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \\ &= \langle \cos(k\Delta x) \rangle + \langle \cos(2kx - 2\omega t + 2\phi_0 + k\Delta x) \rangle, \quad \Delta x = B_{ij} \sin \theta' \\ &= \cos\left(\frac{2\pi}{\lambda} B_{ij} \sin \theta'\right) \sim \cos\left(\frac{2\pi}{\lambda} B_{ij} \theta'\right) \end{aligned}$$

1D linear radio interferometer



$$E_A = E_0 \cos(kx - \omega t + \phi_0)$$

With correlators, we can evaluate each of the following terms:

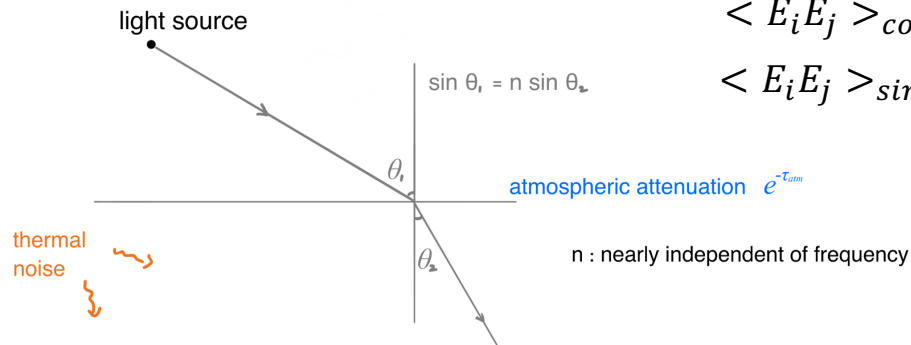
$$\begin{aligned} & \langle (E_A + E_B + E_C + E_D + E_E + \dots)^2 \rangle \\ &= \langle E_A^2 \rangle + \langle E_B^2 \rangle + \langle E_C^2 \rangle + \langle E_D^2 \rangle + \langle E_E^2 \rangle \\ & \quad + \langle 2E_A E_B \rangle + \langle 2E_A E_C \rangle + \langle 2E_A E_D \rangle + \langle 2E_A E_E \rangle \\ & \quad + \langle 2E_B E_C \rangle + \langle 2E_B E_D \rangle + \langle 2E_B E_E \rangle \\ & \quad + \langle 2E_C E_D \rangle + \langle 2E_C E_E \rangle \\ & \quad + \langle 2E_D E_E \rangle \end{aligned}$$

General expression for the cross (cosine) correlation of antennae i and j:

$$\begin{aligned} & \langle 2E_0^2 \cos(kx - \omega t + \phi_0) \cos(k(x + \Delta x) - \omega t + \phi_0) \rangle \\ &= \langle \cos(k\Delta x) \rangle + \langle \cos(2kx - 2\omega t + 2\phi_0 + k\Delta x) \rangle, \\ &= \cos\left(\frac{2\pi}{\lambda} B_{ij} \sin \theta'\right) \sim \cos\left(2\pi \frac{B_{ij}}{\lambda} \theta'\right) \equiv \cos(2\pi u_{ij} \theta') \end{aligned}$$

$$\Delta x = \underbrace{B_{ij}}_{\text{Separation of antennae i and j}} \sin \theta'$$

1D linear radio interferometer



Evaluating Complex visibility of a point source at θ'
(i.e., Dirac delta function) taken from antennae i and j

$$\langle E_i E_j \rangle_{\cos} \cong \tilde{P}(\theta') \cos\left(\frac{2\pi}{\lambda} B_{ij} \theta'\right) = \int \delta(\theta - \theta') \tilde{P}(\theta) \cos(2\pi u_{ij} \theta) d\theta$$

$$\langle E_i E_j \rangle_{\sin} \cong \tilde{P}(\theta') \sin\left(\frac{2\pi}{\lambda} B_{ij} \theta'\right) = \int \delta(\theta - \theta') \underbrace{\tilde{P}(\theta)} \sin(2\pi u_{ij} \theta) d\theta$$

Response function of each single-dish reflector

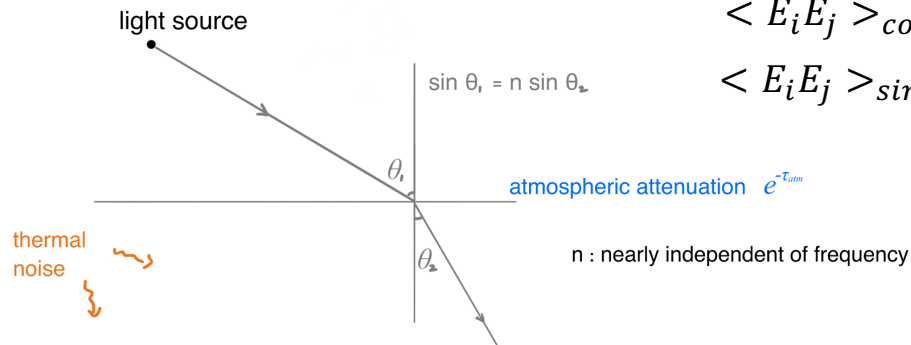
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Separation of
antennae i and j

1D linear radio interferometer



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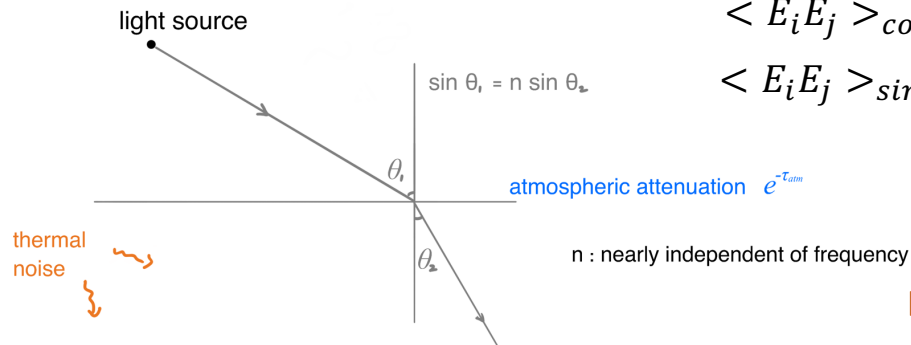
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1D linear radio interferometer



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(i.e., Dirac delta function) taken from antennae i and j

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Evaluating Complex visibility for continuous intensity distribution

$$V_{ij} = \int A(\theta) \overline{P(\theta)} e^{i2\pi u_{ij} \theta} d\theta$$

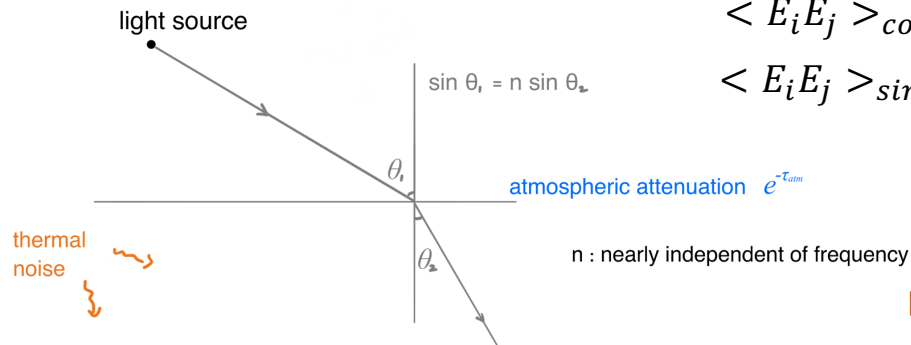


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1D linear radio interferometer



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Evaluating Complex visibility for continuous intensity distribution

$$V_{ij} = \int A(\theta) \overline{P(\theta)} e^{i2\pi u_{ij} \theta} d\theta$$

Mathematical form of fourier transform

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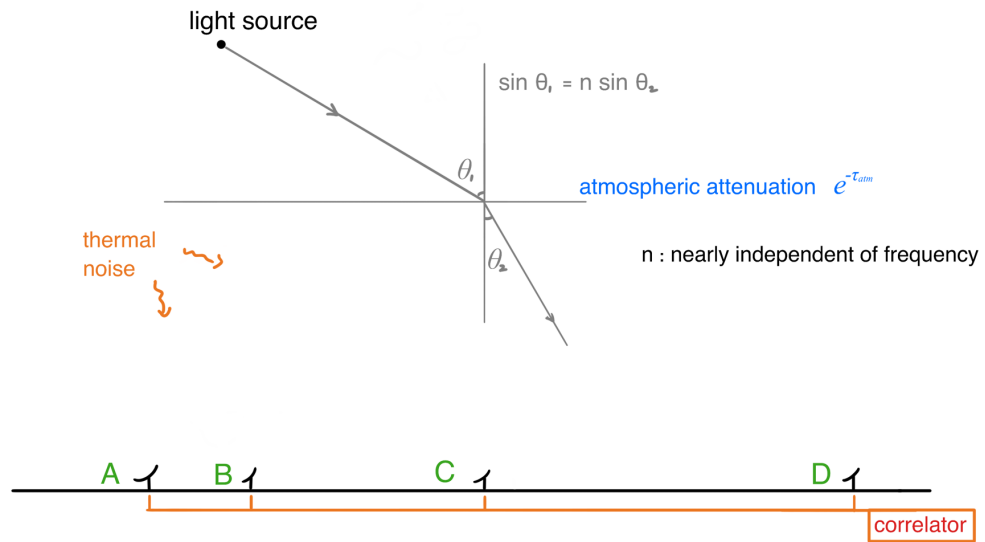
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1D linear radio interferometer

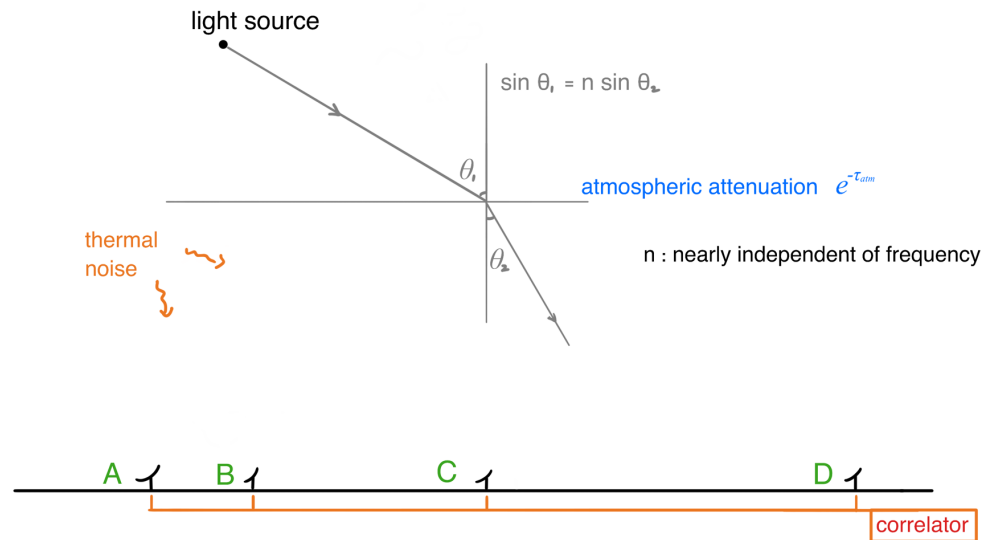
Evaluating Complex visibility for continuous intensity distribution

$$V_{ij} = \int A(\theta) \overline{P(\theta)} e^{i2\pi u_{ij}\theta} d\theta$$



1D linear radio interferometer

Evaluating Complex visibility for continuous intensity distribution



$$V(u) = \int A(\theta) \underbrace{\overline{P(\theta)} e^{i2\pi u \theta}}_{\text{Mathematical form of fourier transform}} d\theta$$

Mathematical form of fourier transform

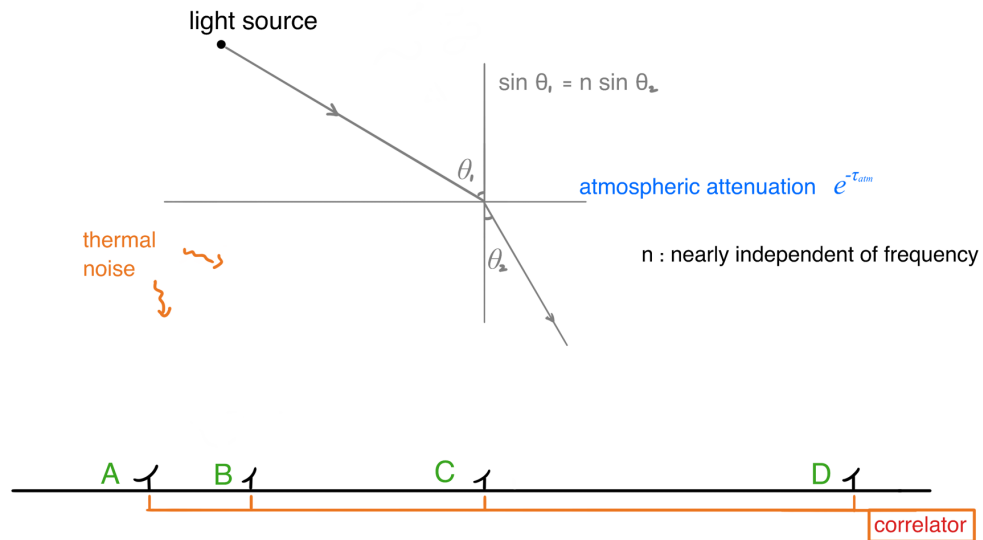
What we directly measure

1D linear radio interferometer

Evaluating Complex visibility for continuous intensity distribution

$$V(u) = \int \underbrace{A(\theta) \overline{P(\theta)}}_{\text{Mathematical form of fourier transform}} e^{i2\pi u \theta} d\theta$$

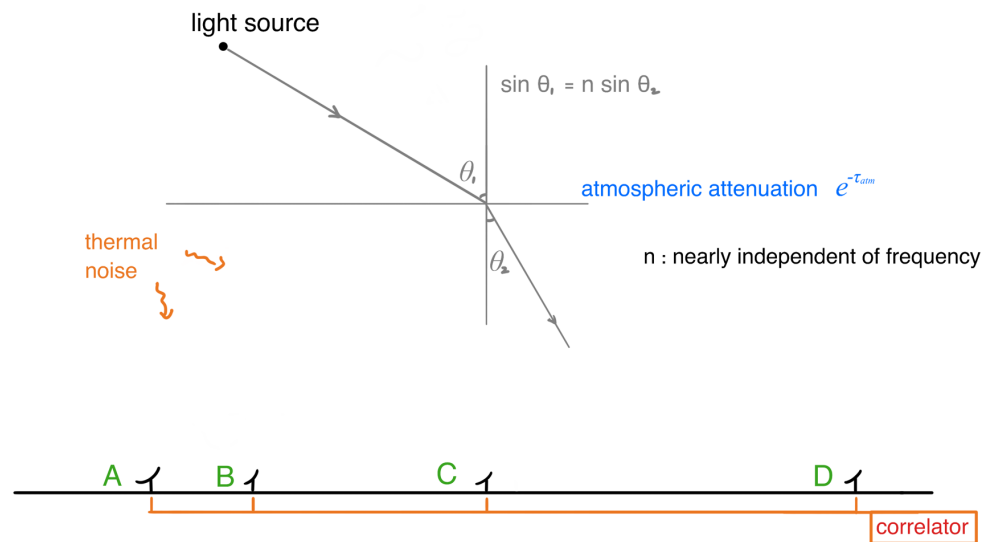
Mathematical form of fourier transform



What we are interested in

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Evaluating Complex visibility for continuous intensity distribution



$$V(u) = \int \underbrace{A(\theta) \overline{P(\theta)}}_{\text{Mathematical form of fourier transform}} e^{i2\pi u \theta} d\theta$$

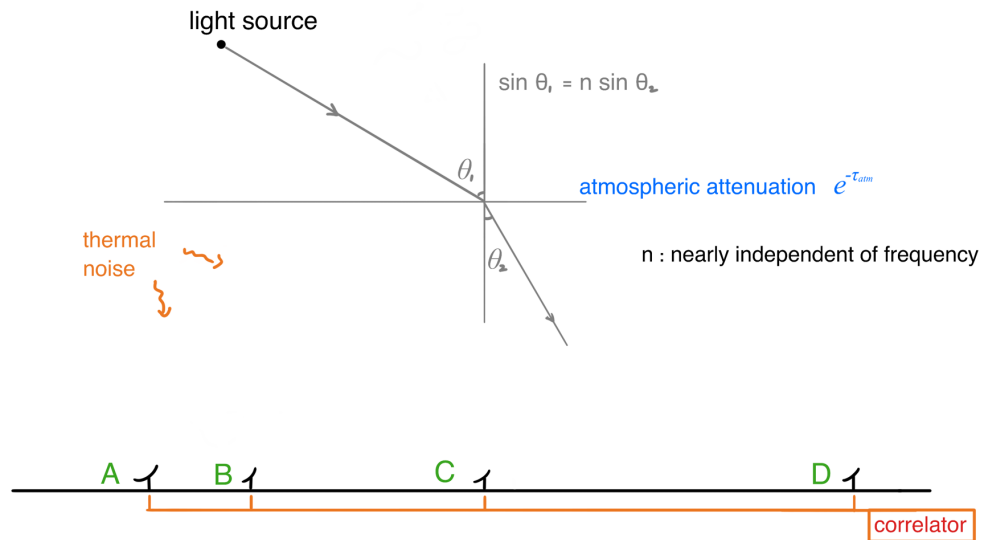
Mathematical form of fourier transform

Inverse fourier transform

$$A(\theta') \overline{P(\theta')} = \int V(u) e^{-i2\pi u \theta'} du$$

If $V(u)$ can be completely sampled
(i.e., measured at any u)

1D linear radio interferometer



If $V(u)$ can be completely sampled
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Evaluating Complex visibility for continuous intensity distribution

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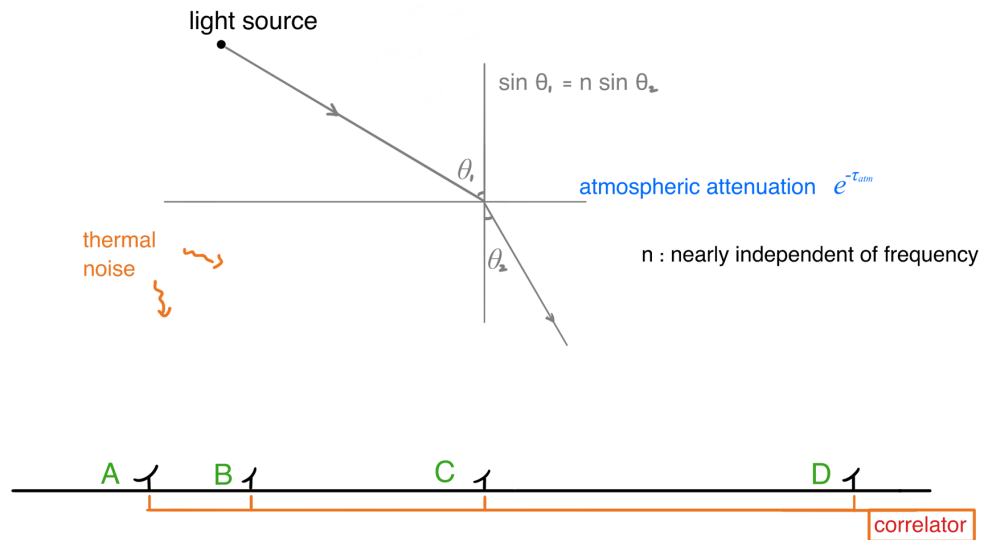
Inverse fourier transform

$$A(\theta') \overline{P(\theta')} = \int V(u) e^{-i2\pi u \theta'} du$$

Requiring measurements at arbitrary antenna-separations, even at infinitely small antenna-separations.

In reality, an interferometric array can only take discretized samples at certain antenna-separations. In addition, the shortest possible (projected) antenna-separation is limited by the sizes of the reflector.

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If $V(u)$ can be completely sampled
(i.e., measured at any u)

Evaluating Complex visibility for continuous intensity distribution

$$V(u) = \int \underbrace{A(\theta) \overline{P(\theta)}}_{\text{Mathematical form of fourier transform}} e^{i2\pi u \theta} d\theta$$

Mathematical form of fourier transform

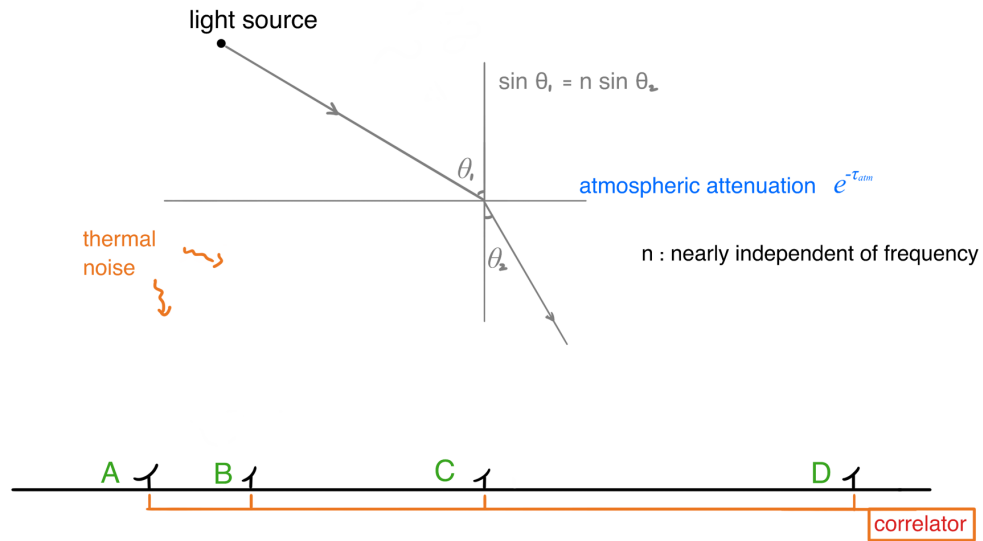
Inverse fourier transform

$$A(\theta') \overline{P(\theta')} = \int V(u) e^{-i2\pi u \theta'} du$$

Sampling function (at $u_k, k = 1, 2, 3, \dots, M$)

$$S(u) \equiv \sum_{k=1}^M \delta(u - u_k)$$

1D linear radio interferometer



If $V(u)$ can be completely sampled
(i.e., measured at any u)

Evaluating Complex visibility for continuous intensity distribution

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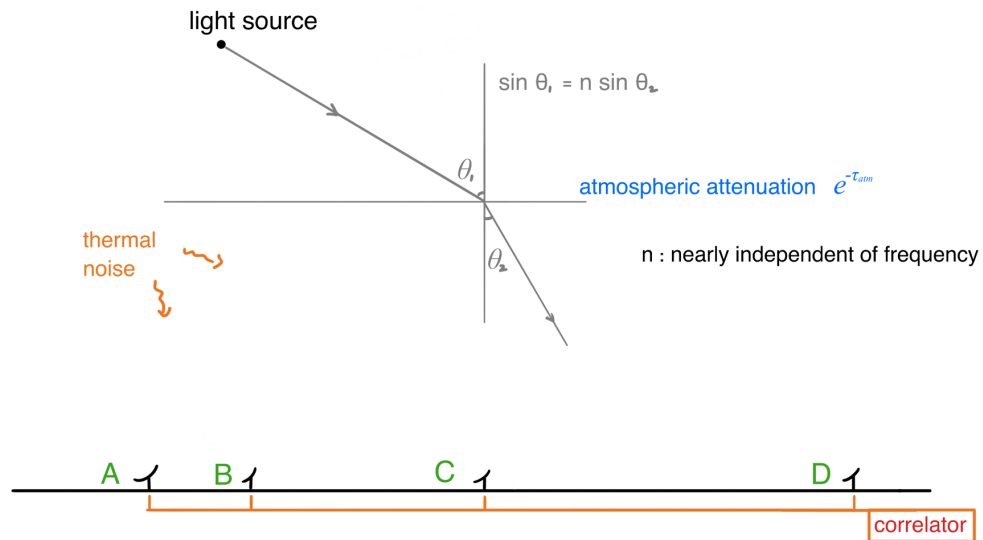
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Naïve inverse fourier transform

$$[A(\theta') \overline{P(\theta')}]^D \equiv \int S(u) V(u) e^{-i2\pi u \theta'} du$$

1D linear radio interferometer



Convolution theorem

$$F.T.(f * g) = F.T.(f) \cdot F.T.(g)$$

$$F.T.(f \cdot g) = F.T.(f) * F.T.(g)$$

Evaluating Complex visibility for continuous intensity distribution

$$V(u) = \int \underbrace{A(\theta) \overline{P(\theta)}}_{\text{Mathematical form of fourier transform}} e^{i2\pi u \theta} d\theta$$

Mathematical form of fourier transform

Inverse fourier transform

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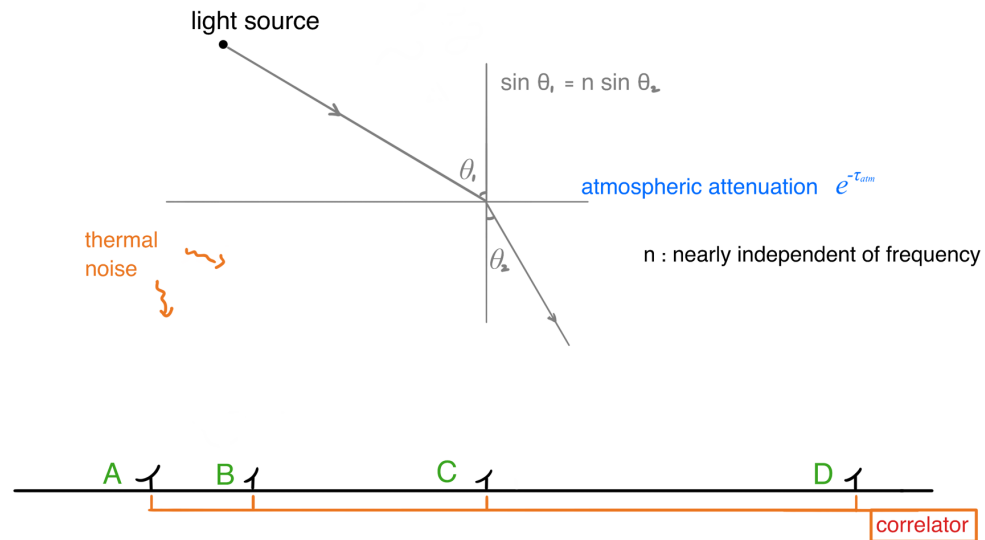
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Naïve inverse fourier transform

$$\underbrace{[A(\theta') \overline{P(\theta')}]^D}_{\text{Dirty image}} \equiv \int S(u) V(u) e^{-i2\pi u \theta'} du$$

1D linear radio interferometer



Evaluating Complex visibility for continuous intensity distribution

$$V(u) = \int \underbrace{A(\theta) \overline{P(\theta)}}_{\text{Mathematical form of fourier transform}} e^{i2\pi u \theta} d\theta$$

Mathematical form of fourier transform

Naïve inverse fourier transform

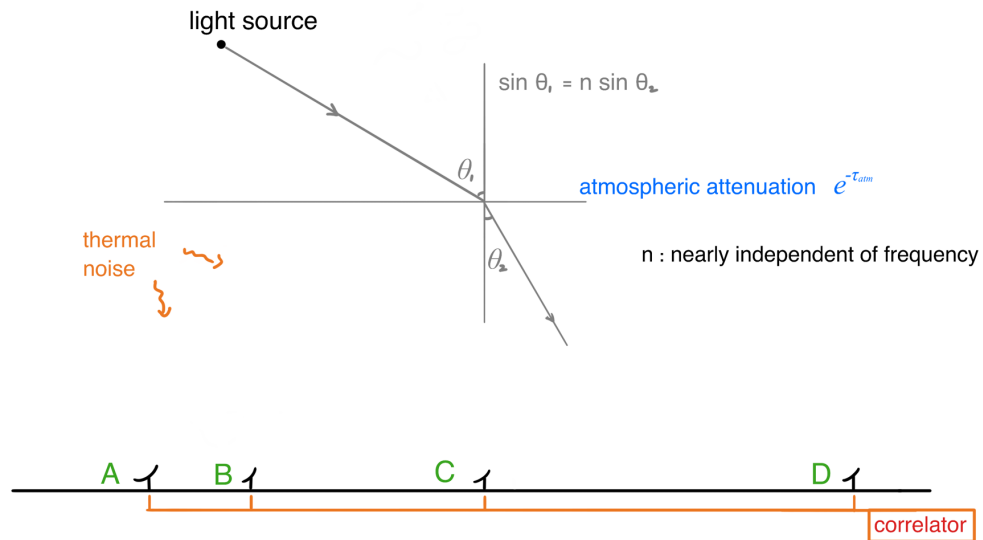
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1D linear radio interferometer



Evaluating Complex visibility for continuous intensity distribution

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Mathematical form of fourier transform

Naïve inverse fourier transform

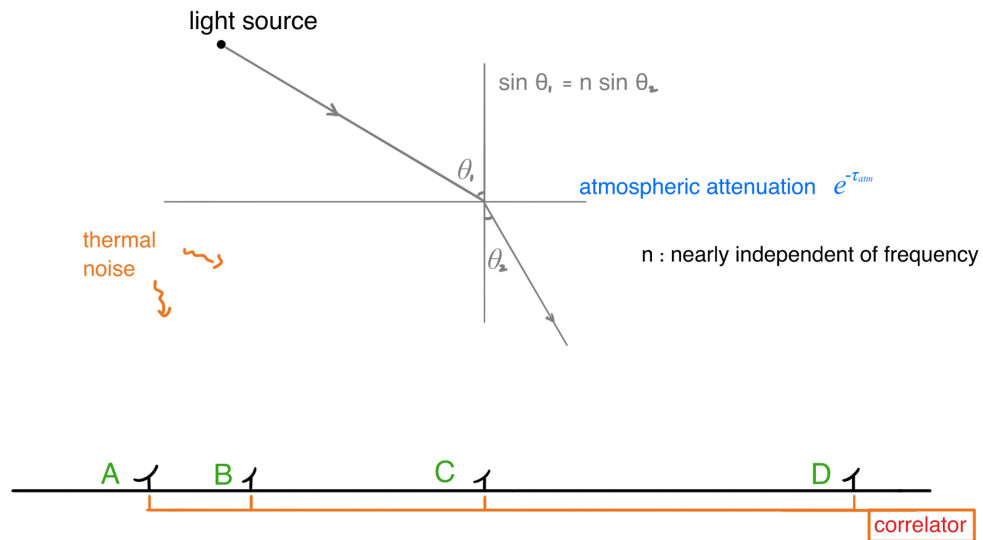
$$\begin{aligned} [A(\theta') \overline{P(\theta')}]^D &\equiv \int S(u) V(u) e^{-i2\pi u \theta'} du \\ &= (\int S(u) e^{-i2\pi u \theta'} du) * (\int V(u) e^{-i2\pi u \theta'} du) \end{aligned}$$

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1D linear radio interferometer



Evaluating Complex visibility for continuous intensity distribution

$$V(u) = \int \underbrace{A(\theta) \widetilde{P(\theta)}}_{\text{Mathematical form of fourier transform}} e^{i2\pi u \theta} d\theta$$

Mathematical form of fourier transform

Naïve inverse fourier transform

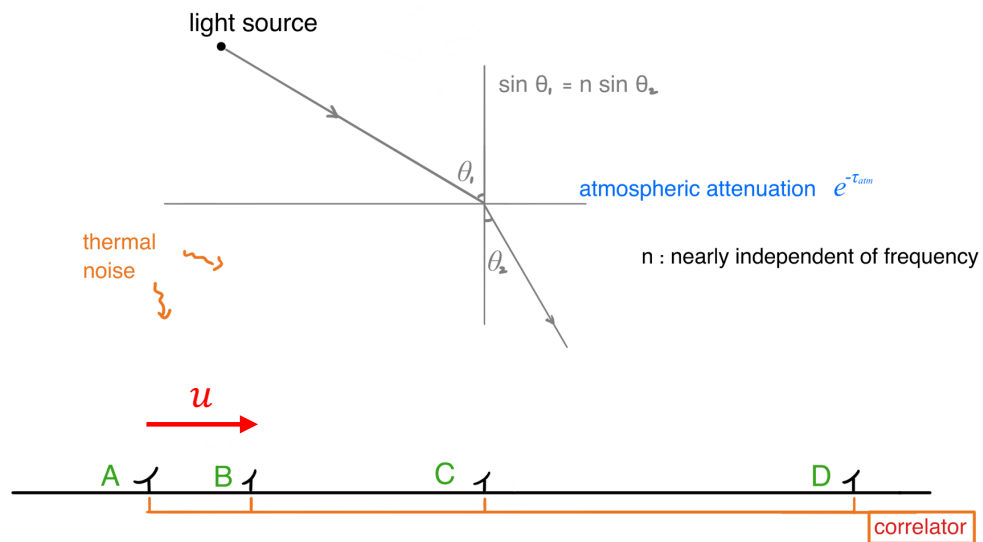
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1D linear radio interferometer



Evaluating Complex visibility for continuous intensity distribution

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Mathematical form of fourier transform

Naïve inverse fourier transform

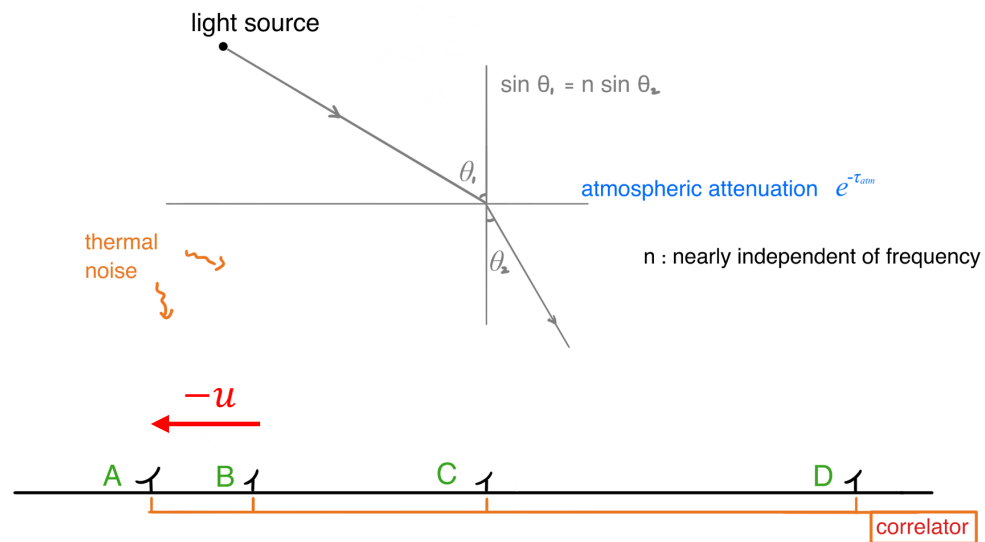
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1D linear radio interferometer



Evaluating Complex visibility for continuous intensity distribution

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Mathematical form of fourier transform

Naïve inverse fourier transform

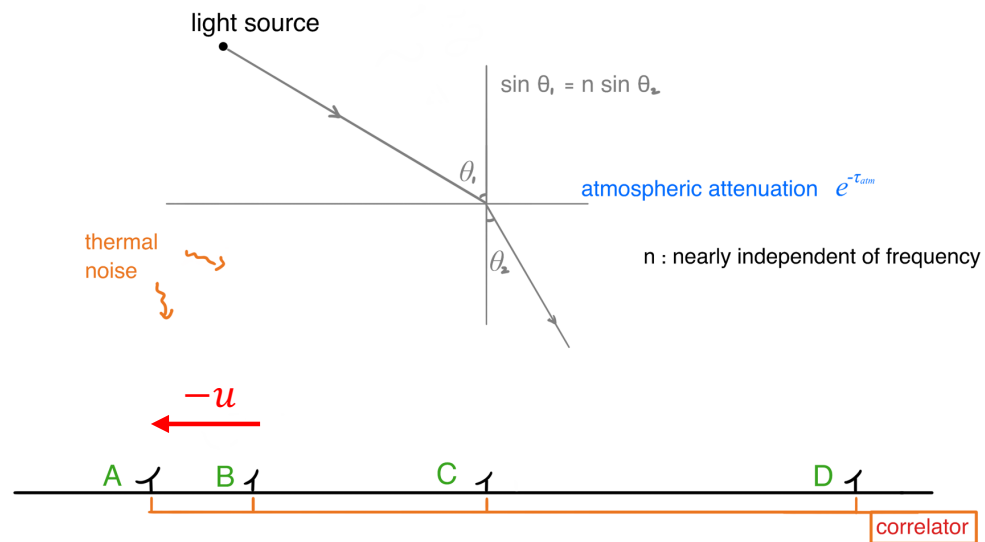
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1D linear radio interferometer



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Evaluating Complex visibility for continuous intensity distribution

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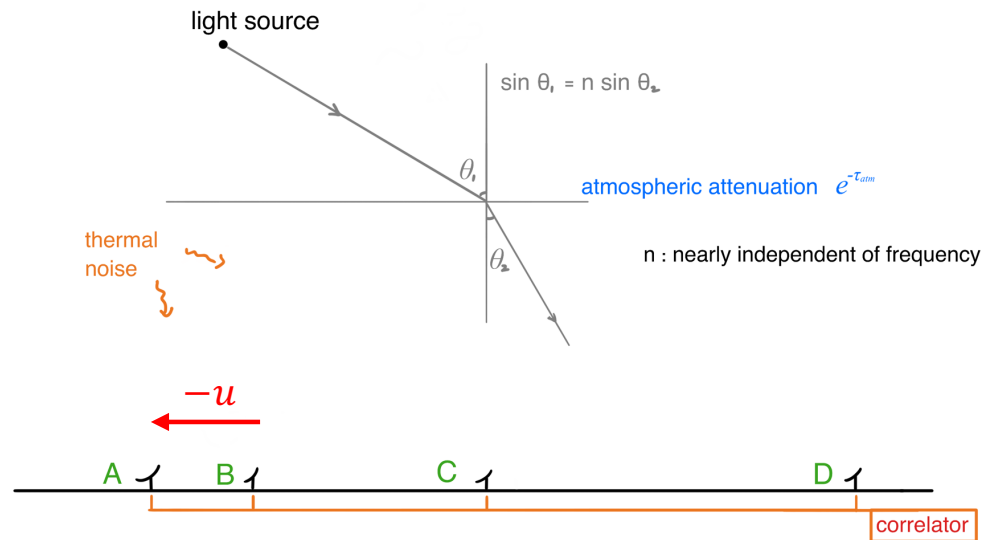
Mathematical form of fourier transform

Naiive inverse fourier transform

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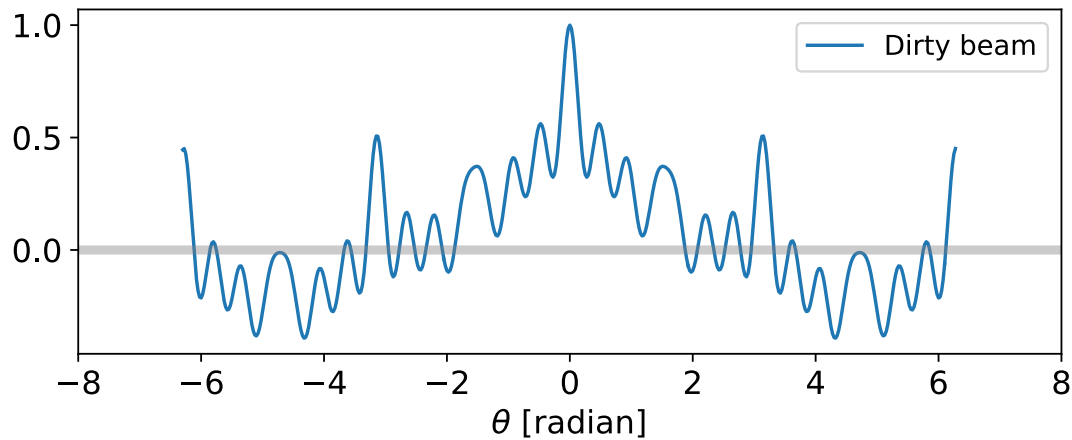
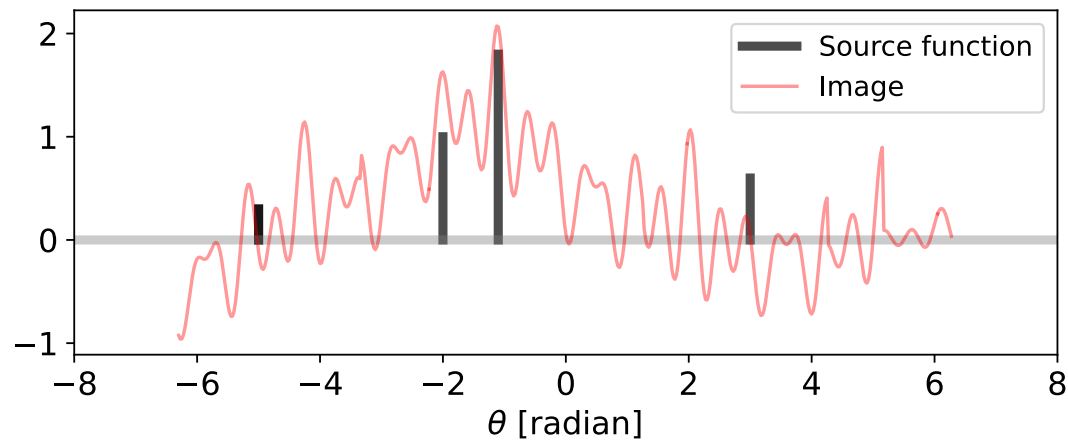
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$$\sum_{k=1}^M \int \delta(u - u_k) \cos(2\pi u \theta') du = \sum_{k=1}^M \cos(2\pi u_k \theta')$$

Resemblance to the diffraction
pattern of the diffraction
grating



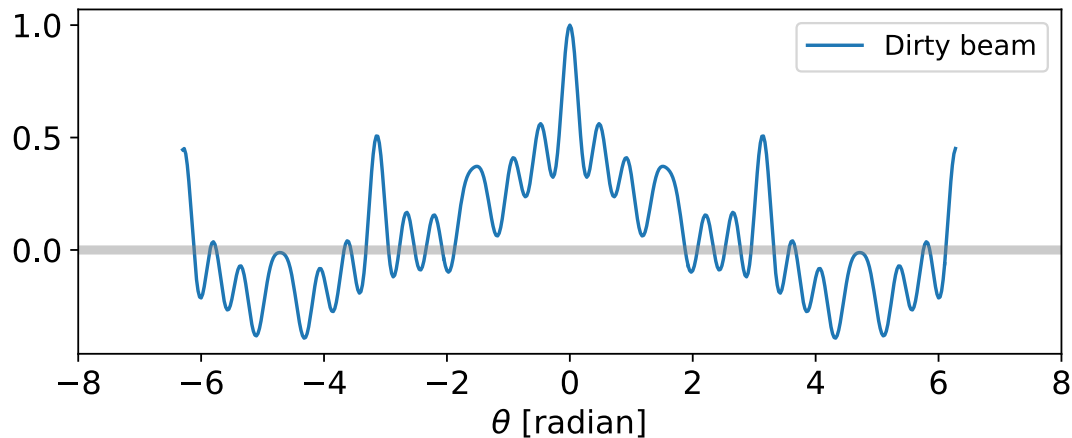
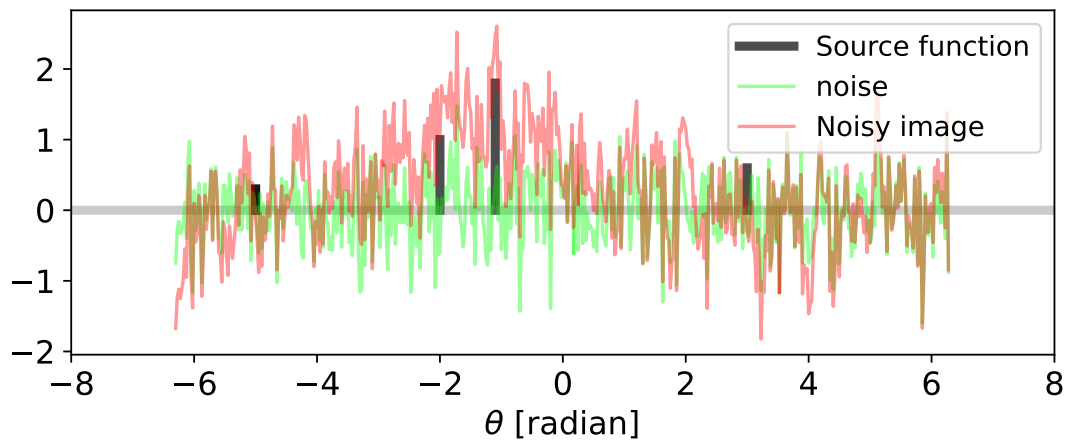
Naiive inverse fourier transform

$$[A(\theta')\widetilde{P(\theta')}]^D \equiv \int S(u)V(u)e^{-i2\pi u\theta'}du$$

$$= \underbrace{\left(\int S(u)e^{-i2\pi u\theta'}du\right)}_{\text{Dirty beam}} * \underbrace{\left(\int V(u)e^{-i2\pi u\theta'}du\right)}_{A(\theta')\widetilde{P(\theta')}}$$



$$\sum_{k=1}^M \int \delta(u - u_k) \cos(2\pi u\theta') du = \sum_{k=1}^M \cos(2\pi u_k\theta')$$



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$$\sum_{k=1}^M \int \delta(u - u_k) \cos(2\pi u\theta') du = \sum_{k=1}^M \cos(2\pi u_k\theta')$$

Need to deconvolve $[A(\theta')\widetilde{P(\theta')}]^D$

The CLEAN algorithm

Ansatz: intensity distribution of target sources can be approximated by point sources and 2D Gaussians with various pre-defined sizes

$$\text{Need to deconvolve } [A(\theta')\overline{P(\theta')}]^D$$

